

Odd-Cycle-Free Latin Squares: Structural Results, Exact Small-Order Obstructions, and Constructed Families

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Abstract

A Latin square is FFF when every two-line permutation in each of its row, column and symbol views has only even cycles. Direct products take the componentwise Boolean OR of the three failure patterns and are group-isotopic exactly when both factors are. Subsquare and congruence obstructions restrict possible constructions. Exact small-order checks exclude orders 6 and 10, while independently checked tables of orders 12 and 14 refute the former power-of-two existence conjecture. Finite-field constructions give additional infinite even-order families. An affine prime-successor family has an exact three-view criterion and parameter classification; a mixed-character-sum estimate gives more than $p/6000$ represented FFF classes at order $p + 1$ for every prime $p \geq 10^{12}$. This does not determine the complete order spectrum.

A row-dependent extension has pattern equal to the OR of its quotient and fibre patterns. When its fibres have no binary three-sorted quotient, all binary quotients descend from the outer square. For an elementary abelian 2-group as outer square, this recovers the layers intrinsically and classifies a fixed family of fibre assignments by affine orbits. A finite audit supplies 1703 pairwise distinct FFF main classes of order 20; the construction and Burnside's lemma then give explicit lower bounds in orders $20 \cdot 2^d$, not censuses of those orders. Direct binary-rank checks establish quotient freeness for these fixed order-20 inputs, independently of the computational order-10 exclusion.

For cell-dependent block substitutions, an exact return-map formula explains how an odd-cycle quotient can nevertheless lift to an FFF table. A common binary-field fibre constructs FFF squares of order $N2^D$ for every odd cubefree N when $\text{ord}_{\text{rad}(N)}(2) \mid D$ and $2^D > 3(N - 1)$. A distinguished affine corner trade preserves all three pattern bits and produces FFF tables with exactly one intercalate, including explicit orders 14 and 20. These constructions and their necessary boundaries do not decide order 18.

An exact position-value coloring formulation supplies a smaller three-view constraint representation with automatic color balance. Proofs, finite computations, and bounded search outcomes are distinguished throughout. Existence at order 18, the complete spectrum, and literature priority for the recent constructions remain unresolved.

Keywords. Latin square; quasigroup; Latin trade; direct product; main class; affine orbit; computational enumeration.

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Status, attribution, and evidence boundary

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The evidence intake for this edition is 28 September 2026 at 10:24 UTC, through register entry C271. It incorporates the preceding review corrections and selected completed results after C251. Work completed after this fixed intake is not silently incorporated. Earlier source packages, claims and numerical records are preserved. This is a new publicly distributable research snapshot, not journal acceptance.

This audit-corrected successor retains that scientific cutoff. It makes the nontrivial-block hypothesis explicit in the compact introduction and displays the already implied 1703-input class bounds. Fresh reruns and failure-path tests strengthen the accompanying audit software; they do not establish literature priority, human peer review, or a decision at order 18.

Theorems with written proofs are distinguished from statements labelled *Computational Theorem*, which require exact finite evidence in addition to a mathematical reduction. Explicitly scoped local exclusions and timeouts are not global theorems. Internal AI-assisted proof review, independent implementations, and machine proof checking are three different forms of checking; none is independent human refereeing. The compact companion imports the order-10 exclusion proved in the full report as an explicit computational dependency. Neither document resolves order 18 or asserts a complete higher-order main-class census.

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1 Introduction

An odd cycle in the permutation induced by two lines of a Latin square supports a particularly simple Latin trade: swap the two lines on that cycle. The move preserves the Latin property. In the row and column views it reverses the conventional row–column Latin sign; in the symbol view it preserves that sign but reverses the two parastrophically transported signs. It is therefore a natural primitive for attempts to construct local sign-reversing involutions in parastrophic versions of the Alon–Tarsi parity setting. The changes of the three individual line-parity products under row, column, and symbol cycle switches are already given by Wanless [Wan04, Proposition 1]. Our pairwise row–column, row–symbol, and column–symbol sign table is the corresponding explicit product calculation, not a claim of priority for the cycle-switch parity rule. Bérczi’s recent experimental program explicitly studies such local corrections as candidate ingredients for global constructions [Bér26]. That work is exploratory and formulates conjectural mechanisms rather than claiming proofs.

This paper isolates the underlying existence question. For each of the row, column, and symbol views, we ask whether some pair of lines induces a permutation with a nontrivial odd cycle. Recording the three answers gives a pattern in $\{F, T\}^3$. A square of pattern FFF has no such local odd-cycle move in any view; equivalently, it avoids this trade primitive for all three pairwise parity products. This distinction matters: a symbol-view odd trade need not reverse the fixed conventional row–column sign. Cycle structures of Latin-square lines have long been useful for parity and symmetry questions; see, for example, Kotlar [Kot12]. The alternating-cycle correspondence for pairwise unions of factors in the associated bipartite factorization is also classical; Wanless and Ihrig [WI05, Section 3] state the symbol-view version. The tripartite statement below applies its coordinate-conjugate versions.

Failure patterns are componentwise monotone from Latin subsquares to their ambient squares. They are also exactly preserved by a Latin quotient when all congruence blocks have size two, while any nontrivial odd congruence-block size forces pattern TTT. It follows that a hypothetical FFF quasigroup of order $2p$, with p an odd prime, must be

congruence-simple in every isotope. Iteration characterizes FFF in the restricted class admitting a prime-factor congruence series: all prime factors in the series must have size two. A twisted binary extension of C_4 also gives an explicit, census-free non-group-isotopic FFF loop of order 8. This excludes every congruence-imprimitive order-10 quasigroup construction. A later one-view reduction separately excludes imprimitive generated line actions; the exhaustive palette masters developed below subsequently exclude the remaining FFF case altogether.

A second subsquare theorem identifies the remaining binary line-incidence defect exactly. Nonuniversal vectors in the left binary line kernel are in bijection with half-order subsquares, and quotienting by the two universal constant relations groups them into complementary four-packs. This gives a closed counting formula and forces maximal binary line rank for every FFF order congruent to 2 modulo 4 and greater than 2. At order 10 this is a sharp necessary condition, but not a decision: the tracked non-FFF diagnostic tables already have the same maximal rank.

Specializing Moorhouse’s 3-net character formula, every additional prime-field line dependency defines a balanced three-sorted cyclic quotient homotopy; after isotopy its fibres form a quasigroup congruence. It follows that every Smith torsion prime divides the square order. Combining this with the congruence-pattern theorem shows, without a primitive-group classification, that one F view at order $2p$ for odd prime p forces the entire integer line-incidence lattice to be saturated. Thus a hypothetical order-10 FFF square has rank 28 over every field, although current non-FFF diagnostics already satisfy this stronger necessary condition.

We additionally derive a cycle-parity condition satisfied by every FFF line pair. It is equivalent to the even-cycle condition at degree 6 and leaves only one odd-cycle type, $5 + 3 + 2$, at degree 10. An exact order-8 scan shows that the condition is genuinely weaker than FFF, but it provides a smaller target for certified order-10 obstruction searches.

The exact even-cycle condition itself admits a second formulation. A permutation has only even cycles exactly when it preserves no odd-cardinality subset. Summing the induced k -subset actions of a Latin line family therefore gives a zero-one matrix for every relevant odd k exactly when the view is F. This yields both a structural lift characterization and a direct proof-producing encoding; at order 10, only the levels $k = 3$ and $k = 5$ are needed.

The local alternating bipartitions also glue globally. On the dual of a canonical three-coloured flag surface, assigning one basis vector to every edge colour defines an $\mathbb{F}_2^3$ -valued cochain whose colour-stratified coboundary detects the complete view pattern. It is a cocycle exactly for FFF squares; then its three coordinate classes sum to the first Stiefel–Whitney class and force equal holonomy fibres and divisibility of every component’s flag count. This gives a topological realization of the slice-wise even-cycle condition rather than another scalar aggregate.

For arbitrary Latin squares L and K we further prove

$$\text{pat}(L \times K) = \text{pat}(L) \text{ OR } \text{pat}(K)$$

componentwise. We also prove

$$L \times K \text{ is group-isotopic} \iff L \text{ and } K \text{ are both group-isotopic.}$$

The proofs are short but useful: the induced permutations are Cartesian products, and their cycle lengths are least common multiples of factor cycle lengths. The group-isotopy statement follows from a basis-family closure criterion.

The computational component includes an exhaustive order-8 census scan. McKay et al. [MMM07] enumerated isotopy and main classes of small Latin squares. The associated

ANU data page supplies one representative of each of the 283657 order-8 main classes [McK]. Scanning those representatives gives the full three-view pattern distribution and exactly 230 FFF main classes. Only 5 are group-isotopic; the remaining 225 are not. The frozen census input was downloaded again from the cited ANU URL and matched byte for byte. Its SHA-256 is recorded in Section 21.

Combining the structural and computational parts yields infinite families. Every order 2^m , $m \geq 3$, realizes all eight patterns, and every such order contains a non-group-isotopic FFF square. The latter existence claim now follows entirely theoretically from the explicit twisted order-8 loop; only the realization of all eight patterns uses the census. We make no claim of unrestricted cancellation for direct products. The later cycle-spectrum and quotient-recovery arguments do preserve distinctions within specified families, with all hypotheses stated explicitly.

Exhaustive generation of all 9408 reduced squares of order 6 finds no FFF square. The generated set agrees exactly with the 9408-square ANU reduced-order-6 file. At order 10, two exhaustive cycle-palette master searches prove nonexistence. Those negative results remain valid, but the former power-of-two existence conjecture is false: an explicit reduced order-12 FFF table passes all 198 three-view line-pair tests. Further tables of orders 14 and 36 pass their corresponding exact tests. The order-12 example alone refutes the conjecture; the next undecided small even non-power-of-two order is 18.

A separate elementary calculation shows that the classical round-robin midpoint Latin square of order $p + 1$ is FFF exactly for primes $p \equiv 3 \pmod{8}$. This supplies infinitely many non-power-of-two FFF orders. The order-20 instance and the independently audited order-10 exclusion give the exact threshold $h(5) = 2$. The underlying perfect 1-factorisation is classical; a broad priority claim for the FFF criterion is not made.

More generally, a three-point patch to a cyclic finite-field lift constructs an FFF square of order $m2^{\text{ord}_m(2)}$ for every odd $m > 1$. The proof controls the row, column, and symbol return translations on every subgroup coset, including a root/inverse choice for composite m ; the order-12 witness supplies the exceptional $m = 3$ case. This establishes an explicit upper bound on the dyadic exponent for each odd part, not the least exponent or a classification of all FFF orders. A literature-priority check and external expert review remain open.

The row-fibred construction permits different inner operations over a common quotient and satisfies the same componentwise pattern rule. When the fibres have no binary three-sorted quotient, every binary quotient descends to the outer square. Over an elementary abelian outer group, its characters recover all layers intrinsically. The main classes within the fixed family are then exactly the affine orbits of the fibre assignments. A complete audit of one order-20 trade family and two further controls supplies 514 source classes for the original numerical lower bounds. The enlarged representative audit and a pattern-preserving affine corner trade now raise this lower bound to 1703. Small binary-rank checks establish quotient freeness for all these fixed inputs without using the order-10 exclusion.

A general block-return formula explains why the quotient-OR identity does not extend to arbitrary cell-dependent blocks. A common binary-field fibre also extends the prime-square construction to every odd cubefree quotient, with explicit size conditions and construction-specific lower bounds. These general results do not settle order 18.

The paper does not claim a proof of the Alon–Tarsi conjecture. It establishes the precise limitations of one local-trade primitive, gives exact finite data, and provides a product construction that converts the order-8 phenomenon into infinite families.

The exposition places the structural theory, products, and finite census before the order-10 structural reductions and exhaustive master searches. The flag-surface and pair-label

constructions are retained as a second research layer rather than as prerequisites for the core FFF results.

2 Definitions and invariance

Let X be a set of size n . A *Latin square* on X is a map $L: X \times X \rightarrow X$ such that every row and every column is a bijection. For $a, c, u \in X$, define the row, column, and symbol-line bijections

$$r_a(c) = L(a, c), \quad q_c(a) = L(a, c), \quad s_u(c) = \text{the unique } a \text{ such that } L(a, c) = u.$$

For two lines in the corresponding view, put

$$\rho_{a,b} = r_b^{-1} r_a, \quad \kappa_{c,d} = q_d^{-1} q_c, \quad \sigma_{u,v} = s_v^{-1} s_u.$$

All three permutations act on an n -element complementary coordinate set.

Lemma 2.1. *If the two lines are distinct, the corresponding induced permutation is fixed-point-free.*

Proof. A fixed point of $\rho_{a,b}$ would give $L(a, c) = L(b, c)$ in one column, contrary to the Latin property. The column case is identical. A fixed point of $\sigma_{u,v}$ would say that the same cell contains the two distinct symbols u and v . $\square$

Definition 2.2. A view of L *fails* if one of its induced two-line permutations has an odd cycle of length greater than one. Write

$$\text{pat}(L) = (\text{row}(L), \text{col}(L), \text{sym}(L)) \in \{F, T\}^3,$$

where T means that the view fails. A square is *FFF* if $\text{pat}(L) = \text{FFF}$.

The terminology “fails” is chosen from the perspective of the local-trade program: a T view supplies an odd-cycle trade, while an F view does not.

Definition 2.3. Fix two rows $a \neq b$ and a union C of cycles of $\rho_{a,b}$. The associated *two-line trade* swaps the entries in rows a, b in precisely the columns in C . Column- and symbol-view trades are defined by permuting the coordinate roles.

Proposition 2.4. *A two-line trade produces another Latin square. Define the three line-sign products*

$$R(L) = \prod_a \text{sgn}(r_a), \quad C(L) = \prod_c \text{sgn}(q_c), \quad S(L) = \prod_u \text{sgn}(s_u),$$

and put

$$\epsilon_{RC} = RC, \quad \epsilon_{RS} = RS, \quad \epsilon_{CS} = CS.$$

The first is the conventional row-column Latin sign, while the other two are its parastrophic companions. If a trade has support size k and $\delta = (-1)^k$, then its sign ratios are

trade view	R'/R	C'/C	S'/S	$\epsilon'_{RC}/\epsilon_{RC}$	$\epsilon'_{RS}/\epsilon_{RS}$	$\epsilon'_{CS}/\epsilon_{CS}$
row	1	δ	δ	δ	δ	1
col	δ	1	δ	δ	1	δ
sym	δ	δ	1	1	δ	δ

Consequently an odd row- or column-view trade reverses the conventional ϵ_{RC} , whereas an odd symbol-view trade preserves ϵ_{RC} and reverses ϵ_{RS} and ϵ_{CS} .

Proof. Write $\pi = \rho_{a,b}$. Since $r_a(c) = r_b(\pi(c))$ and C is π -invariant, the two rows contain the same symbol set on C . Swapping those entries therefore preserves both row symbol sets, while each affected column merely swaps two entries. Thus the result is Latin.

Extend $\pi|_C$ by the identity outside C . The two changed row permutations are obtained by precomposition with π and π^{-1} , so their sign changes cancel and $R'/R = 1$. Each of the $|C|$ affected column permutations is composed with a transposition, so $C'/C = (-1)^{|C|}$. The same support contains the same $|C|$ symbols in the two rows; each associated symbol-position permutation swaps those two rows, giving $S'/S = (-1)^{|C|}$. This proves the first table row. Exchanging rows and columns proves the second.

For a symbol-view trade, each support column swaps the two selected symbols, and the cycle relation pairs the affected cells in exactly the same number of rows. Hence $R'/R = C'/C = (-1)^{|C|}$. The two changed symbol-position permutations differ from their predecessors by mutually inverse permutations on the support, so their sign changes cancel and $S'/S = 1$. Multiplying the appropriate line-sign ratios proves the last three columns. $\square$

Proposition 2.5 (Tripartite link formulation). *Regard the cells of L as the triangle decomposition*

$$\{\{a, c, L(a, c)\} : a, c \in X\}$$

of the complete tripartite graph on the row, column, and symbol copies of X . The links of the vertices in each part form a 1-factorization between the other two parts. An induced two-line cycle of length ℓ corresponds bijectively to an alternating cycle of length 2ℓ in the union of the two associated link factors. Hence L is FFF if and only if every two-factor union in all three link factorizations has only cycle lengths divisible by four.

Furthermore, the total number of 4-cycles among the two-factor unions is the same in all three views and equals the number of intercalates of L .

Proof. For a row a , its incident triangles give the perfect matching $M_a = \{\{c, r_a(c)\} : c \in X\}$ between columns and symbols. For distinct rows a, b , alternately following an edge of M_a and an edge of M_b sends a column c to $r_b^{-1}r_a(c) = \rho_{a,b}(c)$. Thus an ℓ -cycle of $\rho_{a,b}$ is exactly one alternating 2ℓ -cycle of $M_a \cup M_b$. Cyclically permuting the three coordinate classes proves the column and symbol statements and the FFF equivalence.

A 4-cycle in $M_a \cup M_b$ is exactly a 2-cycle of $\rho_{a,b}$, hence a 2-by-2 Latin subsquare on two rows, two columns, and two symbols. This is an intercalate. The same intercalate is counted once by its row pair, once by its column pair, and once by its symbol pair, proving the three equalities. $\square$

Lemma 2.6 (Global three-view sign identity). *For every Latin square of order n ,*

$$R(L)C(L)S(L) = (-1)^{\binom{n}{2}}.$$

Proof. Give each coordinate copy of X the same fixed order, and let $\tau(x, y) = (y, x)$ on X^2 . The $\binom{n}{2}$ off-diagonal pairs are the transpositions of τ , so $\text{sgn}(\tau) = (-1)^{\binom{n}{2}}$.

The row-entry map $A(r, c) = (r, L(r, c))$ is block diagonal by rows and has sign $R(L)$. The map $B(r, c) = (c, L(r, c))$ is a coordinate transposition followed by the block-diagonal family of column permutations, so its sign is $(-1)^{\binom{n}{2}}C(L)$. Likewise, $D(c, s) = (s, s_s(c))$ has sign $(-1)^{\binom{n}{2}}S(L)$. Since $A = \tau \circ D \circ B$, taking signs gives

$$R(L) = (-1)^{3\binom{n}{2}}S(L)C(L) = (-1)^{\binom{n}{2}}S(L)C(L),$$

which is equivalent to the claimed identity. $\square$

For later comparison, let $C_{v,k}$ denote the total number of k -cycles among all induced line-pair permutations in view v .

Proposition 2.7 (Aggregate cycle-profile baseline). *Let n be even and let p be prime. If*

$$\sum_{k=2}^n a_k x_k + b = 0 \pmod{p}$$

for every cycle-count vector $(x_2, \dots, x_n)$ of a fixed-point-free permutation of degree n , then, separately for each view v ,

$$\sum_{k=2}^n a_k C_{v,k} + \binom{n}{2} b = 0 \pmod{p}.$$

Moreover,

$$C_{\text{row},2} = C_{\text{col},2} = C_{\text{sym},2}$$

and

$$\sum_{v \in \{\text{row}, \text{col}, \text{sym}\}} \sum_{\substack{2 \leq k \leq n \\ k \text{ even}}} C_{v,k} = \frac{n(n-1)}{2} \pmod{2}.$$

Proof. The aggregate profile in one view is the sum of the $\binom{n}{2}$ fixed-point-free line-pair profiles, proving the first identity. The second identity is the intercalate bijection from Proposition 2.5. For the last identity, the sign of a fixed-point-free permutation of even degree is $(-1)^e$, where e is its number of even cycles. Multiplying relative-permutation signs over all line pairs in a view gives the product of its line signs. Lemma 2.6 gives the product over the three views as $(-1)^{n(n-1)/2}$, which proves the congruence. $\square$

Proposition 2.8 (Line-pair sign cut). *Fix one view and write its line bijections as λ_i . For $P_{ij} = \lambda_j^{-1} \lambda_i$, the pairs for which P_{ij} is odd form a cut of the complete graph on the lines. If n is even and the view is F , then the pairs for which P_{ij} has an odd number of cycles form the same cut. Consequently, for even $n > 2$, no view can have every induced line-pair permutation equal to one n -cycle.*

Proof. Put $\varepsilon_i = \text{sgn}(\lambda_i)$. Since sign is a homomorphism,

$$\text{sgn}(P_{ij}) = \varepsilon_i \varepsilon_j.$$

Thus P_{ij} is odd exactly when the endpoint signs differ. These pairs are the edges of the cut between positive and negative lines.

In an F view of even degree, every cycle of P_{ij} has even length and therefore negative sign. Hence $\text{sgn}(P_{ij}) = (-1)^{c(P_{ij})}$, where $c(P_{ij})$ is the number of cycles. Finally, if all relative permutations were n -cycles, every edge would be negative. A triangle of three lines would then cross the sign cut three times, whereas every cut meets a cycle evenly. $\square$

An *isotopy* independently permutes rows, columns, and symbols. A *parastrophy* permutes the three coordinate roles. Their combined orbits are the *main classes* (also called species).

Proposition 2.9. *Under isotopy, each of the three view bits is invariant. Under parastrophy, the three bits are permuted. Consequently, being FFF and having at least one failing view are main-class invariants.*

Proof. Let an isotopy act by permutations α, β, γ of rows, columns, and symbols. The transformed row maps satisfy $r'_{\alpha(a)} = \gamma r_a \beta^{-1}$, and hence

$$(r'_{\alpha(b)})^{-1} r'_{\alpha(a)} = \beta(r_b^{-1} r_a) \beta^{-1}.$$

Thus row cycle lengths are unchanged. The column calculation conjugates by α , and the symbol calculation conjugates by β . Parastrophy only permutes which coordinate pair is called row, column, or symbol. $\square$

3 Elementary structural results

3.1 Odd orders and group tables

Proposition 3.1. *Every Latin square of odd order $n > 1$ has pattern TTT.*

Proof. By Lemma 2.1, every induced permutation between two distinct lines is fixed-point-free on an odd set. If all of its nontrivial cycles were even, their lengths would have even sum. Hence it has an odd cycle of length greater than one. This argument applies separately to any pair of distinct rows, columns, and symbols. $\square$

Let G be a finite group and let $L_G(x, y) = xy$ be its Cayley table.

Proposition 3.2. *The row permutation induced by rows a, b of L_G is left multiplication by $b^{-1}a$. The column permutation induced by columns c, d is right multiplication by cd^{-1} . The symbol permutation induced by symbols u, v is right multiplication by $u^{-1}v$. Each cycle length is therefore the order of the corresponding group element.*

Proof. The row identity is $r_b^{-1} r_a(x) = b^{-1} ax$. Likewise $q_d^{-1} q_c(x) = xcd^{-1}$. Finally, $s_u(c) = uc^{-1}$ and $s_v^{-1}(a) = a^{-1}v$, whence $s_v^{-1} s_u(c) = cu^{-1}v$. Left or right multiplication by g has orbits equal to cosets of $\langle g \rangle$, all of length $\text{ord}(g)$. $\square$

Corollary 3.3. *The Cayley table of a finite group G is FFF if and only if G is a 2-group.*

Proof. If G is a 2-group, every nonidentity element has even order, so Proposition 3.2 gives no nontrivial odd cycle. Conversely, if $|G|$ has an odd prime divisor, Cauchy's theorem supplies a nonidentity element of odd order; the corresponding induced permutations give a failing view. $\square$

3.2 A closure criterion for group isotopy

Fix a base row b of a Latin square L and define

$$H_L(b) = \{r_b^{-1} r_x : x \in X\} \subseteq \text{Sym}(X).$$

Proposition 3.4 (Basis-family closure criterion). *A Latin square L is isotopic to a group table if and only if $H_L(b)$ is closed under composition. This condition is independent of the chosen base row.*

Proof. Suppose first that $H = H_L(b)$ is closed. It contains the identity, has $|X|$ distinct elements, and is a finite submonoid of a symmetric group; hence it is a group. For each $z \in X$, the values $(r_b^{-1} r_x)(z)$, as x varies, are all distinct, since a column of L contains every symbol once. Thus H acts regularly on X .

Fix $z_0 \in X$. Relabel rows by $x \mapsto h_x = r_b^{-1} r_x$. Use the regular action to identify every column z with the unique $h_z \in H$ satisfying $h_z(z_0) = z$, and identify every relabeled

symbol $r_b^{-1}(u)$ in the same way. The entry in row h_x and column h_z is represented by the unique element sending z_0 to $h_x h_z(z_0)$, namely $h_x h_z$. The resulting isotope is therefore the Cayley table of H .

Conversely, for a group table the family at base row b is the full left regular representation $\{\lambda_{b^{-1}x} : x \in G\}$ and is closed. Isotopy conjugates the family and relabels its members, preserving closure. The same argument for any base row proves base independence. $\square$

The same criterion can equivalently be written in the column or symbol view, by parastrophy.

4 Subsquare and congruence obstructions

The FFF property is hereditary for Latin subsquares in a stronger, componentwise sense.

Theorem 4.1 (Subsquare monotonicity). *If M is a Latin subsquare of L , then*

$$\text{pat}(M) \leq \text{pat}(L)$$

componentwise, with $F < T$.

Proof. Let the rows, columns, and symbols of M be R_0, C_0, S_0 . For $a, b \in R_0$, the ambient row permutation $r_b^{-1}r_a$ preserves C_0 : if $c \in C_0$, then $L(a, c) \in S_0$, and row b contains this symbol at a unique column of C_0 . Its restriction is precisely the induced row permutation of M . Thus every odd row cycle of M is also an odd row cycle of L . Interchanging rows and columns proves the column statement.

For $u, v \in S_0$ and $c \in C_0$, the row $s_u(c)$ lies in R_0 . In that row, the unique occurrence of v lies in a column $d \in C_0$. Hence $s_v^{-1}s_u$ preserves C_0 and restricts to the symbol permutation of M . This proves the symbol statement. $\square$

Corollary 4.2. *Every Latin subsquare of an FFF square is FFF. In particular, an FFF square contains no subsquare of odd order greater than one.*

Proof. The first assertion is Theorem 4.1 with $\text{pat}(L) = \text{FFF}$. The second follows from Proposition 3.1. $\square$

Together with Computational Theorem 7.1, the same implication excludes order-6 subsquares; that additional exclusion inherits the computational status of the order-6 theorem.

There is an exact linear-algebraic description of all half-order subsquares. Let M_L be the $3n$ -by- n^2 binary incidence matrix whose rows are the row, column, and symbol lines and whose columns are the occupied cells of L . Let

$$U = \{(\alpha \mathbf{1}_R, \beta \mathbf{1}_C, \gamma \mathbf{1}_S) : \alpha + \beta + \gamma = 0\} \subseteq \ker M_L^\top$$

and put

$$\delta_2(L) = (3n - 2) - \text{rank}_{\mathbb{F}_2} M_L.$$

Theorem 4.3 (Binary line kernel and half-order subsquares). *Individual embedded half-order Latin subsquares of L are in bijection with*

$$\ker M_L^\top \setminus U.$$

Equivalently, the nonzero classes in $\ker M_L^T/U$ parametrize their complementary four-packs. Thus their number is exactly

$$4(2^{\delta_2(L)} - 1). \quad (1)$$

If $n = 2m$, where $m > 1$ is odd, then any one F -view forces

$$\text{rank}_{\mathbb{F}_2} M_L = 3n - 2. \quad (2)$$

In particular, (2) holds for every FFF square of order congruent to 2 modulo 4 and greater than 2.

Proof. A left-kernel vector is a triple (a, b, c) satisfying

$$a_r + b_j + c_{L(r,j)} = 0. \quad (3)$$

If one coordinate, say a , is constant, then fixing a column and varying its rows shows that c is constant; fixing a row then makes b constant. Hence every vector outside U has all three coordinates nonconstant.

Fix a row r . Since that row permutes the symbols, (3) identifies the multiset of c -values with that of $a_r + b$. Choosing rows with $a_r = 0$ and $a_r = 1$ gives

$$\text{wt}(c) = \text{wt}(b) = n - \text{wt}(b).$$

Thus $n = 2m$ and b, c each have weight m . The analogous fixed-column argument gives $\text{wt}(a) = m$. For their binary fibres A_x, B_y, C_z , equation (3) now gives

$$L(A_x, B_y) \subseteq C_{x+y} \quad (x, y \in \mathbb{F}_2).$$

Every set has size m ; Latinness makes each of the four displayed blocks an order- m Latin subsquare.

Adding an element of U complements zero or two of the three partitions and selects a different block in the same four-pack. Conversely, if $R_0 \times C_0$ is a half-order subsquare with symbol set S_0 , row and column counting forces the complementary pattern

$$R_0 \times C_0, R_1 \times C_1 \longrightarrow S_0, \quad R_0 \times C_1, R_1 \times C_0 \longrightarrow S_1.$$

The three complement indicators satisfy (3), and the chosen block recovers them uniquely. This proves both bijections. Since U has dimension two and the quotient has dimension $\delta_2(L)$, counting its nonzero classes proves the counting formula.

Finally suppose that $m > 1$ is odd and $\delta_2(L) > 0$. The preceding construction gives an order- m subsquare. Two distinct rows inside it induce a fixed-point-free permutation preserving its m columns. Such a permutation on an odd set has a nontrivial odd cycle, contradicting an F row view. The two parastrophes give the same conclusion for either other view, which proves the rank formula. The exception $m = 1$ is real: the order-2 FFF square has four singleton subsquares and $\delta_2 = 1$. $\square$

The complete finite audit independently constructs both sides of the bijection on 9778 tables and obtains exact set equality in every case. In particular, the 230 order-8 FFF representatives have

$$\delta_2 = 0, 1, 2, 3 \quad \text{with multiplicities} \quad 63, 156, 10, 1,$$

which explains their half-order-subsquare distribution $0^{63}4^{156}12^{10}28^1$. A hypothetical order-10 FFF square must have binary line rank 28 and no order-5 subsquare. This condition is

not sufficient: all 135 tracked order-10 diagnostic tables already satisfy it, none is FFF, and that corpus is not exhaustive.

The same kernel equation has a uniform interpretation over every prime field, now in terms of quotient homotopies rather than individual subsquares. Moorhouse’s character formula for loop-coordinatized 3-nets gives the general identity $\text{rank}_p M_L = 3n - 2 - \dim \text{Hom}(G, \mathbb{F}_p)$ and identifies its defect with elementary-abelian quotient characters [Moo91, Theorem 4.2]. We record a direct three-sorted proof in our notation because the individual cyclic quotient is the input needed for the FFF corollary below. The author’s publication page flags an erratum for the original article; the reference entry therefore points to the author-hosted article copy and the separate erratum listing. The direct proof given here, rather than an unexamined appeal to the original pagination, is the logical basis for the specialization used below.

Theorem 4.4 (Prime-field line quotients). *Let p be prime and let $(a, b, c) \in \ker_{\mathbb{F}_p} M_L^T$ be nonconstant modulo the two-dimensional constant subspace. Then $p \mid n$, each of a, b, c takes every value exactly n/p times, and their fibres satisfy*

$$L(R_x, C_y) \subseteq S_{-x-y} \quad (x, y \in \mathbb{F}_p).$$

After a suitable isotopy these three fibre partitions become one quasigroup congruence with p blocks of size n/p and quotient isotopic to C_p .

Over characteristic zero, M_L has rank $3n - 2$. Every prime dividing a nonzero Smith invariant factor of the integer matrix M_L divides n .

Proof. If one of a, b, c is constant, fixing a line in either other sort and using Latinness makes the remaining two constant. Let $\mu_b(t)$ and $\mu_c(t)$ be value multiplicities. Since every row permutes the symbols, equation (3), now read in $\mathbb{F}_p$, gives

$$\mu_c(t) = \mu_b(-a_r - t).$$

Comparing rows with $a_r - a_{r'} \neq 0$ makes μ_b invariant under a nonzero translation. That translation generates the additive group of $\mathbb{F}_p$, so μ_b is uniform. Thus $p \mid n$ and both b, c are balanced; the column argument balances a . The displayed fibre rule is the kernel equation itself.

Choose bijections from one partitioned n -set to the row, column, and symbol sets, matching equally labelled fibres. Transporting L by these three bijections gives an isotope whose block operation is $x * y = -x - y$ on $\mathbb{F}_p$. It is Latin and isotopic to addition, so block equivalence is compatible with multiplication and both divisions and is a quasigroup congruence.

In characteristic zero a finite nonempty multiset cannot be invariant under a nonzero translation, because its iterated translates would have infinite support. Hence only the two constant dependencies remain. Finally, if a prime divides a nonzero Smith invariant factor, reduction modulo that prime lowers the rank; the first part then forces the prime to divide n . $\square$

An eight-worker audit checks this theorem over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7, \mathbb{F}_{11}$ on 9778 tables. It verifies all 1160 nonuniversal quotient classes and a separate implementation reconstructs 48890 modular ranks, with no errors.

We next use congruences in the quasigroup algebra with multiplication and the two division operations. All congruence blocks have equal size and form a Latin quotient.

Theorem 4.5 (Congruence obstructions). *Let L be a finite Latin square.*

- (i) If L has a congruence with odd block size greater than one, then $\text{pat}(L) = \text{TTT}$.
- (ii) If the congruence blocks have size two and the quotient is Q , then $\text{pat}(L) = \text{pat}(Q)$ componentwise.

Proof. For (i), choose distinct rows x, x' in one congruence block A and any column block B . The two left translations map B bijectively onto the same product block. Thus $L_{x'}^{-1}L_x$ preserves B and is fixed-point-free there. Since $|B|$ is odd, its restriction has a nontrivial odd cycle. Applying the same argument to the two relevant parastrophes proves the column and symbol bits.

For (ii), label every two-element block by $(i, 0), (i, 1)$, where $i \in Q$. The unique form of a binary Latin square gives a function $f: Q^2 \rightarrow \mathbb{F}_2$ such that

$$(i, a) * (j, b) = (i * j, a + b + f(i, j)).$$

Rows in the same quotient block induce only transpositions inside the column fibres. For rows in different quotient blocks, the lift above a quotient cycle of length ℓ is either two cycles of length ℓ or one cycle of length 2ℓ , according to its binary voltage. Thus an ambient odd cycle can occur only above an odd quotient cycle.

Conversely, suppose quotient rows i, i' induce an odd cycle C of length $\ell > 1$. Above a quotient column $j \in C$, the lifted row permutation comparing (i, a) with (i', a') has the form

$$(j, b) \mapsto (\pi(j), b + a + a' + f(i, j) + f(i', \pi(j))).$$

Its voltage around C is $a + a' + S$, because ℓ is odd, where S is independent of a, a' . Choose $a + a' = S$. The voltage is then zero, so the lift of C consists of two cycles of the same odd length ℓ . Hence a quotient row witness lifts to an ambient row witness. This proves equality of the row bits. The same complete argument in the parastrophes proves equality of the column and symbol bits. $\square$

Corollary 4.6 (Order- $2p$ simplicity). *Let p be an odd prime. Every quasigroup of order $2p$ with at least one F view is congruence-simple, and every isotope of it is congruence-simple.*

Proof. A nontrivial congruence has block-size/quotient-order pair $(p, 2)$ or $(2, p)$. The first is excluded by Theorem 4.5(i). In the second, the quotient has pattern TTT by Proposition 3.1, so (ii) excludes it. Both congruence cases force TTT, contradicting even one F view. Finally, the full view pattern is isotopy invariant by Proposition 2.9, and the same argument applies to every isotope. $\square$

Corollary 4.7 (Order- $2p$ line-incidence saturation). *Let $p > 1$ be an odd prime and let L have order $2p$. If even one view of L is F , then M_L has rank $6p - 2$ over every field, all its nonzero Smith invariant factors are 1, and*

$$\mathbb{Z}^{6p} / \text{im}_{\mathbb{Z}} M_L \cong \mathbb{Z}^2.$$

In particular, this holds for every hypothetical order-10 FFF square.

Proof. By Theorem 4.4, a torsion prime must divide $2p$. A characteristic-two defect yields, after isotopy, a congruence with two blocks of odd size p , so Theorem 4.5(i) gives pattern TTT. A characteristic- p defect yields blocks of size two; part (ii) identifies the pattern with that of the odd-order quotient, again TTT by Proposition 3.1. Isotopy invariance contradicts the assumed F view in either case. Thus no Smith prime remains, proving saturation and the displayed cokernel. $\square$

Thus a hypothetical order-10 FFF square must be isotopically congruence-simple. This removes every congruence-imprimitive quasigroup construction, but it does not rule out simple quasigroups and therefore does not decide order 10.

The binary equality iterates through congruence series.

Corollary 4.8 (Prime-factor congruence series). *Suppose that a quasigroup admits a chain of congruence quotients ending in the one-element quasigroup, with every congruence block size prime. It is FFF if all those primes are 2, and it has pattern TTT if at least one is odd.*

Proof. A binary step preserves the entire pattern by Theorem 4.5(ii). At an odd step, Theorem 4.5(i) gives TTT independently of the lower quotient. Iterating proves both statements. $\square$

This characterizes FFF within this congruence-decomposable class. It does not imply the former global power-of-two conjecture, which is refuted by the order-12 witness in Computational Theorem 8.1.

The same theorem also gives a census-free nongroup example.

Theorem 4.9 (Explicit nongroup FFF loop). *On $\mathbb{Z}_4 \times \mathbb{F}_2$, define*

$$(i, a) * (j, b) = (i + j, a + b + \mathbf{1}_{(i,j)=(1,1)}),$$

with coordinates taken modulo 4 and 2, respectively. This is an FFF loop of order 8 that is not group-isotopic.

Proof. The identity is $(0, 0)$, and the first-coordinate fibres form a congruence with quotient C_4 . The quotient is FFF by Corollary 3.3, so binary pattern equality makes the loop FFF. It is not associative because

$$((1, 0) * (1, 0)) * (2, 0) = (0, 1), \quad (1, 0) * ((1, 0) * (2, 0)) = (0, 0).$$

At the identity base row, the basis family in Proposition 3.4 is the family of left translations. Closure would force $L_x L_y = L_{x*y}$ and hence associativity. The displayed failure is therefore a closure failure, so the loop is not group-isotopic. $\square$

5 Direct products

Let L be a Latin square on X and K a Latin square on Y . Their direct product $P = L \times K$ is the Latin square on $X \times Y$ defined by

$$P((x, y), (x', y')) = (L(x, x'), K(y, y')).$$

Lemma 5.1 (Product-cycle formula). *Let π and τ be permutations. If x lies on a π -cycle of length m and y lies on a τ -cycle of length n , then (x, y) lies on a cycle of length $\text{lcm}(m, n)$ under $\pi \times \tau$.*

Proof. The pair returns after $t > 0$ steps exactly when both $m \mid t$ and $n \mid t$. The least such t is $\text{lcm}(m, n)$. $\square$

Theorem 5.2 (Pattern product theorem). *For arbitrary Latin squares L and K ,*

$$\text{pat}(L \times K) = \text{pat}(L) \text{ OR } \text{pat}(K)$$

componentwise.

Proof. Rows of P are indexed by (a, α) , and their row maps are $r_{(a, \alpha)}^P = r_a^L \times r_\alpha^K$. Hence the induced permutation between (a, α) and (b, β) is

$$\rho_{(a, \alpha), (b, \beta)}^P = \rho_{a, b}^L \times \rho_{\alpha, \beta}^K.$$

If one coordinate of the two product rows is equal, the corresponding factor is the identity. The identical calculation for column maps gives $\kappa^P = \kappa^L \times \kappa^K$.

For the symbol view,

$$s_{(u, \eta)}^P(c, d) = (s_u^L(c), s_\eta^K(d)),$$

and therefore $\sigma^P = \sigma^L \times \sigma^K$. Thus the same argument applies in all three views.

If a factor has a nontrivial odd cycle in one view, keep the other factor line fixed. The identity contributes cycle length 1, so the product contains the same odd cycle. Conversely, suppose a product induced permutation has an odd cycle of length $\ell > 1$. By Lemma 5.1, $\ell = \text{lcm}(m, n)$ for factor cycle lengths m, n . Both are odd, and at least one is greater than one. That factor already has a nontrivial odd cycle in the same view. This proves the Boolean OR formula in each view. $\square$

Corollary 5.3. *$L \times K$ is FFF if and only if both L and K are FFF. In particular, product with a finite 2-group table preserves the full pattern of the other factor.*

Proof. The first assertion is the all- F case of Theorem 5.2. For the second, combine the same theorem with Corollary 3.3. $\square$

Theorem 5.4 (Group-isotopy product theorem). *$L \times K$ is group-isotopic if and only if both L and K are group-isotopic.*

Proof. At a product base row (b, c) , the basis family factors as

$$H_{L \times K}(b, c) = H_L(b) \times H_K(c).$$

If both factor families are closed, their Cartesian product is closed. Conversely, both factor families contain the identity. If the product family is closed, then $(h_1, \text{id})(h_2, \text{id})$ lies in it for all $h_1, h_2 \in H_L(b)$, proving closure of $H_L(b)$; using (id, k_i) proves closure of $H_K(c)$. Apply Proposition 3.4. $\square$

Corollary 5.5. *Let L be a non-group-isotopic FFF Latin square of order n . For every $m \geq 0$, the product $L \times C_2^m$ is a non-group-isotopic FFF Latin square of order $n2^m$.*

Proof. The table of C_2^m is FFF by Corollary 3.3; apply Corollary 5.3 and Theorem 5.4. $\square$

6 A cycle-parity obstruction

There is a useful necessary condition weaker than FFF. For a fixed-point-free permutation π of even degree n , let

$$q(\pi) = \#\{\text{cycles of length } 2 \pmod{4}\} \pmod{2}.$$

Proposition 6.1 (Cycle-parity obstruction). *If π has only even cycles, then*

$$q(\pi) = n/2 \pmod{2}.$$

Consequently, every FFF Latin square satisfies this identity for every pair of lines in all three views.

Proof. Write the cycle lengths as $2d_1, \dots, 2d_s$. Dividing their sum by two gives

$$n/2 = d_1 + \dots + d_s \pmod{2}.$$

The summand d_i is odd precisely when the corresponding cycle has length 2 modulo 4. This proves the identity. Induced two-line permutations are fixed-point-free by Lemma 2.1, so the assertion applies independently to every row, column, and symbol pair. $\square$

Call a square *QQQ* when every line-pair permutation in all three views satisfies Proposition 6.1. This is auxiliary notation, not a replacement for the failure pattern. At degree 6, the fixed-point-free partitions are

$$6, \quad 4 + 2, \quad 3 + 3, \quad 2 + 2 + 2.$$

The Q condition rejects exactly $3 + 3$, so QQQ is equivalent to FFF at order 6. At degree 10, the Q-inadmissible partitions are

$$7 + 3, \quad 5 + 5, \quad 4 + 3 + 3, \quad 3 + 3 + 2 + 2.$$

Among the admissible partitions, $5 + 3 + 2$ is the only one with odd cycles. Equivalently, Q-admissibility in degree 10 excludes precisely

$$\text{fix}(\pi^3) = 6, \quad \text{fix}(\pi^5) = 10, \quad \text{fix}(\pi^7) = 7.$$

The exact order-8 census contains 1285 QQQ representatives: all 230 FFF representatives and 1055 further representatives. Thus QQQ is strictly weaker than FFF. Nevertheless, a certified QQQ-unsatisfiability result at order 10 would immediately exclude FFF and uses a smaller cycle classification than the full even-cycle coloring model.

7 Computational results

The following statements are exhaustive finite computations. Their inputs, programs, commands, and outputs are specified in Section 21.

Computational Theorem 7.1 (Small orders). *The unique reduced Latin square of order 2 is FFF, and all four reduced Latin squares of order 4 are FFF. None of the 9408 reduced Latin squares of order 6 is FFF. More precisely, their patterns are*

$$6600 \text{ TTT}, \quad 936 \text{ TFF}, \quad 936 \text{ FTF}, \quad 936 \text{ FFT}.$$

The generator constructs reduced Latin squares by exact cover and loads no cached square list. Every generated square was checked to be Latin and reduced, all 9408 table strings were distinct, and the generated set agrees exactly with the ANU reduced-order-6 data file [McK]. Proposition 2.9 therefore extends the nonexistence statement from reduced squares to all Latin squares of order 6.

For all order-8 record references below, “local source line” means the one-based physical line number in the frozen file consumed by the scanner. The ANU catalogue numbers its records from zero, so local source line j corresponds to ANU record $j - 1$.

Computational Theorem 7.2 (Order-8 pattern census). *Across the 283657 order-8 main-class representatives, the pattern distribution is*

<i>Pattern</i>	<i>Count</i>	<i>Pattern</i>	<i>Count</i>	<i>Pattern</i>	<i>Count</i>	<i>Pattern</i>	<i>Count</i>
FFF	230	FFT	81	FTF	40	FTT	315
TFF	133	TFT	209	TTF	232	TTT	282417

Among the 230 FFF main classes, exactly 5 are group-isotopic and 225 are not group-isotopic.

Computational Theorem 7.3 (Subsquare census of the FFF classes). *Among the 230 FFF representatives, the number of intercalates ranges from 2 to 112. Each endpoint is attained by exactly one representative: local source line 10776 (ANU record 10775) is the unique minimum class and is not group-isotopic, while local source line 1 (ANU record 0) is the unique maximum class and is group-isotopic. No representative contains an order-3 subsquare. The distribution of the number of order-4 subsquares is*

$$0^{63}, \quad 4^{156}, \quad 12^{10}, \quad 28^1,$$

where exponents count representatives.

The cycle-parity obstruction of Proposition 6.1 is also scanned on the complete order-8 input. Exactly 1285 representatives are QQQ: the 230 FFF representatives and 1055 non-FFF representatives. This is an exact census fact and demonstrates that QQQ is necessary but not sufficient for FFF.

Computational Theorem 7.4 (Affine cycle-profile saturation). *Over each of $\text{GF}(2)$, $\text{GF}(3)$, $\text{GF}(5)$, and $\text{GF}(7)$, every affine relation among the aggregate cycle profiles of all 283657 order-8 main-class representatives lies in the span of the relations in Proposition 2.7.*

Incremental finite-field row reduction includes a constant coordinate and processes the complete census directly. The analogous order-6 audit matches the baseline over the three odd fields but has three additional Latin-specific relations over $\text{GF}(2)$. Conversely, three extra binary relations visible in the tracked 135-table order-10 partial corpus are refuted by explicit Latin squares obtained from a deterministic intercalate-trade walk on the D_{10} table. Thus those order-10 relations were sampling artifacts. This does not constitute an exhaustive order-10 classification.

The associated validator checks every order-2, order-3, and order-4 subsquare and finds no violation of Theorem 4.1. The multiples of four in the order-4 distribution also follow directly: an order- m subsquare inside an order- $2m$ square forces the three complementary blocks to be Latin subsquares, so half-order subsquares occur in complementary quartets.

The non-symmetric labels in the table refer to the particular representatives in the cited ANU file. Parastrophy permutes the three bits, so, for example, an ordered label such as FFT is not itself a main-class invariant. The FFF count, the existence of every pattern, and the group-isotopy split are unaffected by this distinction.

The scan tests every unordered pair of lines in all three views. A separate closure-family computation reproduces the 5/225 split using Proposition 3.4. The input is the ANU/McKay main-class file, whose stated size and equivalence convention are given on the data page [McK]; our frozen copy is an exact byte match to the file served by that URL on August 9, 2026.

Corollary 7.5. *For every $m \geq 3$, Latin squares of order 2^m realize all eight patterns. Every such order also contains a non-group-isotopic FFF Latin square.*

Proof. Computational Theorem 7.2 supplies one order-8 square of each pattern. Take direct products with the FFF group table of C_2^{m-3} . The pattern statement follows from Theorem 5.2. For the second assertion, instead start from the explicit loop in Theorem 4.9; non-group-isotopicity is preserved by Theorem 5.4. $\square$

This is an existence result only. It neither counts higher-order main classes nor asserts that direct products of distinct order-8 classes remain distinct.

As an additional check, Wanless' complete online list of order-8 species with transitive autotopism group contains 11 squares [Wan]. The frozen local input is a byte-for-byte match to the online file (SHA-256 1c777328c099e3c242a49f9f2019d83314d7ab800505c7d9291bdb16ef029146). All 11 have pattern FFF; 5 are group-isotopic and 6 are not. This secondary computation is not needed for Corollary 7.5.

8 Explicit non-power-of-two witnesses and a dyadic lift bound

The computational nonexistence result at order 10 does not extend to all even non-powers of two. The following finite claims are checked by direct three-view cycle scans, not by an uncompleted solver search.

Computational Theorem 8.1 (Explicit non-power-of-two witnesses). *There are FFF Latin squares of orders 12, 14, and 36. The displayed order-12 table is reduced. Thus the former conjecture that FFF squares of order greater than one exist only at powers of two is false.*

L_{12}	0	1	2	3	4	5	6	7	8	9	10	11
0	0	1	2	3	4	5	6	7	8	9	10	11
1	1	0	3	2	5	4	7	6	10	11	8	9
2	2	3	0	1	7	6	5	4	9	8	11	10
3	3	2	1	0	6	7	4	5	11	10	9	8
4	4	7	5	6	8	11	9	10	3	0	2	1
5	5	6	4	7	9	10	8	11	2	1	3	0
6	6	5	7	4	10	9	11	8	1	2	0	3
7	7	4	6	5	11	8	10	9	0	3	1	2
8	8	9	10	11	3	2	1	0	6	7	4	5
9	9	8	11	10	0	1	2	3	4	5	6	7
10	10	11	8	9	2	3	0	1	5	4	7	6
11	11	10	9	8	1	0	3	2	7	6	5	4

Table 1: An explicit reduced order-12 FFF witness. Every row and column is a permutation; all 198 induced line-pair permutations across the three views have only even cycles.

For odd $m > 1$, let $h(m)$ be the least $a \geq 0$ for which an FFF square of order $m2^a$ exists. The odd-order obstruction, the order-6 computation, and Computational Theorem 8.1 give $h(3) = 2$ and $h(7) = 1$; the order-36 table gives $h(9) \leq 2$. Nothing here decides existence at order 18.

The classical polygon-and-centre round-robin factorisation gives a separate, particularly simple family. Cheng and Sgueglia [CS26, Introduction] recall this factorisation and its prime-order perfectness; the following three-view FFF criterion is proved directly, without claiming priority for the construction or an exhaustive literature classification.

Theorem 8.2 (Prime round-robin family). *Let p be an odd prime and $X = \mathbb{F}_p \cup \{\infty\}$. Define M by $M(\infty, \infty) = \infty$, $M(\infty, x) = M(x, \infty) = x$ for finite x , $M(x, x) = \infty$ for finite x , and $M(x, y) = (x + y)/2$ for distinct finite x, y . Then M is Latin and is FFF if and only if $p \equiv 3 \pmod{8}$.*

Proof. In each finite row a , the affine map $y \mapsto (a + y)/2$ takes the value a only at $y = a$. Replacing that entry with ∞ and placing a at column ∞ makes the row a permutation. The infinity row is the identity. Symmetry gives the Latin property for columns as well.

Write R_a for a row permutation and $d = \text{ord}_p(2)$. R_∞ is the identity, while R_a swaps a and ∞ and acts, after translating a to zero, as multiplication by $1/2$ on $\mathbb{F}_p^*$. Hence an infinity/finite row pair has type $2^{d(p-1)/d}$. For distinct finite a, b , put $P = R_b^{-1}R_a$. Apart from a , $2b - a$, and ∞ , the map is translation $x \mapsto x + a - b$. At the exceptional inputs,

$$P(a) = b, \quad P(2b - a) = \infty, \quad P(\infty) = 2a - b,$$

and $P(b) = a$. In the single p -cycle of the translation, the segment $2b - a \rightarrow b \rightarrow a \rightarrow 2a - b$ is replaced by the 2-cycle (a, b) and the connection $2b - a \rightarrow \infty \rightarrow 2a - b$. The remaining cycle has length $p - 1$, including when $p = 3$. Thus all finite/finite row pairs have type $(2, p - 1)$, and the row view is F exactly when d is even. Symmetry gives the identical column condition.

For a symbol s , let T_s send a column to the row containing s in it. Then T_∞ is the identity; for finite s ,

$$T_s(\infty) = s, \quad T_s(s) = \infty, \quad T_s(x) = 2s - x \quad (x \in \mathbb{F}_p \setminus \{s\}).$$

Thus each T_s is a fixed-point-free involution. An infinity/finite symbol pair has type $2^{(p+1)/2}$. Affine relabelling of the finite points reduces any two distinct finite symbols to $0, 1$. The relative permutation T_1T_0 has precisely the cycles

$$(0, 1, 3, 5, \dots, p - 2), \quad (\infty, 2, 4, 6, \dots, p - 1),$$

both of length $(p + 1)/2$. Hence the symbol view is F exactly when $p \equiv 3 \pmod{4}$.

For $p \equiv 7 \pmod{8}$, the supplement $(2/p) = 1$ and Euler's criterion make d divide the odd number $(p - 1)/2$, so the row and column views are T. For $p \equiv 3 \pmod{8}$, $(2/p) = -1$ makes d even; the symbol view is also F. Primes $p \equiv 1 \pmod{4}$ already fail in the symbol view. This proves the equivalence. $\square$

By Dirichlet's theorem, infinitely many primes satisfy $p \equiv 3 \pmod{8}$; apart from $p = 3$, their orders $p + 1$ are not powers of two. At $p = 19$ the table has order 20. Together with the odd-order obstruction Proposition 3.1 and the separately audited computational exclusion of order 10, this gives the sharp threshold $h(5) = 2$. This last equality depends on the exact order-10 computation, whereas Theorem 8.2 is algebraic.

A second class at order 20. The classical no-hexagon STS(19), with explicit starters in Erskine–Griggs [EG24, Section 3.2] and classification context in Colbourn et al. [CFG⁺10], yields a Steiner loop whose 570 two-line permutations have only even cycles. Its three-view cycle profile has no 18-cycle, whereas the midpoint table at $p = 19$ has type $(2, 18)$ in every row pair. Since the combined cycle-type profile is invariant under paratopy, these two examples are not in the same main class. This initial two-class control does not reproduce the external STS classification. The separate trade-family audit in Section 11 strengthens the represented class lower bound to 514.

Theorem 8.3 (Three-point lift bound). *For every odd integer $m > 1$, there is an FFF Latin square of order $m2^{\text{ord}_m(2)}$. Consequently*

$$h(m) \leq \text{ord}_m(2),$$

and FFF squares exist at every order $m2^a$ with $a \geq \text{ord}_m(2)$.

Proof. The case $m = 3$ is supplied by the order-12 table above, since $\text{ord}_3(2) = 2$. Suppose $m > 3$. Set $e = \text{ord}_m(2)$, take $K = \mathbb{F}_{2^e}$, and choose $r \in K^\times$ of multiplicative order m . Its existence follows from $m \mid (2^e - 1)$ and cyclicity of $K^\times$.

For coefficients $s_i \in K$, $i \in \mathbb{Z}_m$, define a multiplication on $\mathbb{Z}_m \times K$ by

$$(i, a) * (j, b) = (i + j, a + r^i b + s_i r^j). \quad (4)$$

The quotient coordinate and the nonzero coefficient r^i make this a Latin square. Put $t_i = r^{-i} s_i$ and regard t_i as a function $t(r^i)$ on $H = \langle r \rangle$. For a coset C of any nontrivial subgroup of H , write

$$S_C = \sum_{x \in C} x t(x), \quad T_C = \sum_{x \in C} x^{-1} t(x).$$

We first verify that pairwise distinct t_i and nonzero S_C, T_C for all such cosets imply FFF in all three views.

Two distinct lines in the same quotient fiber have relative permutations that are nonzero translations of K : for rows the translation is $r^{-i}(a + a')$, for columns it is $r^i(b + b')$, and for symbols it is $r^{-(k-j)}(u + u')$ in each column fiber. Each is an involution without fixed points. For distinct quotient rows i, k , put $d = i - k$ and let ℓ be the order of d in $\mathbb{Z}_m$. Directly composing their row maps gives

$$R_{(k, a')}^{-1} R_{(i, a)}(j, b) = (j + d, r^d b + r^{-k}(a + a') + r^{j-k}(s_i + r^d s_k)).$$

The quotient shifts by d ; its ℓ -step fiber return is

$$b \mapsto b + r^{j-i}(s_i + r^d s_k) = b + r^j(t_i + t_k).$$

For distinct quotient columns j, l , the quotient shift is $j - l$ and the one-step map, with $d = j - l$, is

$$C_{(l, b')}^{-1} C_{(j, b)}(i, a) = (i + d, a + r^i(b + b') + r^j s_i + r^l s_{i+d}).$$

Its return on the associated coset C is

$$a \mapsto a + (r^j + r^l) S_C.$$

For distinct quotient symbols $(k, u), (l, v)$, use the symbol-line map

$$S_{(k, u)}(j, b) = (k - j, u + r^{k-j} b + s_{k-j} r^j).$$

With $d = l - k$, one step of $S_{(l, v)}^{-1} S_{(k, u)}$ is

$$(j, b) \mapsto (j + d, b + r^{j-k}(u + v) + (1 + r^d) r^j t_{k-j}).$$

Summing over a quotient cycle gives the return

$$b \mapsto b + (1 + r^{l-k}) r^k T_C.$$

In each calculation the line-label terms vanish because a nontrivial quotient cycle has odd length $\ell > 1$ and the relevant geometric sum of an ℓ th root other than one is zero. Every displayed multiplier outside S_C or T_C is nonzero. A nonzero characteristic-two translation has order two, so each lifted cycle has even length 2ℓ . This establishes the asserted three-view criterion.

It remains to choose the coefficients. Put $D = r + 1$, $u = D^{-1}$, and start with the injective function $t_0(1) = 1$ and $t_0(x) = (x + 1)^{-1}$ for $x \in H \setminus \{1\}$. Its values at $1, r, r^2$ are $1, u, u^2$, which are distinct for $m > 3$. Cyclically permute these three values:

$$t(1) = u, \quad t(r) = u^2, \quad t(r^2) = 1, \quad t(x) = (x + 1)^{-1} \quad (x \notin \{1, r, r^2\}),$$

and set $s_i = r^i t(r^i)$. Injectivity is retained.

Let $C = aJ$ be a coset of a subgroup J of odd order $\ell > 1$, and write $c = a^\ell$. If $1 \notin C$, then $\prod_{x \in C} (X + x) = X^\ell + c$ and logarithmic differentiation at $X = 1$ gives $\sum_{x \in C} (1 + x)^{-1} = (1 + c)^{-1}$. Also $\sum_{x \in C} x^{-1} = 0$. Any coset untouched by the patch therefore has

$$S_C = \frac{c}{1 + c}, \quad T_C = \frac{1}{1 + c},$$

both nonzero.

Now let J be proper, set $z = r^\ell$, and let $A = (\ell - 1)/2 \pmod{2}$, interpreted as an element of K . Its three affected cosets J, rJ, r^2J are distinct. The value changes relative to t_0 are $u + 1$, r/D^2 , and r^2/D^2 , respectively. Pairing x with x^{-1} on $J \setminus \{1\}$ gives $\sum_{x \in J \setminus \{1\}} (1 + x)^{-1} = A$. Including $t_0(1) = 1$, the unpatched moments on J are $S_J(t_0) = A + 1$ and $T_J(t_0) = A$. The other two cosets use the preceding unpatched formula. Adding the appropriate value changes yields

C	S_C	T_C
J	$A + u$	$A + 1 + u$
rJ	$\frac{z + r^2}{(1 + z)D^2}$	$\frac{z + r^2}{(1 + z)D^2}$
r^2J	$\frac{[z(r^2 + r + 1) + r^2]^2}{(1 + z^2)D^2}$	$\frac{z^2 + r^2}{(1 + z^2)D^2}$

All denominators are nonzero. The J entries are nonzero because $u \notin \{0, 1\}$. The rJ numerator could vanish only if $\ell \equiv 2 \pmod{m}$, and the last T numerator only if $\ell \equiv 1 \pmod{m}$; neither is possible for a proper divisor $1 < \ell < m$. The only possible failure is therefore

$$r^\ell(r^2 + r + 1) = r^2. \quad (5)$$

At most one exponent modulo m satisfies this equation. If it holds for a proper divisor ℓ , replace r by r^{-1} . A bad proper divisor j for the inverse would require $r^2 + r + 1 = r^j = r^{2-\ell}$, so $j = m + 2 - \ell > m/2$. Every proper divisor of odd m is at most $m/3$; there is no such j . Thus either r or its inverse makes every proper coset moment nonzero.

For the full-group coset H , all three patched points lie together. With $A = (m - 1)/2 \pmod{2}$, direct summation gives

$$S_H = 1 + A + r^2 + \frac{r}{r + 1}, \quad T_H = A + \frac{r}{r + 1}.$$

When $m \equiv 3 \pmod{4}$, these are $r(r^2 + r + 1)/(r + 1)$ and $1/(r + 1)$; the first could vanish only if r had order three. When $m \equiv 1 \pmod{4}$, they are $(r^3 + r^2 + 1)/(r + 1)$ and $r/(r + 1)$. The cubic is irreducible over $\mathbb{F}_2$ and its roots have order seven, incompatible with $m \equiv 1 \pmod{4}$. Hence both moments are nonzero. The three-view criterion proves that (4) is FFF.

Finally, multiplying by the table of an elementary abelian 2-group preserves FFF by Corollary 5.3, giving every exponent $a \geq e$. $\square$

Remark 8.4 (The exceptional affine base case). At $m = 3$, over the prescribed minimal field $K = \mathbb{F}_{2^{\text{ord}_3(2)}} = \mathbb{F}_4$, the affine formula (4) cannot be FFF, although Table 1 gives an order-12 FFF square. Put $t_i = r^{-i}s_i$. Row-F requires distinct t_i ; column-F and symbol-F require $S = t_0 + rt_1 + r^2t_2 \neq 0$ and $T = t_0 + r^2t_1 + rt_2 \neq 0$. Common translation and nonzero scaling give $(t_0, t_1) = (0, 1)$. Since $r^2 + r + 1 = 0$, $t_2 = r$ forces $T = 0$, whereas $t_2 = r^2$ forces $S = 0$. This argument concerns $\mathbb{F}_4$, not the same formula over all larger binary fields. As a scope control, over $\mathbb{F}_{16} = \mathbb{F}_2[x]/(x^4 + x + 1)$ take $r = x^2 + x$, $(t_0, t_1, t_2) = (0, 1, x)$, and $s_i = r^i t_i$. Direct three-view checks give an FFF square of order 48. This is not a new order-48 existence claim: products of the order-12 witness already supply that order.

The finite checks in the associated companion run evaluate all subgroup coset moments for $m = 5, 7, 9, 15, 21$ and all line-pair cycles at orders 56, 80, and 240. They control the implementation; the universal quantifier in Theorem 8.3 rests on the argument above. The theorem does not determine $h(m)$ in general, resolve order 18, classify higher-order main classes, or establish literature priority.

9 A common binary fiber over prime-square quotients

The preceding bound may require a larger binary exponent when the odd part is a prime square. The following construction uses a single binary fiber over a two-dimensional odd quotient. It is an internally audited research result awaiting independent mathematical review, not a claim of literature priority.

Theorem 9.1 (Quadratic-norm prime-square lift). *Let p be an odd prime, $e = \text{ord}_p(2)$, and D a positive multiple of e . If $q = 2^D > 3(p^2 - 1)$, then there is an FFF Latin square of order p^2q . Every relative two-line cycle in every view has length 2 or $2p$. In particular, $2^e > 3(p^2 - 1)$ implies $h(p^2) \leq e$.*

Proof. Let $G = \mathbb{F}_p^2$, choose a nonsquare $\nu \in \mathbb{F}_p$, and put $B(x, y) = x_1y_1 - \nu x_2y_2$ and $Q(x) = B(x, x)$. The norm Q is anisotropic: if $Q(d) = 0$ and $d_2 \neq 0$, then $\nu = (d_1/d_2)^2$, a contradiction; the case $d_2 = 0$ is immediate. Choose ω of order p in $K = \mathbb{F}_q$ and write $\chi(t) = \omega^t$. For a twist array $c : G \times G \rightarrow K$, define

$$(x, a) * (y, b) = (x + y, \chi(2B(x, y))a + \chi(Q(x))b + c(x, y)). \quad (6)$$

Both fiber coefficients are nonzero, so this is Latin for every c .

We calculate the relative action of two lines with distinct quotient coordinates. For rows x, z , put $d = x - z$ and $y_t = y + td$. For columns y, z , put $d = y - z$ and $x_t = x + td$. For symbols k, l , put $d = l - k$, $y_t = y + td$, and $x_t = k - y_t$. In each case the quotient orbit has length p . Solving (6) and then the inverse line map gives a fiber step

$$w_{t+1} = \chi(D_t)w_t + \chi(E_t)a + \chi(F_t)b + \chi(U_t)c(C_{1,t}) + \chi(V_t)c(C_{2,t}).$$

The slope exponents and the two twist cells are

$$\begin{aligned} D_t^{\text{row}} &= Q(x) - Q(z), & (C_{1,t}, C_{2,t}) &= ((x, y_t), (z, y_t + d)), \\ D_t^{\text{col}} &= 2B(x_t, y) - 2B(x_t + d, z), & (C_{1,t}, C_{2,t}) &= ((x_t, y), (x_t + d, z)), \\ D_t^{\text{sym}} &= 2B(x_t, d), & (C_{1,t}, C_{2,t}) &= ((x_t, y_t), (x_t, y_t + d)). \end{aligned}$$

Every $\chi(U_t)$ and $\chi(V_t)$ is nonzero. For clarity in the symbol case, the line map at symbol (k, a) sends a column (y, w) to the unique row $(k - y, u)$, where

$$u = \chi(Q(k - y) - 2B(k - y, y))w + \chi(-2B(k - y, y))(a + c(k - y, y)).$$

Taking the inverse of the second symbol map yields the third line above.

The sum of the p slope exponents is zero in each view: it is constant for rows and affine in t for columns and symbols, and $\sum_{t \in \mathbb{F}_p} t = 0$. The exponent of a label contribution from step t gains the later slopes $\sum_{s > t} D_s$. Expanding the quadratic form gives the following two accumulated exponents, where x, y denote the starting quotient row and column in the respective line:

	first line label	second line label
row	$2B(x, y) - Q(x) + tQ(d)$	$2B(z, y + d) - Q(x) - tQ(d)$
col	$Q(x) - 2B(x, y) - tQ(d)$	$Q(x) - 2B(x, z) + (t + 1)Q(d)$
sym	$-Q(x) - tQ(d)$	$-Q(x) - 2B(x, d) + tQ(d)$

Since $Q(d) \neq 0$, summing the character of any one of these affine progressions over $t \in \mathbb{F}_p$ gives zero. Thus both line-label coefficients cancel, the p -step slope is one, and the fiber return is $w \mapsto w + \Lambda_C(c)$ for a linear form in the twist cells.

The two cells at each step lie on different quotient lines. Within a line the p visited cells are distinct, also for symbols because a cell has a unique quotient symbol. Thus each Λ_C contains exactly $2p$ distinct variables with nonzero coefficients. In particular it is not the zero form. A fixed cell belongs to exactly $p^2 - 1$ such forms in each of the three views: one quotient orbit for each other quotient line. Its total incidence is $3(p^2 - 1)$.

Order the p^4 cells. Enforce $\Lambda_C(c) \neq 0$ when the last cell in its support is assigned. Previously assigned cells then forbid precisely one value for this cell. At most $3(p^2 - 1)$ conditions close at any cell, so $q > 3(p^2 - 1)$ leaves an allowed value. Induction makes every return a nonzero translation of K . Consequently different quotient lines have cycles of length $2p$. Two distinct lines in the same quotient fiber induce a nonzero translation on every complementary fiber, giving only 2-cycles. The argument applies to rows, columns, and symbols, proving FFF. $\square$

Corollary 9.2. *For all but $O((\log X)^2)$ primes $p \leq X$, with $e = \text{ord}_p(2)$, one has $h(p^r) \leq \lceil r/2 \rceil e$ for every $r \geq 1$.*

Proof. Theorem 9.1 supplies the prime-square factor; the order- $p2^e$ instance of Theorem 8.3 supplies an odd residual factor. Direct products preserve FFF, giving the stated bound for each r . An exceptional prime satisfies $2^e \leq 3(p^2 - 1) < 4p^2$, hence $e < 2 \log_2 p + 2$. If $p \leq X$ and $N = \lfloor 2 \log_2 X + 2 \rfloor$, then p divides $\prod_{j=1}^N (2^j - 1)$. This product has fewer than $\sum_{j=1}^N j = N(N + 1)/2$ distinct prime factors. $\square$

At $p = 11$, $e = 10$ and $3(p^2 - 1) = 360 < 1024$, giving an order-123904 construction. A separate standard-library checker validates all 239580 quotient-return conditions of a frozen compact twist certificate; it does not materialize that Latin table. Fresh smaller controls cover $p = 3, q = 64$ and $p = 5, q = 256$. The finite checks support the computation, whereas the universal quantifiers rest on the proof above. No claim is made about minimal exponents, order 18, or literature priority.

10 Affine prime-successor loops

For an odd prime p and $a \in \mathbb{F}_p \setminus \{0, 1\}$, put $b = 1 - a$. Define $Q_p(a)$ on $\mathbb{F}_p \cup \{\infty\}$ by

$$\infty * x = x * \infty = x, \quad x * x = \infty \ (x \in \mathbb{F}_p), \quad x * y = ax + by \ (x \neq y, \ x, y \in \mathbb{F}_p).$$

In each finite row or column, the affine bijection fixes the line label. Replacing that value by ∞ and placing the old label in the new position proves that this is a Latin loop. Write $[u]_p$ for the least positive residue of a nonzero field element.

Theorem 10.1 (Three-view affine criterion). *The loop $Q_p(a)$ is FFF if and only if*

$$\begin{aligned} [a]_p, [a^{-1}]_p, [b^{-1}]_p \text{ are even,} \\ \text{ord}_p(a), \text{ord}_p(b), \text{ord}_p(-b/a) \text{ are even.} \end{aligned} \quad (7)$$

Proof. Use a generic row parameter t , with $c = 1 - t$. A finite row relative to the identity row consists of a two-cycle and $(p - 1)/\text{ord}_p(c)$ cycles of length $\text{ord}_p(c)$. Translations and nonzero scalings of finite labels are automorphisms and are two-transitive on them. It remains to compare rows $0, 1$. Their relative map $P = r_1^{-1}r_0$, with $D = -t/c$, satisfies

$$P(\infty) = D, \quad P(0) = 1, \quad P(c^{-1}) = \infty, \quad P(y) = y + D \quad \text{at every other finite } y.$$

Number the translation cycle by jD , $j \in \mathbb{Z}_p$, and put $k = [-t^{-1}]_p$. Then $c^{-1} = kD$ and $1 = (k + 1)D$. The redirected arrows give cycles of lengths $p - k$ and $k + 1$, both even exactly when $[t^{-1}]_p$ is even.

The column view has parameter b . A finite symbol z , viewed as a map from columns to rows, swaps z, ∞ and otherwise sends y to $a^{-1}z - (b/a)y$. Its parameter is a^{-1} and its other coefficient is $-b/a$; the infinity symbol line is the identity. Applying the row calculation to these three parameters proves (7) in both directions. $\square$

Computational Theorem 10.2 (A residue-class boundary control). *The explicitly defined loop $Q_{97}(36)$ is FFF of order 98. Thus $n \equiv 2 \pmod{8}$, $n > 2$, is not a universal FFF obstruction.*

Indeed, $b = 62$, $36^{-1} = 62$, $62^{-1} = 36$, $-62/36 = 36$ in $\mathbb{F}_{97}$, and $36^3 = 62^3 = -1$. The orders and the three least residues in (7) are even. Primality and these identities are exact finite checks. The materialized table also passes all $3\binom{98}{2} = 14259$ line-pair checks. This is not an order-18 construction.

10.1 A uniform large-prime bound

Theorem 10.3 (Eventual prime-successor existence). *For every prime $p \geq 10^{12}$, more than $p/1000$ parameters produce FFF loops $Q_p(a)$.*

The analytic input is Katz's published mixed-character-sum estimate [Kat08, Theorem 1.5]. Its application and the explicit cutoff are checked below, not inferred from finite tables.

Proof. Put $U = \mathbb{F}_p \setminus \{0, 1\}$. For $p \equiv 3 \pmod{4}$, take the quadratic character χ and the sufficient mask

$$W(a) = \frac{1}{4}(1 - \chi(a))(1 - \chi(1 - a)).$$

For $p \equiv 1 \pmod{4}$, take a character of exact order four and use

$$W(a) = \frac{1}{8}(1 - \chi(a)^2)(1 - \chi(1 - a) + \chi(1 - a)^2 - \chi(1 - a)^3).$$

These are zero-one indicators. In the first case $a, b, -b/a$ are nonsquares. In the second, write $p - 1 = 2^s m$ with m odd. The primitive-root exponent of a is odd, that of b is 2 modulo 4, and that of $-b/a$ is odd. All three multiplicative orders are therefore even in either case. The mask's character expansion has coefficient absolute sum one and constant coefficient at least $1/8$.

Let $\psi(x) = \exp(2\pi ix/p)$ and let χ have order $q \in \{2, 4\}$. Except for the all-zero tuple, the sums

$$S(j, k; u, v, w) = \sum_{a \in U} \chi(a)^j \chi(1-a)^k \psi(ua + v/a + w/(1-a)), \quad 0 \leq j, k < q,$$

have absolute value at most $4\sqrt{p}$. For $u \neq 0$, apply the cited theorem with polynomial ua , rational remainder $v/a + w/(1-a)$, and multiplicative polynomial $a^j(1-a)^k$. All finite poles are simple; the pole order at infinity is below one, and every nonpole zero has nontrivial character multiplicity. With t endpoint poles and e other endpoint zeros, the estimate is $(2t+e)\sqrt{p}$. Deleting the remaining $2-t-e$ endpoints changes the sum by at most that number, so the result is at most $4\sqrt{p}$. For $u=0, v \neq 0$, substitute $z=1/a$; for $u=v=0, w \neq 0$, substitute $z=1/(1-a)$. Each is a bijection on U , gives a nonzero linear phase plus at most one simple finite pole, and changes the multiplicative factor to powers of $z, z-1$, up to a constant. The same argument applies. With zero additive phase, the nontrivial multiplicative sums are Jacobi sums or one-character endpoint sums, bounded by $\sqrt{p}$. The all-zero tuple has sum $p-2$.

Let $E(x)$ indicate nonzero even least residues. Geometric summation gives its normalized Fourier coefficients

$$\hat{E}(0) = \frac{p-1}{2p}, \quad |\hat{E}(h)| = \frac{1}{2p|\cos(\pi h/p)|} \quad (h \neq 0).$$

Pairing $h, p-h$ and using $\sin x \geq 2x/\pi$ on $[0, \pi/2]$ bounds the nonzero coefficient sum by $\sum_{r=0}^{(p-3)/2} (2r+1)^{-1} \leq 1 + \frac{1}{2} \log p$. The full absolute sum is at most $L(p) = 3/2 + \frac{1}{2} \log p$. Expanding $W(a)E(a)E(a^{-1})E((1-a)^{-1})$ therefore bounds its parameter count by

$$N_p \geq \frac{(p-2)(p-1)^3}{64p^3} - 4\sqrt{p}L(p)^3. \quad (8)$$

The first term is at least $(p-5)/64 \geq p/65$ for $p \geq 325$. The function $\sqrt{x}/L(x)^3$ increases once $L(x) > 3$. At $x = 10^{12}$, $\log 10 < 2.303$ gives $L(x) < 15.318$ and $260(15.318)^3 < 935000 < 10^6$, exact rational inequalities. It follows that $N_p > (1 - 0.935)p/65 = p/1000$. For example, the finite rational inequality $\sum_{k=0}^9 (2303/1000)^k/k! > 10$ proves the logarithm bound. $\square$

For $p > 3$ these loops are not group-isotopic. Associativity at $(0, 0, 1)$ would force $b^2 = 1$, hence $b = -1$, and at $(1, 0, 0)$ would force $a = -1$, contradicting $a + b = 1$. A group-isotopic loop is a group by the basis-family closure criterion: since its identity row is the identity permutation, closure gives $L_x L_y = L_{xy}$ after evaluating at the identity, hence associativity.

10.2 Exact classification inside the affine family

Theorem 10.4 (Affine-family main classes). *For $p \geq 5$, the loops $Q_p(a), Q_p(a')$ are main-class-equivalent exactly when*

$$a' \in \mathcal{A}(a) = \{a, 1-a, 1/a, 1/(1-a), a/(a-1), (a-1)/a\}.$$

Distinct parameters are never isomorphic; each loop has automorphism group $\text{AGL}(1, p)$.

Proof. Swapping the inputs and solving the relation for either input give the displayed parameters, including all triples involving ∞ . Thus these relations suffice.

The types of row pairs involving infinity and of two finite rows are

$$(2, \text{ord}_p(b)^{(p-1)/\text{ord}_p(b)}), \quad ([a^{-1}]_p, p+1 - [a^{-1}]_p).$$

They coincide only if $\text{ord}_p(b) = p-1$ and $[a^{-1}]_p \in \{2, p-1\}$. The column version exchanges a, b . Outside $E = \{2, 1/2, -1\}$ both identity lines are intrinsic: each is the unique line with all p incident pair types the same, whereas a finite line has both types. An isotopy between such loops must fix the two identity lines; the loop equations make its three maps equal. Its finite-field bijection f satisfies

$$f(ax + by) = a'f(x) + (1 - a')f(y),$$

also at $x = y$. Subtract $f(0)$ and set one argument to zero. Inverting a, b in the resulting identities proves additivity of $h(x) = f(x) - f(0)$. On a prime field h is multiplication by a nonzero scalar; the identity then forces $a = a'$.

For a paratopy first perform its coordinate permutation, replacing a by $t \in \mathcal{A}(a)$. The set E is a single orbit. If exactly one of t, a' is in E , either its special and finite pair types coincide in some view, giving an invariant mismatch, or both identity lines remain intrinsic and the same argument forces equality. If both are in E , they already lie in the same displayed orbit. This proves necessity. For an isomorphism the identity is fixed without exceptions, so the additive argument always gives $a = a'$ and $f(x) = ux + v$. Conversely every such map with $u \neq 0$, fixing ∞ , is an automorphism. $\square$

Let S_p be the full parameter set satisfying (7). The anharmonic orbits have size six, apart from E of size three and the two roots of $a^2 - a + 1 = 0$ when present. Let e_p, h_p indicate whether these exceptional orbits lie in S_p . The exact number of represented FFF main classes is

$$C_p^{\text{aff}} = (|S_p| + 3e_p + 4h_p)/6.$$

Indeed, the three involutions fix only $1/2, 2, -1$, and the order-three elements fix exactly the two polynomial roots. For $p \geq 5$ these exceptional sets are disjoint. Consequently, Theorem 10.3 gives more than $p/6000$ FFF main classes at every prime-successor order $p+1$ with $p \geq 10^{12}$.

Finite controls construct all 253 parameters for the 13 odd primes through 43, compare all views with the criterion, and check parameter transformations and orbit counts. They do not prove the analytic estimate or its uniform cutoff. At $p = 17$ the criterion excludes only the scalar affine family, not all order-18 squares. The classification is within this family, not a census of all Latin squares. Literature priority and external expert review of the analytic application remain open.

11 Row-fibred extensions and represented main classes

The following construction allows the inner operation to depend on an outer row. Related index-two constructions already occur in the loop literature, for example Johnson's dihedral extension [Joh06]. No priority claim for the construction, its FFF consequences, or the bounds below is made here.

11.1 An exact three-view pattern formula

Let $B : Y \times Y \rightarrow Y$ be Latin and, for each $y \in Y$, let $Q_y : X \times X \rightarrow X$ be Latin. Define

$$M((r, y), (c, z)) = (Q_y(r, c), B(y, z)). \quad (9)$$

All sets are finite and nonempty. The fibre squares need not be isotopic.

Theorem 11.1 (Row-fibred pattern formula). *The table M is Latin, and in the three named views*

$$\text{pat}(M) = \text{pat}(B) \text{ OR } \bigvee_{y \in Y} \text{pat}(Q_y).$$

In particular, M is FFF if and only if B and every Q_y are FFF.

Proof. In a fixed row, both coordinate maps in (9) are bijections. In a fixed column, the second output determines y and then the first determines r , proving the Latin property.

For rows $(r, y), (r', y')$ with $y = y'$, the induced permutation fixes the outer column z and restricts on each such layer to the corresponding row-pair permutation of Q_y . With $y \neq y'$, it projects to the row-pair permutation of B on outer columns. For columns $(c, z), (c', z')$ with $z = z'$, the analogous permutation fixes y and restricts to a column-pair permutation of Q_y ; with $z \neq z'$ it projects to the column-pair permutation of B . For symbols $(s, t), (s', t')$ with $t = t'$, the outer column z is fixed, and the outer row is the unique y such that $B(y, z) = t$. The induced map on inner columns is the symbol-pair permutation of Q_y . Every y occurs for one z . With $t \neq t'$, the outer columns instead follow the symbol-pair permutation of B .

Every fibre witness therefore embeds in the same view of M . In every cross-layer case an upstairs cycle has length divisible by its projected cycle length. Thus, if both the quotient and all fibres are F in that view, no odd cycle is possible upstairs. To finish the equality, a quotient odd cycle must be lifted without increasing its length by an even factor. This requires a choice of inner coordinates.

For quotient rows y, y' , fix an inner column c and an inner row r . Choose r' uniquely so that $Q_y(r, c) = Q_{y'}(r', c)$. The induced upstairs row-pair map fixes c , so every quotient cycle lifts with its original length. For quotient columns z, z' , choose upstairs columns with a common inner column c and fix an inner symbol s . At each outer row y on the cycle, choose r_y by $Q_y(r_y, c) = s$. The induced map carries (r_y, y) around precisely the quotient cycle. Finally, for quotient symbols t, t' , take a common inner symbol s and an inner column c . The symbol-pair map keeps c fixed and follows the quotient symbol cycle on z : in the relevant row y , the inner row is determined by $Q_y(r, c) = s$. Hence the quotient odd cycle survives in this view as well. $\square$

For $|X|, |Y| > 1$, the construction cannot produce FFF of order 18: one factor of every such factorization of 18 is odd and greater than one. The order-one square is vacuously FFF and is not an odd-order obstruction.

11.2 Cycle spectra and the order-20 input

For an unordered pair of distinct lines in a view, sort its cycle lengths as a partition λ . Let $\Sigma_L(\lambda)$ count all such pairs, summed over the three views. Isotopy conjugates or inverts pair permutations; parastrophy also permutes the views. Thus Σ_L is a main-class invariant. Sorting each partition before comparison is essential.

Proposition 11.2 (Product spectra and intercalates). *Let $I(L)$ denote the number of intercalates in a Latin square of order n , and let K have order m . Then*

$$I(L \times K) = m^2 I(L) + n^2 I(K) + 4I(L)I(K).$$

If L is FFF, and D duplicates each part of a partition, then

$$\Sigma_{L \times C_2} = 4D_* \Sigma_L + 3n[2^n]. \quad (10)$$

In particular, distinct spectra remain distinct after any number of products with C_2 .

Proof. A two-cycle in a row-pair permutation is one intercalate. If product rows differ only in the L coordinate, the common K row has m choices and each two-cycle has m copies, giving $m^2 I(L)$. The other one-coordinate case gives $n^2 I(K)$. If both coordinates differ, a pair of factor two-cycles has two alignments of the product rows, each producing two two-cycles, giving the last term. Distinct Latin lines have no fixed points, so there are no other ways to produce a two-cycle.

For (10), in each view the n product-line pairs with equal L coordinate have type 2^n . Each unordered distinct L line pair has four product-line pairs. A cycle of length a times a cycle of length b has $\gcd(a, b)$ cycles of length $\text{lcm}(a, b)$. Since every L part is even, both multiplication by the identity on two points and by its transposition duplicate every part. Summing the three views proves the formula. The map D is injective on partitions, so its pushforward is injective. $\square$

Computational Theorem 11.3 (Frozen order-20 family). *One specified cycle-union trade family of an FFF Steiner-loop table of order 20 represents exactly 512 main classes within that family. Two separately specified quadratic tables represent further classes. Consequently at least 514 FFF main classes of order 20 exist.*

The finite evidence enumerates all $2^{10} = 1024$ masks of the ten transpositions in one row pair. Every resulting table is Latin and FFF under two independent physical three-view scanners. Exactly 512 sorted spectra occur, each at complementary masks. Complementing a mask swaps the two rows globally, so it supplies the upper bound of 512 classes as well as the invariant's lower bound. The two quadratic controls have distinct spectra outside that family; their intercalate counts are 190 and 19, whereas the trade-family counts range from 542 to 676. These are exact table comparisons, not a census of all order-20 squares. Equation (10) alone propagates the lower bound 514 to orders $20 \cdot 2^d$.

11.3 Recovering quotient layers

A *binary three-sorted quotient* of Q is a triple of surjective maps a, b, d from its row, column and symbol sets to $\mathbb{F}_2$ satisfying $d(Q(r, c)) = a(r) + b(c)$. The independent labelling of the three sorts is part of the definition. The corresponding partitions, not their chosen bit labels, are intrinsic.

Lemma 11.4 (Binary quotient descent). *Suppose no Q_y in (9) has a binary three-sorted quotient. Every such quotient of M is constant on each inner layer and descends to a binary quotient of B ; conversely, each binary quotient of B pulls back to one of M .*

Proof. First, any three binary maps satisfying $d(Q(r, c)) = a(r) + b(c)$ are either all constant or all surjective. If a is constant, fixing c and varying r shows that d is constant, and then b is constant. The argument with a, b exchanged is the same; if d is constant, varying either input shows both others constant.

Restrict a quotient triple of M to the block of rows at y , columns at z , and symbols at $B(y, z)$. It satisfies this equation on Q_y . Quotient freeness makes all three restrictions constant. Running over y, z makes the full maps constant on every inner layer, and their constants satisfy the quotient equation on B . Surjectivity is preserved by descending or pulling back. $\square$

For $B = E = (\mathbb{F}_2)^d$, every binary quotient has the form

$$a(y) = \chi(y) + a(0), \quad b(z) = \chi(z) + b(0), \quad d(t) = \chi(t) + a(0) + b(0),$$

where $\chi : E \rightarrow \mathbb{F}_2$ is a nonzero linear character. This follows by evaluating the quotient equation at $(y, 0)$ and $(0, z)$ and subtracting the constants. All $2^d - 1$ characters separate points of E . Their common refinement therefore recovers each inner layer intrinsically, in all three sorts.

Theorem 11.5 (Affine-orbit classification of a fixed family). *Fix representatives $Q_1, \dots, Q_m$ of pairwise distinct main classes of order n , none admitting a binary three-sorted quotient. For each function $f : E \rightarrow \{1, \dots, m\}$ define*

$$M_f((r, y), (c, z)) = (Q_{f(y)}(r, c), y + z).$$

Two members M_f, M_g are main-class-equivalent if and only if $g(Ay + u) = f(y)$ for some $A \in \text{GL}(d, 2)$ and $u \in E$.

Proof. By Lemma 11.4, every main-class equivalence preserves the intrinsic layers, up to a permutation of their three sorts, and induces a paratopism of the outer table of E . Its blocks are colored by the main classes $[Q_{f(y)}]$. A nonconstant coloring is constant along an outer row but not along an outer column or symbol line. Thus a paratopism between nonconstant members must carry the source row sort to the target row sort. A constant coloring can match only the same constant coloring, since the input main classes are distinct.

Every paratopism of $y + z = t$ acts affinely on each sort with a common invertible linear part. Indeed, permuting its three coordinates preserves this relation in characteristic two. After undoing that permutation, write $\gamma(y + z) = \alpha(y) + \beta(z)$. Setting $u = \alpha(0)$, $v = \beta(0)$ gives $\alpha(y) = Ay + u$, $\beta(z) = Az + v$ and $\gamma(t) = At + u + v$, with A additive and bijective. The transported block colors now give $g(Ay + u) = f(y)$.

Conversely, this equality makes the three maps

$$(r, y) \mapsto (r, Ay + u), \quad (c, z) \mapsto (c, Az + v), \quad (s, t) \mapsto (s, At + u + v)$$

an explicit isotopism for any $v \in E$. The fixed choice of the representatives is used in this sufficiency argument. $\square$

Corollary 11.6 (Constructed class counts). *Let $c(h)$ be the number of cycles of h on E . The fixed family in Theorem 11.5 represents exactly*

$$T_d(m) = \frac{1}{|\text{AGL}(d, 2)|} \sum_{h \in \text{AGL}(d, 2)} m^{c(h)}, \quad |\text{AGL}(d, 2)| = 2^d \prod_{i=0}^{d-1} (2^d - 2^i) \quad (11)$$

main classes. If the inputs are FFF, these are FFF classes and hence give a lower bound for the total at order $n2^d$. A weaker bound is $\binom{m+2^d-1}{2^d}$, obtained by distinguishing input multisets.

Proof. Burnside's lemma counts colorings fixed by each affine permutation: they are precisely the colorings constant on its cycles, numbering $m^{c(h)}$. Affine orbits refine the input multisets. The FFF assertion follows from Theorem 11.1. $\square$

Lemma 11.7 (A direct certificate for binary quotient freeness). *For an order- n Latin square Q , let A_Q be its $3n$ by n^2 line-cell incidence matrix: the column for (r, c) has ones on row line r , column line c , and symbol line $Q(r, c)$, and zeros elsewhere. Then Q has no binary three-sorted quotient if and only if $\text{rank}_{\mathbb{F}_2} A_Q = 3n - 2$. A nonsingular $(3n - 2)$ -square minor certifies this equality.*

Proof. The two independent line vectors $(\mathbf{1}, \mathbf{1}, \mathbf{0})$ and $(\mathbf{1}, \mathbf{0}, \mathbf{1})$ lie in $\ker(A_Q^T)$, so the rank is at most $3n - 2$. Their span consists exactly of the constant triples $(a, b, d) = (\alpha, \beta, \alpha + \beta)$. The kernel equation for any triple of binary line maps is $a(r) + b(c) + d(Q(r, c)) = 0$ for every cell. As in the proof of Lemma 11.4, any nonconstant solution has all three maps surjective: if one were constant, fixing one input and using the Latin bijections would force the other two to be constant as well. Thus an additional kernel vector is exactly a binary quotient. Rank-nullity proves the equivalence. A nonsingular minor supplies the matching lower bound on the rank. $\square$

Computational Theorem 11.8 (Specialization from order 20). *Direct 58 by 58 binary minors certify quotient freeness for all 1024 trade-mask tables and the two quadratic controls of Computational Theorem 11.3. In particular, its 514 input main classes are binary-quotient-free. Therefore there are at least $T_d(514)$ FFF main classes of order $20 \cdot 2^d$. These earlier certified bounds remain valid. The first four values are:*

Order	Proven lower bound, not a census
40	132355
80	2942384005
160	3625556867870760586
320	73586711420426563009097417622853741216

Proof. The certificates use all 20 row lines, the first 19 column lines, and the first 19 symbol lines, with 58 specified cell columns for each table. A separate binary elimination checks each minor and its binding to the exact table. All 1026 minors are nonsingular. Lemma 11.7 therefore proves quotient freeness for these inputs. Theorem 11.5 and Equation (11) give the construction count; exact affine-group enumeration through dimension four gives the displayed integers. No physical enumeration of that many output squares is asserted. $\square$

This certificate argument concerns the specified input family, not every order-20 FFF square. It removes C157 from these particular class bounds, not from order-10 nonexistence or $h(5) = 2$. The stronger statement that *every* FFF square of order 20 is binary-quotient-free would still use C157: a binary quotient has balanced fibres of size 10 and hence an FFF subsquare of order 10, which C157 excludes.

Two checked order-160 outputs even have the same input multiset and the same sorted cycle spectrum, but their four-point color sets are an affine plane and an affine tetrahedron. Their affine spans have different dimensions, so Theorem 11.5 separates their main classes. This demonstrates a limitation of the spectrum invariant, not its completeness.

The count is exact only within the fixed family. It does not classify all higher-order FFF squares, justify arbitrary direct-product cancellation, or identify all extensions obtained by changing the chosen representatives within their main classes. The binary-quotient-free hypothesis must be checked on the inputs; it cannot be iterated blindly on outputs that already have binary quotients. Computational Theorem 13.3 below supplies a larger independently rank-checked input family and strengthens the lower bound to $T_d(1703)$; the displayed historical $T_d(514)$ values are not claimed to be the strongest current bounds.

For clarity, evaluating the same formula on the 1703 certified inputs gives the following stronger bounds. A fresh enumeration of the affine groups through dimension four checks the integer evaluations; the construction theorem, not a scan of all output squares, supplies their interpretation as lower bounds on distinct main classes.

Order $20 \cdot 2^d$	Current lower bound $T_d(1703)$
40	1450956
80	351703270490
160	52641008907323966249185
320	15517575138479887787890371386037048688772562656

These remain family-based lower bounds, not higher-order censuses or a claim that all possible product and extension constructions are distinct.

12 Block return maps and a common-fibre construction

The quotient-OR identity for the row-fibred construction must not be extended to arbitrary cell-dependent blocks. The distinction is already visible in the order-12 counterexample.

Theorem 12.1 (Cycle lengths in a Latin block substitution). *Let Q be Latin on A , and let F_{ab} be Latin on a common set B for every $(a, b) \in A^2$. The operation*

$$H((a, x), (b, y)) = (Q(a, b), F_{ab}(x, y))$$

is Latin. In each view, same-outer-label line pairs restrict to block line pairs. For different outer labels, a quotient cycle of length d has a bijective inner return map Ω . Each e -cycle of Ω lifts to one cycle of length de . Consequently a view fails exactly when some block fails in that view, or some odd quotient cycle has an odd return-map cycle for a lifted line pair.

Proof. In a fixed row, the desired outer symbol determines the outer column; the block bijection then determines the inner column. The column argument is identical. For rows $(a, x), (a', x')$, equality of outer symbols sends b to b' with $Q(a, b) = Q(a', b')$. The inner step is $F_{a'b'}(x', -)^{-1}F_{ab}(x, -)$. If $a = a'$, then $b = b'$ and this is exactly a row-pair permutation in one block. Otherwise it projects to the quotient row-pair map. Exchanging the two inputs proves the column statement.

For a symbol (s, z) , a column (b, y) determines the unique outer row a with $Q(a, b) = s$; the inner row is obtained by inverting the symbol line of F_{ab} . Equal outer symbols leave b fixed and give a block symbol-pair permutation. Distinct outer symbols follow the quotient symbol-pair map. Every block and every pair of its lines occur as these labels vary, in all three views.

Above a quotient cycle $(b_0, \dots, b_{d-1})$, write the successive inner bijections as $h_0, \dots, h_{d-1}$. A first return to the starting fibre is $\Omega = h_{d-1} \cdots h_0$. An e -cycle of this map closes after exactly e returns and hence de steps; it cannot close before an outer return. This accounts for all cycles and proves the parity claim. $\square$

Both frozen order-12 examples consist of nine FFF blocks of order 4 over a TTT quotient of order 3. Fresh checks verify the return formula on every physical pair in each view. Thus a failing quotient can be masked by even inner returns. This neither contradicts the row-fibred theorem nor the special two-element-kernel equality.

Theorem 12.2 (Odd cubefree common-fibre lift). *Let $N > 1$ be odd and cubefree, $M = \text{rad}(N)$, and $e = \text{ord}_M(2)$. If $D > 0$ is divisible by e and $2^D > 3(N - 1)$, there exists an FFF square of order $N2^D$. Its two-line cycles have length 2 or $2h$, where h is the additive order of a nonzero quotient difference.*

Proof. Write $N = \prod_p p^{r_p}$, with $r_p \in \{1, 2\}$, and put $G = \prod_p \mathbb{F}_p^{r_p}$ and $R = \prod_p \mathbb{F}_p$. Choose $B_p(x, y) = xy$ in rank one and $B_p(x, y) = x_1y_1 - \nu_p x_2y_2$ in rank two, with ν_p a nonsquare.

Let $B = (B_p)_p$ and $Q(x) = B(x, x)$. Anisotropy gives $Q_p(d) \neq 0$ exactly when $d_p \neq 0$. Hence $Q(d)$ in the additive group of R and d in G have the same odd order $h(d)$.

The condition on D gives a primitive M th root in $K = \mathbb{F}_{2^D}$. The additive CRT identification of R with $\mathbb{Z}/M\mathbb{Z}$ gives an injective character $\chi : R \rightarrow K^*$. Define, for a twist array $c : G^2 \rightarrow K$,

$$(x, a) * (y, b) = (x + y, \chi(2B(x, y))a + \chi(Q(x))b + c(x, y)).$$

The two fibre coefficients are nonzero, so every twist gives a Latin square. The row, column and symbol relative-step formulas are precisely those in the proof of Theorem 9.1, now evaluated componentwise in R over a quotient orbit of length $h = h(d)$.

Here is why replacing p by composite h is legitimate. Each slope exponent is constant or affine in the step index. Its nonzero coordinates are supported where $d_p \neq 0$; each such odd prime divides h , and $\sum_{t=0}^{h-1} t = h(h-1)/2 = 0$ in that component. Thus the total slope is one. Composing the later slopes into each of the two label coefficients gives the same displayed exponents as in that proof, of the form $A + tQ(d)$ or $A - tQ(d)$. This follows by expanding the quadratic form in each active component; in an inactive component $d_p = 0$ and the step is constant. The element $\chi(Q(d))$ has exact order h , so

$$\sum_{t=0}^{h-1} \chi(A \pm tQ(d)) = 0.$$

Both line labels therefore cancel in every view. Every inner return is $w \mapsto w + \Lambda_C(c)$.

Choose one orientation and starting point per unordered quotient pair and orbit; conjugate returns have the same nonvanishing condition. The form Λ_C has $2h$ distinct twist cells with nonzero coefficients. For symbols, the two sets of cells have distinct quotient symbols; this ensures distinctness just as different quotient rows or columns do in the other views. A cell lies in one return form for each other quotient line in each view, a total of $3(N-1)$ forms. Assign twist cells in a fixed order and enforce a form to be nonzero when its last cell is assigned. Each closing form forbids one field value. The field-size inequality leaves an allowed choice at every step.

For different quotient labels the return is now a nonzero translation of a characteristic-two field, hence has only two-cycles; physical cycles have length $2h$. For equal quotient labels and distinct fibre labels a, b , the row, column and symbol translations respectively have nonzero constants

$$\chi(2B(x, y) - Q(x))(a + b), \quad \chi(Q(x) - 2B(x, y))(a + b), \quad \chi(-Q(x))(a + b).$$

They give two-cycles. Every physical pair in all three views is covered. $\square$

For example, $N = 11^2 \cdot 41^2 = 203401$ has $\text{ord}_{451}(2) = 20$ and $2^{20} > 3(N-1)$, giving $h(N) \leq 20$. The product of the two separate prime-square constructions gave the upper bound $10 + 20 = 30$. This improves that particular construction, not a known minimum; the order-213281406976 table is not materialized. The compact finite records cover $N = 9, 15, 45, 75$ and direct return-map controls. Their checks do not replace the general proof.

12.1 Limits of the bilinear method

The finite-field zero-count arguments below are elementary special cases of the classical Chevalley–Warning method [Che35, War35]; their needed details are included.

Proposition 12.3 (Scalar quotient boundary). *Let G be a finite odd abelian group and let $\beta : G^2 \rightarrow K^*$ be a symmetric bicharacter over a characteristic-two finite field. If some twist makes*

$$(x, a) * (y, b) = (x + y, \beta(x, y)^2 a + \beta(x, x)b + c(x, y))$$

row- F , every Sylow subgroup of G is elementary abelian of rank at most two. Conversely every such group admits an FFF instance over a sufficiently large field. This is a boundary of this formula only.

Proof. If $d \neq 0$ and $\beta(d, d) = 1$, composition around a row quotient orbit of odd length h has slope one. Its two label coefficients are nonzero scalar multiples of $\sum_{t=0}^{h-1} \beta(d, d)^{\pm t} = 1$. Whatever the twist, one may choose the second row label to cancel the return translation. This creates an odd physical h -cycle. Thus no nonzero d can have $\beta(d, d) = 1$.

An element u of order p^2 violates this condition at $d = pu$. On a rank-three elementary p -subgroup, the exponent of $\beta(d, d)$ is a quadratic form over $\mathbb{F}_p$ (or is identically zero). Such a form has a nonzero zero: $\sum 1 - Q(d)^{p-1}$ over $\mathbb{F}_p^3$. Each monomial has a variable of exponent below $p - 1$, so its sum is zero modulo p . The number of zeros is divisible by p and includes zero, hence is at least p . Necessity follows; Theorem 12.2 proves sufficiency for the stated groups. $\square$

Proposition 12.4 (Commuting-matrix fibre bound). *Let p be an odd prime, $r \geq 1$ and $D \geq 0$. In the same formula let $G = \mathbb{F}_p^r$, let the fibre be $V = \mathbb{F}_2^D$, and let β be a symmetric bicharacter with commuting image in $\text{GL}(V)$. Row- F requires $D \geq \text{ord}_p(2) \lceil r/2 \rceil$. If $2^{\text{ord}_p(2)} > 3(p^2 - 1)$, this dimension is attained by an FFF construction within this ansatz.*

Proof. Write $e = \text{ord}_p(2)$. The row-return computation uses only commutativity: the label coefficients are invertible matrices multiplied by $S(q) = \sum_{t=0}^{p-1} q^t$ and $S(q^{-1})$, where $q = \beta(d, d)$. If $q = I$, both sums are I , and a choice of the second row label cancels the return. Thus $\beta(d, d) \neq I$ for every nonzero d is necessary.

The commuting matrices of exponent p diagonalize simultaneously over $\mathbb{F}_{2^e}$. Each nontrivial eigencharacter has a Frobenius orbit of length e , so there are at most $t = \lfloor D/e \rfloor$ such orbits. One representative from each yields a quadratic exponent form Q_i on $\mathbb{F}_p^r$. If $r > 2t$, summing $\prod_i (1 - Q_i(d)^{p-1})$ shows, by the same degree argument as above, that their common zero count is divisible by p . A nonzero common zero would give $\beta(d, d) = I$, a contradiction. Hence $r \leq 2t$. For attainment, split G into rank-two summands and at most one rank-one summand. Apply Theorem 12.2 to each over $\mathbb{F}_{2^e}$ and take their direct product. The resulting block diagonal bicharacter has exactly the required fibre dimension. $\square$

These propositions concern the exact scalar or commuting-matrix formula, not nonlinear blocks, arbitrary quotient operations, or the unrestricted minimum $h(m)$. They provide no new exclusion of order 18. In particular, having a fixed vector is not the same as being the identity matrix and cannot replace the condition used in the last proof.

13 A pattern-preserving affine corner trade

Latin squares with a unique intercalate are an established subject; see Wanless [Wan01]. The result here concerns the three-view FFF pattern of a particular affine trade. Neither the existence of unique-intercalate squares nor priority for this formula is claimed to be new.

For an odd prime p and $a \in \mathbb{F}_p \setminus \{0, -1\}$, define P_a on $\mathbb{F}_p \cup \{\infty\}$ by

$$P_a(r, c) = \begin{cases} \infty & c = ar, \\ r + c & c \neq ar \end{cases} \quad (r, c \in \mathbb{F}_p),$$

with $P_a(r, \infty) = (a+1)r$, $P_a(\infty, c) = (a+1)c/a$, and $P_a(\infty, \infty) = \infty$. This is the standard one-point prolongation of addition along its affine transversal. Let P'_a swap the two entries in each of rows $0, \infty$ at columns $0, \infty$. Those four cells form an intercalate.

Theorem 13.1 (Corner trade). *Both tables are Latin and $\text{pat}(P'_a) = \text{pat}(P_a)$ component-wise. If $p \geq 5$ and $a \notin \{1, -2, -1/2\}$, then $I(P_a) = p$ and $I(P'_a) = 1$.*

Proof. The prolongation is Latin, and an intercalate swap preserves this property. Put $b = (a+1)/a$, $d = \text{ord}_p(b)$ and let a_0 be the integer representative of a in $\{1, \dots, p-2\}$. The autotopisms

$$(r, c, s) \mapsto (r+t, c+at, s+(a+1)t)$$

fix infinity. Together with the scaling autotopisms $(r, c, s) \mapsto (ur, uc, us)$ for $u \neq 0$, they make all distinct finite-finite row pairs have the same cycle type. For rows $(0, t)$ with $t \neq 0$, scaling columns by t^{-1} gives the relative map

$$0 \mapsto a, \quad \infty \mapsto -1, \quad a+1 \mapsto \infty, \quad x \mapsto x-1 \quad (x \neq 0, a+1).$$

Its cycles containing 0 and infinity are separate, of lengths $a_0 + 1$ and $p - a_0$. For rows (∞, t) , the ordinary finite map is $x \mapsto bx - t$. Prolongation replaces its fixed point at by (at, ∞) ; the remaining cycles have length d . If $t \neq 0$, the point 0 belongs to one of those d -cycles.

Writing $\tau = (0, \infty)$ on columns, the trade precomposes rows $0, \infty$ with τ . The pair $(0, \infty)$ is conjugated and keeps its type. For $(0, t)$ the transposition joins the two cycles into one $(p+1)$ -cycle. For (∞, t) it joins the two-cycle containing infinity to a separate d -cycle, giving lengths $d+2, d, \dots, d$. Pairs of distinct nonzero finite rows are unchanged. Thus row-F holds after the trade exactly when $a_0 + 1, p - a_0, d$ are all even, which is also its condition before the trade. Necessity uses an unchanged nonzero finite pair and the pair $(0, \infty)$; these exist also for $p = 3$.

Transposition gives exactly $P_a^T = P_{a^{-1}}$, preserving the corner trade. In the symbol view, take symbols as lines and columns as positions, and relabel finite columns by $c' = -c$. Solving the table equation gives precisely the same construction with parameter $-a/(a+1)$, again taking the corner to the corner. Applying the row calculation to these two parameters proves column and symbol equality.

Each two-cycle in a row-pair permutation is one intercalate. In P_a the p pairs involving infinity each contribute their distinguished two-cycle. Additional two-cycles occur exactly when $d = 2$, $a_0 + 1 = 2$ or $p - a_0 = 2$, that is, $a = -1/2, 1, -2$. Outside these exceptions the count is p . After the trade, the complete pair list above leaves only the distinguished two-cycle of $(0, \infty)$, giving one. $\square$

Corollary 13.2 (One intercalate and no standard two-point contraction). *For every prime $p \geq 10^{12}$ there exists a non-group-isotopic FFF square of order $p+1$ with exactly one intercalate. It is not isotopic or parastrophic to a standard two-transversal prolongation of an order- $(p-1)$ square with all retained cells unchanged.*

Proof. The isotopy $x = (a+1)r$, $y = (a+1)c/a$, with symbols unchanged, takes P_a to the loop of Section 10 with parameter $\alpha = 1/(a+1)$. The inverse correspondence $a = (1-\alpha)/\alpha$

is bijective. Theorem 10.3 gives more than $p/1000 > 3$ FFF parameters; at most three are exceptional for Theorem 13.1. This proves existence. A group table of even order n with t nonidentity involutions has $tn^2/4$ intercalates: each involution gives $n/2$ unordered row pairs with $n/2$ two-cycles. Here $t \geq 1$ and $n \geq 4$, so this is greater than one. Intercalate count is isotopy invariant.

The unique intercalate of a generic P'_a has rows, columns and symbols $\{0, \infty\}$. Delete them. For each $r \neq 0$, the retained position $(r, -r)$ is a hole because its symbol was zero. Its possible values must lie in

$$\{r, (a+1)r\} \cap \{-r, -(a+1)r/a\}.$$

The four possible equalities respectively force characteristic two, $a = -1/2$, $a = -2$ or $a = -1$, all excluded. Thus the intersection is empty and even one such hole forbids a Latin completion with retained entries fixed. A standard two-transversal prolongation has an order-two corner whose reverse operation is just such a completion. Uniqueness of the intercalate and invariance under all three-sort relabellings and coordinate permutations prove the final assertion. $\square$

This corollary inherits the published character-sum estimate used in Theorem 10.3; it is not a physical computation at enormous orders. Independently scanned small examples are $(p, a) = (13, 3)$ and $(19, 7)$, of orders 14 and 20. The finite checker tests pattern preservation at all 253 admissible parameters over primes through 43 and the intercalate and empty-domain formulas at all 216 generic parameters. Every one of the 15 affine cases at $p = 17$ remains non-FFF. No arbitrary order-18 claim follows.

Computational Theorem 13.3 (Expanded explicit class lower bounds). *There are at least 1703 FFF main classes at order 20, at least $T_d(1703)$ at order $20 \cdot 2^d$, and in particular at least 1450956 at order 40.*

Proof. The frozen enlargement of the Steiner input tests every trade mask for each of its 22 all-two-cycle row pairs. Its historical enumeration records 22528 masks, 19480 FFF labelled mask instances and 1700 distinct exact aggregate spectra. Two quadratic controls add two further spectra. For the present lower bound it suffices to reconstruct one specified table per spectrum: the fresh companion checker reconstructs all 1702, validates Latinness and all three views by two methods, and compares the spectra as exact tuples, not merely hashes. Each has at least 19 intercalates; the checked P'_7 at $p = 19$ has one and supplies another class.

For all 1703 selected tables, fresh binary elimination produces a 58-square minor, and a separate explicit matrix elimination checks its nonsingularity and binding to the table. Lemma 11.7 establishes binary quotient freeness without C157. Apply Theorem 11.5 and Equation (11); at $d = 1$ the count is $1703 \cdot 1704/2 = 1450956$. $\square$

The fresh representative audit proves these lower bounds without repeating the complete 22528-mask search. It does not prove an exact number of all FFF classes, completeness of the spectrum invariant, or literature priority. The older 514-input bounds remain valid but are no longer the strongest bound in this edition.

14 Odd-subset lift characterization

The even-cycle condition also has an exact formulation without auxiliary cycle colours. This formulation will be useful both structurally and for proof-producing computation.

Lemma 14.1 (Invariant odd subsets). *A permutation of a finite set has only even cycles if and only if it preserves no subset of odd cardinality.*

Proof. The support of an odd cycle is an invariant subset of odd cardinality. Conversely, every invariant subset is a disjoint union of cycles. A union of even cycles has even cardinality, so an invariant subset of odd cardinality contains an odd cycle. $\square$

Let $\Lambda = \{\lambda_i : U \rightarrow V : i \in I\}$ be one line family of a Latin square. Thus, in the row view, U is the column set, V is the symbol set, and λ_i is row i as a bijection. The column view uses bijections from rows to symbols, and the symbol view uses the position bijections from columns to rows. In every case $|I| = |U| = |V| = n$, and distinct lines are pointwise disjoint.

For $1 \leq k \leq n$, let $\rho_k(\lambda_i)$ be the permutation matrix of the induced bijection from the k -subsets of U to the k -subsets of V , and put

$$B_k(\Lambda) = \sum_{i \in I} \rho_k(\lambda_i). \quad (12)$$

Its (A, B) entry counts the lines satisfying $\lambda_i(A) = B$.

Theorem 14.2 (Odd-subset collision characterization). *Let Λ be a view of a Latin square of even order n . The view is F if and only if $B_k(\Lambda)$ is a zero-one matrix for every odd k with $3 \leq k \leq n/2$. If $n/2$ is odd, it is enough at level $n/2$ to test one member of each complementary pair of subsets.*

Consequently, a Latin square of even order is FFF if and only if this collision-free condition holds for the row, column, and symbol line families.

Proof. For distinct i, j and a k -subset $A \subseteq U$, the two induced images collide precisely when

$$\lambda_i(A) = \lambda_j(A) \iff (\lambda_j^{-1}\lambda_i)(A) = A.$$

Thus an entry of $B_k(\Lambda)$ exceeds one precisely when a relative two-line permutation preserves the corresponding k -subset.

By Lemma 14.1, a relative permutation has an odd cycle if and only if it preserves an odd-cardinality subset. Since two distinct Latin-square lines have no common point, their relative permutation is fixed-point-free, so $k = 1$ is automatic. Because n is even, a subset is invariant if and only if its complement is invariant, and the two sets have the same odd parity. It therefore suffices to retain sizes at most $n/2$; when $k = n/2$ is odd, one set from each complementary pair is enough. This proves the claim for one view. Applying the same argument to the three line families proves the FFF statement. $\square$

Every matrix $B_k(\Lambda)$ has constant row and column sum n . Hence, in an F-view, it is the biadjacency matrix of a simple n -regular bipartite graph on the two k -subset sets. At order 10, the theorem requires exactly the levels $k = 3$ and $k = 5$, with one representative of each complementary pair at level five. At orders 6 and 8, only $k = 3$ is needed.

The characterization has a direct CNF consequence. If $P_{ij}(x, y)$ denotes the exact permutation matrix of $\lambda_j^{-1}\lambda_i$, then for every tested odd subset A one may add the clause

$$\bigvee_{x \in A, y \notin A} P_{ij}(x, y).$$

Because P_{ij} is a permutation matrix, this clause says exactly that $P_{ij}(A) \neq A$. Theorem 14.2 proves that the resulting three-view encoding is equisatisfiable with FFF. Formula size alone does not imply computational superiority; the bounded order-10 pilots in Section 21 remain undecided.

15 A primitive-group reduction at order ten

Order 10 admits a stronger reduction than congruence-simplicity. Normalize the ten line permutations in one view at a base line, obtaining

$$H = \{h_0 = 1, h_1, \dots, h_9\} \subseteq \text{Sym}(10).$$

Sharp transitivity is preserved, and the relative permutation between two members is $h_j^{-1}h_i$. Thus an F-view makes H a clique of size ten in the graph joining g, h when $h^{-1}g$ has only even cycles.

Computational Theorem 15.1 (Primitive one-view reduction). *Every normalized F line family of degree 10 is primitive. Assuming the complete standard classification of primitive permutation groups of degree 10, the group generated by that family is A_{10} or S_{10} .*

Proof. The only nontrivial equal-block partitions of ten points have two blocks of size five or five blocks of size two. For the first case, fix a point x in a block B of size five. Sharp transitivity gives exactly five members of H sending x into B . Each preserves B setwise, since the generated group permutes the blocks. One is the identity, so choose another such member h . Sharp transitivity against the identity makes h fixed-point-free. Its restriction to the odd set B has an odd cycle of length at least three, contrary to the F-condition. This excludes the first case directly.

For the second case, write the points as (i, t) with $i \in \{0, \dots, 4\}$ and $t \in \mathbb{F}_2$. Every block-preserving member g has the form

$$g(i, t) = (\phi_g(i), t + f_g(i)).$$

The projection $g \mapsto \phi_g$ is injective on the normalized F-family. Indeed, if distinct g, h had the same projection, then $h^{-1}g$ would fix each two-point block setwise. Since it is fixed-point-free, it flips every block. Choose a third family member k . The two relative maps $k^{-1}g$ and $k^{-1}h$ induce the same permutation ψ of the five blocks. The odd set of blocks forces ψ to have an odd cycle D , possibly a fixed point. The binary fibre voltages of the two relative maps around D differ by $|D| \equiv 1 \pmod{2}$. Exactly one voltage is therefore zero, and its lift consists of two odd cycles of length $|D|$ (fixed points when $|D| = 1$), contradicting the F-condition.

Sharp transitivity now makes the ten distinct projections a two-fold sharply transitive subset of S_5 , with binary fibre voltages. The resulting exact finite formula is unsatisfiable; its DRAT proof has been independently checked. Hence the generated group is primitive.

The primitive-group classification [BL96, HLRDR26] lists nine degree-10 actions. A frozen GAP 4.16.0/PrimGrp 4.0.3 query reproduces the catalogue list and versions the external classification boundary. The actions have orders

$$60, 120, 360, 720, 720, 1440, |A_{10}|, |S_{10}|.$$

The first seven are the actions of A_5, S_5 on 2-subsets and the natural projective-line actions of $\text{PSL}(2, 9)$, $\text{PGL}(2, 9)$, $\text{P}\Sigma\text{L}(2, 9) \cong S_6$, M_{10} , and $\text{P}\Gamma\text{L}(2, 9)$; the extension notation agrees with the ATLAS [WPNB]. Direct construction verifies their orders, closure, transitivity, and primitivity. The exact maximum sizes of F-cliques including the identity are, in the same order,

$$1, 2, 1, 2, 2, 2, 4.$$

Two bounds follow from complete enumeration with no eligible nonidentity element. Each of the other five upper bounds is an UNSAT CNF with an independently checked DRAT trace. None can contain the required clique of size ten, so only A_{10} and S_{10} remain. $\square$

The three views are further coupled by signs. If k_R, k_C, k_S count the odd row-, column-, and symbol-line permutations, then

$$k_R + k_C + k_S \equiv 1 \pmod{2}. \quad (13)$$

Indeed, on the n^2 cells the row-entry, column-entry, and symbol-position bijections differ by coordinate transposition. Taking signs gives

$$R_{\text{sign}} C_{\text{sign}} S_{\text{sign}} = (-1)^{n(n-1)/2},$$

which is negative at $n = 10$.

Corollary 15.2. *In every hypothetical FFF Latin square of order 10, the normalized generated group in each view is A_{10} or S_{10} , and at least one view generates S_{10} .*

Proof. If a normalized view group lies in A_{10} , all unnormalized line permutations have the same sign. Its number of odd lines is therefore 0 or 10, hence even. If all three view groups were A_{10} , Equation (13) would fail. Computational Theorem 15.1 leaves S_{10} as the only alternative. $\square$

This result by itself excludes neither remaining group and therefore does not settle order 10; Computational Theorem 17.3 does so later by a different exhaustive argument. The primitive-group reduction replaces an unrestricted primitive search by a three-view compatibility problem inside A_{10} and S_{10} . As a bounded diagnostic, all 135 distinct valid order-10 partial tables already frozen in the computational corpus were audited independently. Of their 257 F-views, 256 generate S_{10} and one generates A_{10} ; 93 of the non-FFF tables already generate S_{10} in all three views. Thus generated-group order alone does not separate FFF squares from current near-models. These figures describe only the frozen corpus and are not a census of order-10 Latin squares.

16 The three-coloured flag surface

The three views can be coupled more tightly than by their separate cycle counts. Fix $r \neq r'$ and a column c , and put

$$s = L(r, c), \quad d = r_r'^{-1}(s), \quad u = L(r, d).$$

The three cells

$$(r, c, s), \quad (r', d, s), \quad (r, d, u)$$

pairwise share a symbol, a column, and a row. We call this triple a *flag*. It determines one induced-cycle atom in each view: the row atom on $\{r, r'\}$ through c, d , the column atom on $\{c, d\}$ through r, r' , and the symbol atom on $\{s, u\}$ through c, d .

Retain the shared coordinate as an edge label: a column–symbol edge is labelled by its row, a row–symbol edge by its column, and a row–column edge by its symbol. This distinction is essential when an intercalate produces parallel edges.

Theorem 16.1 (Flag surface and Gram identities). *The $n^2(n-1)$ flags of a Latin square form the triangular faces of a closed, possibly disconnected, three-coloured delta-complex $K(L)$. A cycle atom of length ℓ has link $C_{2\ell}$. If $C_{V,\ell}$ is the number of such atoms in view V , then*

$$\chi(K(L)) = \sum_{V \in \{R, C, S\}} \sum_{\ell=2}^n C_{V,\ell} - \frac{n^2(n-1)}{2}.$$

Every intercalate is an isolated projective-plane delta-component. If L is FFF, every other nonempty component has negative Euler characteristic.

Let B be the flag-by-cell incidence matrix, P the flag-by-atom incidence matrix, Q the cell-by-atom support matrix, and A the adjacency matrix of the Latin-square graph on cells. Then

$$B^\top B = 3(n-1)I + 2A, \quad B^\top P = 3Q.$$

For $n > 3$, $B^\top B$ is invertible and

$$P^\top P - 9Q^\top (B^\top B)^{-1} Q \succeq 0.$$

Proof. Latinness gives $c \neq d$ and $s \neq u$, and the coloured incidences recover r, r', c , so the stated flags are distinct. A cycle atom of length ℓ is reached from its two line orientations and its ℓ cycle points, hence has 2ℓ incident flags.

A labelled pair of atoms in different views specifies one repeated diagonal of a two-by-two rectangle. Its two orientations are exactly the two incident flags. Thus every labelled edge lies in two faces. Around an atom, the two edge types that preserve it alternate along the induced cycle, giving the link $C_{2\ell}$. Hence $K(L)$ is closed. With $F = n^2(n-1)$ and $E = 3F/2$, the Euler formula follows. A 2-cycle in any view is one intercalate; its three atoms, six labelled edges, and four faces form an isolated nonorientable component of Euler characteristic one, hence a projective plane. In an FFF residual component all vertex degrees are at least eight. Since $3F = \sum_v \deg(v)$ and $E = 3F/2$, one obtains $\chi = V - F/2 \leq -V/3 < 0$.

Every cell lies in $3(n-1)$ flags. Two different cells occur together in two flags exactly when they share one row, column, or symbol, proving the first matrix identity. A cell in an atom support occurs in exactly three flags incident with that atom, proving the second. The Latin-square graph has eigenvalues $3(n-1)$, $n-3$, and -3 , so $B^\top B$ has eigenvalues $9(n-1)$, $5n-9$, and $3(n-3)$ and is invertible for $n > 3$. The final assertion is the Schur complement of the positive-semidefinite Gram matrix $[B \ P]^\top [B \ P]$. $\square$

Let $\alpha_R, \alpha_C, \alpha_S$ be the fixed-point-free involutions on flags that cross the labelled edge opposite the indicated atom colour.

Proposition 16.2 (Dual-word traces). *For every $k \geq 1$,*

$$\text{Fix}((\alpha_C \alpha_S)^k) = 2 \sum_{\ell|k} \ell C_{R,\ell},$$

and cyclically in the other views. If $\beta = \alpha_R \alpha_C \alpha_S$, then

$$\text{Fix}(\beta) = 4I(L),$$

where $I(L)$ is the intercalate count. Every $\text{Fix}(\beta^k)$ is a main-class invariant.

Proof. Around a row atom of length ℓ , the first product has two orbits of length ℓ , giving the divisor sum. A flag is fixed by β precisely when crossing the three edge colours closes its two-by-two rectangle, which means that the second diagonal also repeats a symbol. This is an intercalate, and each intercalate contributes its four flags. Isotopy merely relabels flags. A permutation of the three coordinate colours sends β to a cyclic rotation of β or β^{-1} ; these have the same fixed-point counts in every power. $\square$

Theorem 16.3 (Exact atom projection). *Let $H = P^\top P$, let v be the number of cycle atoms, and let c be the number of connected components of $K(L)$. Then*

$$\text{rank}_{\mathbb{Q}} H = v - 2c.$$

Let N be the cell-by-coordinate incidence matrix, put $U = Q^\top Q$, $W = Q^\top N N^\top Q$, and let ℓ be the vector of atom lengths. Assume $n > 3$ for the following Schur-complement assertions. For

$$D = (n-1)(n-3)(5n-9),$$

the scaled Schur complement has the integral form

$$DS = DH + 6(n-1)W - 3(n-1)(5n-9)U - 32\ell\ell^\top \succeq 0.$$

Its nullity is at least $2c + 1$.

Proof. If $Pz = 0$, subtracting the equations of two triangular faces sharing an edge of colours, say, B and C equates the values at their two opposite A -vertices. Along an arbitrary dual path, crossing an edge of colour A therefore replaces the A -vertex by one with the same value, whereas crossing an edge of colour B or C leaves the A -vertex literally unchanged because it belongs to the shared primal edge. Dual connectivity consequently makes z constant on all A -vertices in a component. The same argument applies cyclically to B and C . Each face equation says that the three resulting constants sum to zero. Thus the kernel has exactly two dimensions per component, which proves the rank formula.

The cell-coordinate matrix satisfies $NN^\top = 3I + A$. Moreover

$$(B^\top B)^{-1} = \frac{5}{3(5n-9)}I - \frac{2}{3(n-3)(5n-9)}A + \frac{8}{9(n-1)(n-3)(5n-9)}J.$$

Substitution in $S = H - 9Q^\top(B^\top B)^{-1}Q$ gives the displayed integer formula. Positive semidefiniteness follows because this is $P^\top$ times the orthogonal projector away from $\text{col}(B)$ times P . The componentwise colour differences give $2c$ kernel vectors. One further vector is constant on all atoms of one colour: its image under P is the all-one flag vector, which lies in $\text{col}(B)$. $\square$

Corollary 16.4 (Topological and sign-cut consequences). *On an orientable flag component every cycle of every odd-length word in the dual involutions has even length. If n is even and L is FFF, then*

$$\chi(K(L)) \equiv \frac{n}{2} \pmod{2}.$$

For even n , every F -view has at least

$$n(n-1) - \left\lfloor \frac{n^2}{4} \right\rfloor$$

cycle atoms. Thus a hypothetical order-10 FFF square has at least 65 atoms in each view and 195 in total.

Proof. Coherent face signs change across every edge, so an odd word reverses sign and can return only after an even number of iterations. For the Euler parity, all atoms are even cycles and the global sign relation of Proposition 2.7 gives $v \equiv n(n-1)/2 \pmod{2}$; moreover $n^2(n-1)/2$ is even, so the face contribution to Euler characteristic is even. Finally, by Proposition 2.8, line pairs with an odd number of cycles form a cut of size at most $\lfloor n^2/4 \rfloor$. Such a pair contributes at least one atom and every other pair at least two. $\square$

The aggregate consequences in this corollary are not sufficient for an order-10 obstruction. An exact numerical repository witness uses the disjoint-union summaries of frozen order-8 FFF flag surfaces 1 and 2 and one isolated projective-plane intercalate component. It has 900 flags, 591 atoms, 176 components and Euler characteristic 141; Theorem 16.3 then forces $\text{rank } H = 239$. In each colour its 169 length-two and 28 length-four atoms can be regrouped into 45 formal degree-10 line-pair partitions, with 25 odd-cycle-count pairs forming the size of a 5|5 cut. The three cut parameters are (5, 5, 5) and therefore have the required odd sum. This is a decoupled aggregate consistency witness only. The formal line-pair labels are not shown compatible with the flag faces, labelled edges, or H incidences; nor does it supply a common set of 100 Latin cells, Q , coordinate-multiplicity, or Schur data. It shows precisely that the listed aggregate and componentwise conditions cannot by themselves decide order 10 by itself. Pair-label/flag-incidence compatibility is an intermediate obstruction layer not excluded by this construction.

Theorem 16.5 (Labelled line-pair Gram reduction). *Assume $n > 3$, and put $m = \binom{n}{2}$. For a flag ϕ , let $\lambda_V(\phi)$ be its unordered line-pair label in colour V , and let Z_V be the corresponding flag-by-pair incidence matrix. Then*

$$Z_V^\top Z_V = 2nI_m, \quad W^{VW} := \frac{1}{2}Z_V^\top Z_W \in \{0, 1, 2\}^{m \times m}.$$

Every row and column of W^{VW} is the weighted edge vector of a two-factor of K_n , and its component lengths are the cycle lengths of the associated two-line permutation. If E is the unsigned vertex-edge incidence matrix of K_n , then

$$EW^{VW} = 2J, \quad W^{VW}E^\top = 2J.$$

Moreover

$$K = \begin{bmatrix} nI & W^{RC} & W^{RS} \\ (W^{RC})^\top & nI & W^{CS} \\ (W^{RS})^\top & (W^{CS})^\top & nI \end{bmatrix} = \frac{1}{2}Z^\top Z \succeq 0.$$

Writing C for the restriction of this block matrix to three copies of $\ker E$, one has

$$\text{rank}_{\mathbb{Q}} K = 3n - 2 + \text{rank}_{\mathbb{Q}} C.$$

The Schur projection from Theorem 16.3, obtained by removing the coarse flag-by-cell incidence satisfies

$$\Sigma \succeq 0 \iff K \succeq 0, \quad \text{rank}_{\mathbb{Q}} \Sigma = \text{rank}_{\mathbb{Q}} K - 1.$$

Finally,

$$t(L) = \text{tr}(W^{RC}W^{CS}(W^{RS})^\top)$$

is a main-class invariant.

Proof. Each line-pair label occurs in $2n$ flags. A cross-codegree counts repeated diagonals in the rectangle selected by two pair labels; its value is zero, one, or two. Equivalently, W^{RC} is the sum over symbols of the permutation matrices induced on unordered row pairs by the corresponding row-column matchings, and cyclically for the other colours. This proves the two-factor statement and the equations with E .

The displayed block matrix is its defining Gram matrix. Since $EE^\top = (n-2)I + J$, one has the orthogonal decomposition

$$\mathbb{Q}^m = \langle \mathbf{1} \rangle \oplus E^\top(\mathbf{1}^\perp) \oplus \ker E.$$

Every W acts by n on constants, annihilates endpoint contrasts, and preserves $\ker E$. The constant block has rank one, the three endpoint blocks have total rank $3(n-1)$, and the rank formula follows.

For the Schur statement, direct flag counting gives $B^\top Z_V = 3N_V E$, where N_V records the V -line of a cell. Substitution of the closed inverse from Theorem 16.3 diagonalizes Σ on the same decomposition: it is zero on the common constant direction, positive definite on the three endpoint-contrast spaces, and equals $2C$ on the cycle spaces. This proves both assertions. Independent row, column, and symbol relabellings conjugate the three pair-label sets, while parastrophy only permutes the colours and transposes blocks. The symmetric triangle contraction $t(L)$ is therefore invariant. $\square$

This theorem isolates the full nontrivial common-Gram information in dimension $3(m-n)$, but it does not characterize realizable Latin squares. In particular, the coarse cell projection cannot supply a stronger pair-matrix obstruction; finer common cell support or another nonlinear compatibility is required.

Proposition 16.6 (Full pair-label tensor). *For unordered row-, column-, and symbol-pair labels e, f, g , let*

$$T(e, f, g) = \#\{\phi : \lambda_R(\phi) = e, \lambda_C(\phi) = f, \lambda_S(\phi) = g\}.$$

Every entry of T lies in $\{0, 1, 4\}$. Its two-dimensional marginals are $2W^{RC}, 2W^{RS}, 2W^{CS}$. More precisely, a marginal entry zero has no completion, an entry two has two distinct completions of multiplicity one, and an entry four has one completion of multiplicity four; the latter is exactly an intercalate.

The multiplicity-one support is a closed three-coloured pair-label pseudocomplex $X_\lambda(L)$. At a line-pair vertex its link is the disjoint union of the cycles $C_{2\ell}$ belonging to the nonintercalate induced cycles of lengths ℓ on that line pair. Consequently L is FFF if and only if every component of every vertex link of $X_\lambda(L)$ has length divisible by four.

For a fixed row r , contracting T over all row-pair labels incident with r and thresholding entries at two gives exactly the two-subset action of the row permutation $c \mapsto L(r, c)$. Cyclic contractions recover the column and symbol line maps. Since the two-subset action of S_n is faithful for $n \geq 3$, the fixed-coordinate map $L \mapsto T(L)$ is injective for $n \geq 3$. This restriction is necessary: the two coordinate-labelled order-2 squares have the same single tensor entry, of value 4.

Proof. Fix a row pair and a column pair. A flag exists exactly when a diagonal of the selected rectangle repeats a symbol. If precisely one diagonal repeats, its two orientations give two flags with different symbol-pair labels; equality of those labels would make the other diagonal repeat as well. If both diagonals repeat, the rectangle is an intercalate and all four flags have the same three pair labels. This proves the local law and every marginal statement, cyclically in the colours.

After deleting multiplicity-four faces, each active two-label edge has the two multiplicity-one completions just described, so it lies in exactly two faces. Around a fixed pair label, the flag-surface links of distinct nonintercalate cycle atoms remain disjoint and are $C_{2\ell}$. Splitting a singular pair-label vertex by these link components recovers the nonintercalate part of the flag surface in Theorem 16.1. Finally, an induced cycle length $\ell > 2$ is even exactly when 2ℓ is divisible by four; the omitted intercalate cycles have length two and are already F-compatible.

For the contraction claim, fix a column pair $\{c, d\}$. The pair $\{L(r, c), L(r, d)\}$ is the unique symbol-pair label of contracted multiplicity at least two: the positive multiplicities

are $(2, 1, 1)$ unless the two diagonal completions use the same partner row, in which case the rectangle is an intercalate and the sole multiplicity is four. Thus the threshold matrix is the edge lift of the row permutation. For distinct $d_1, d_2 \neq c$, the intersection of the two image edges of $\{c, d_1\}$ and $\{c, d_2\}$ is the singleton $\{L(r, c)\}$, proving faithfulness and reconstruction. The cyclic cases are identical. $\square$

Theorem 16.7 (Smith torsion and total unimodularity). *Fix one pair-label coordinate of the tensor T in Proposition 16.6, and let T_e be the resulting integral matrix. Delete its identically zero rows and columns and denote the active square matrix by $\hat{T}_e$. If the associated two-line permutation has t cycles of length two, o odd cycles, and q even cycles of length at least four, then, up to unit Smith factors,*

$$\text{coker}(\hat{T}_e) \cong \mathbb{Z}^q \oplus (\mathbb{Z}/2\mathbb{Z})^o \oplus (\mathbb{Z}/4\mathbb{Z})^t.$$

In particular,

$$o = \text{rank}_{\mathbb{Q}}(T_e) - \text{rank}_{\mathbb{F}_2}(T_e \bmod 2) - \#\{(f, g) : T_e(f, g) = 4\}.$$

Consequently a Latin square is FFF if and only if this expression is zero for every slice in all three tensor directions.

The ranks in the displayed formula are unchanged when zero rows and columns are deleted; the full cokernel merely has the corresponding additional free factors.

Equivalently, define $N_e(f, g) = 1$ when $T_e(f, g) > 0$ and zero otherwise. A Latin square is FFF if and only if every N_e in all three directions is totally unimodular; equivalently, every such matrix is balanced. We use the standard terminology of Camion and Berge [Cam65, Ber72]; the equivalence asserted here is for these special two-regular slice blocks, not for arbitrary zero-one matrices.

Proof. The slice normal form from Proposition 16.6 is a direct sum of $[4]$ over the length-two cycles and $I_\ell + P_\ell$ over all longer cycles. The cokernel of the latter block has generators x_i with relations $x_i + x_{i+1} = 0$. Hence every generator is $(-1)^i x_0$, and closing the cycle leaves a free generator when ℓ is even and imposes $2x_0 = 0$ when ℓ is odd. This proves the Smith and rank formulas; the entries equal to four are precisely the intercalates.

After normalization, the blocks are $[1]$ and $I_\ell + P_\ell$. For odd ℓ the latter has determinant 2 and is not totally unimodular. For even ℓ it is the unsigned vertex-edge incidence matrix of a bipartite cycle; changing signs on one vertex class makes it an oriented incidence matrix, which is totally unimodular. Direct sums preserve this property. The balanced-matrix statement is the same forbidden odd-cycle condition for these two-regular incidence blocks. $\square$

Theorem 16.8 (Triple endpoint contraction). *Let E_R, E_C, E_S be the unsigned vertex-edge incidence matrices of the complete graphs on the row, column, and symbol sets, respectively, and put*

$$D = (E_R \otimes E_C \otimes E_S)T, \quad H = D \bmod 2.$$

Every one-dimensional line of D has sum $8(n-1)$. If $x = (r, c, L(r, c))$ is a Latin cell and $\iota(x)$ is the number of intercalates containing it, then

$$D(r, c, L(r, c)) = 3(n-1) + \iota(x), \quad \sum_{s \neq L(r, c)} D(r, c, s) = 5(n-1) - \iota(x).$$

Consequently H lies in the tensor product of the three even-sum point spaces over $\mathbb{F}_2$, each of its mode ranks is at most $n-1$, and its values on the Latin incidence positions recover the cellwise intercalate parity after the table is reconstructed by Theorem 16.31.

More precisely, write $T = X + 4Y$ with $X = \mathbf{1}_{T=1}$ and $Y = \mathbf{1}_{T=4}$. If $A = (E_R \otimes E_C \otimes E_S)X$ and $B = (E_R \otimes E_C \otimes E_S)Y$, then on every Latin cell x ,

$$B(x) = \iota(x), \quad A(x) = 3(n - 1 - \iota(x)).$$

In particular $0 \leq \iota(x) \leq n - 1$.

Proof. Fix row and column endpoints (r, c) . Summing over the symbol endpoint contributes two endpoints for every symbol edge. Proposition 16.6 and Theorem 16.5 therefore give

$$\sum_s D(r, c, s) = 4 \sum_{e \ni r, f \ni c} W^{RC}(e, f) = 8(n - 1).$$

The other two directions follow cyclically. Reduction modulo two proves the even-line and rank statements.

Every cell lies in exactly $3(n - 1)$ flag faces. A nonintercalate flag has exactly its three face cells in its endpoint box, so it contributes at x exactly when the face contains x . The four flags of an intercalate have one common triple of pair labels. All four endpoint boxes contain each of the four intercalate cells, whereas only three of the four faces contain a fixed cell. Hence every intercalate through x contributes one extra unit, which proves the first displayed cell formula. Subtracting it from the line sum proves the complementary formula.

Finally, every multiplicity-four entry of T is one intercalate and its endpoint box contains precisely its four Latin cells. Thus $B(x) = \iota(x)$; the formula for A follows from $D = A + 4B$. Its nonnegativity gives the local upper bound. $\square$

Theorem 16.9 (Colour-parity holonomy). *Let $K^*(L)$ be the dual cellulation of the flag surface. Assign to every dual edge of colour $V \in \{R, C, S\}$ the corresponding standard basis vector $e_V \in \mathbb{F}_2^3$, and denote the resulting one-cochain by ω . Then ω is a cocycle if and only if L is FFF.*

More precisely, the coordinate cochain ω_R is closed exactly when the column and symbol views are F, and cyclically. On every connected FFF component,

$$[\omega_R] + [\omega_C] + [\omega_S] = w_1(K(L)).$$

If the three colour classes span a subspace of dimension h in $H^1(K(L); \mathbb{F}_2)$, then the number of flags in that component is divisible by 2^{3-h} .

On every intercalate component, which is a projective plane, the three colour classes are equal to the same nonzero class. Finally, if L is FFF and $n \equiv 2 \pmod{4}$, at least one component has nontrivial colour-parity holonomy.

Proof. An atom in the row view with induced cycle length ℓ has a dual boundary containing ℓ column-coloured and ℓ symbol-coloured edges. Therefore

$$(\delta\omega)(a) = \ell(e_C + e_S),$$

and the two cyclic analogues hold in the other views. This vanishes exactly when the atom length is even. Thus the three colour-stratified coboundaries locate every odd-cycle atom and retain the full three-view pattern. By contrast, the three booleans recording whether the coordinate cochains are globally closed are only the pairwise conjunctions of the F-view bits and do not distinguish patterns with at least two failing views.

Orient every primal flag triangle first by the fixed colour order (R, C, S) . The two induced orientations on a shared labelled edge agree, so the transition cochain is one on

every dual edge and represents w_1 . Since $\omega_R + \omega_C + \omega_S$ is precisely this constant-one cochain, the displayed cohomology identity follows.

Let $H \leq \mathbb{F}_2^3$ be the holonomy subgroup obtained by integrating ω around closed dual paths. Its dimension is h . Path integration labels flags by $\mathbb{F}_2^3/H$. A word in the three dual edge involutions acts bijectively on the component and translates every label by the same element. Because the three basis vectors generate the quotient, all 2^{3-h} label fibres have equal size, proving divisibility.

The dual graph of an intercalate component is K_4 , with its three colour classes as the three perfect matchings. Every triangle uses all three colours, so each coordinate evaluates nontrivially on the unique nonzero mod-two homology class; all three equal $w_1 \neq 0$.

For the final statement, Corollary 16.4 gives $\chi(K(L)) \equiv n/2 \pmod{2}$ for even-order FFF squares. If $n \equiv 2 \pmod{4}$, the total Euler characteristic is odd. Since every closed orientable surface has even Euler characteristic, some component is nonorientable. The displayed identity then forces nonzero colour-class span on that component. $\square$

Definition 16.10 (Exact masks and compatible potentials). Let F be a connected component of the dual flag cellulation. For $m = (m_R, m_C, m_S) \in \mathbb{F}_2^3$, put

$$\omega_m = m_R \omega_R + m_C \omega_C + m_S \omega_S.$$

The mask m is *exact on F* if there is a binary zero-cochain p_m on the flags of F with $\delta p_m = \omega_m$, equivalently

$$p_m(\alpha_V f) = p_m(f) + m_V \quad (V \in \{R, C, S\}).$$

The exact masks form a vector space M_F . A *compatible family of potentials* is a linear choice $m \mapsto p_m$ for $m \in M_F$, modulo componentwise constants. Such a family is obtained by choosing potentials for a basis of M_F , normalizing them at one base flag, and extending linearly.

Three-digit masks are written in tuple order (R, C, S) . When integer mask codes are used below, the code of m is $m_R + 2m_C + 4m_S$; hence codes 3, 5, 6 correspond respectively to tuples 110, 101, 011. In particular, an integer's usual binary display must not be read as an (R, C, S) tuple.

The componentwise multiset of flag count, Euler characteristic, orientability, and exact linear relations among the three colour classes is preserved by isotopy. Parastrophy applies one global permutation of the three colours. Sorting these tuples and minimizing over the six global colour permutations therefore gives a main-class invariant.

Computational Theorem 16.11 (Order-8 holonomy profiles). *On the 230 frozen order-8 FFF main-class representatives, the canonical component-holonomy profile takes exactly 152 values. Exactly 109 values are singleton fibres; the maximum fibre size is five. The collision histogram by fibre size is*

$$1 : 109, \quad 2 : 22, \quad 3 : 12, \quad 4 : 4, \quad 5 : 5.$$

This invariant records the global gluing of the alternating bipartitions described by Theorem 16.9. The collisions in the computational theorem show that it is not a complete classifier, and no injectivity in higher orders is asserted.

Theorem 16.12 (Oriented holonomy-to-cell lift). *Let I be an intercalate with rows r, r' , columns c, c' , and symbols s, t , displayed as*

	c	c'
r	s	t
r'	t	s

On its four-flag component, the exact colour masks are 000, 011, 101, 110. For each nonzero exact mask m , choose a binary potential on the flags, replace its values by signs, and push those signs to the four Latin cells by summing over incident flags. Up to an independent global sign in each mask, the resulting cell vectors are

$$u_{011} = (1, 1, -1, -1), \quad u_{101} = (-1, 1, -1, 1), \quad u_{110} = (1, -1, -1, 1).$$

Thus u_{011} has zero column and symbol marginals, u_{101} has zero row and symbol marginals, and u_{110} has zero row and column marginals. The remaining marginal is in each case twice an oriented endpoint difference, and

$$u_{011}u_{101}u_{110} = -\mathbf{1}_I$$

pointwise after the displayed gauge choice.

If these vectors are assembled as cell-by-intercalate matrices, their three entrywise absolute values coincide with the intercalate incidence matrix J . Consequently

$$J\mathbf{1}(x) = \iota(x) = \text{endpoint}(Y)(x)$$

on every Latin cell x , where Y is the multiplicity-four layer in Theorem 16.8.

Proof. Every flag of the intercalate contains exactly three of its four cells, and each cell x is omitted by a unique flag f_x . A potential for a nonzero exact mask has two positive and two negative signs: its two selected colour matchings form a four-cycle in the dual K_4 , and the potential alternates around that cycle. Their total is zero, so the pushforward at x equals the negative of the sign at f_x . It therefore has absolute value one on all four cells. Choosing the three potential gauges gives the displayed vectors, from which all marginal and pointwise-product identities follow directly.

The common absolute-value column is precisely the incidence vector of I . Summing these columns counts the intercalates through x . The final equality is the multiplicity-four endpoint formula of Theorem 16.8. $\square$

The theorem supplies an oriented lift of the complete multiplicity-four attachment layer. It imposes no condition on the multiplicity-one layer X and hence is not, by itself, an order-10 obstruction.

Theorem 16.13 (Exact-mask cell pushforwards). *Let F be any connected flag-surface component, let $m = (m_R, m_C, m_S)$ be an exact colour mask on F , and choose a binary potential p_m on its flags. Define*

$$u_m(x) = \sum_{\substack{f \in F \\ x \in f}} (-1)^{p_m(f)}$$

on Latin cells. If $m_V = 1$, then every line marginal of u_m in view V is zero. Thus an exact mask kills all marginal families indexed by its set bits. Complementing the potential changes u_m only by a global sign.

Proof. Order the three cells in every flag by the defining flag rectangle. For a fixed view V and cell role, there is a fixed-point-free involution preserving the V -line of that role. It is either the dual involution α_V or a conjugate $\alpha_W\alpha_V\alpha_W$: row roles zero and two use α_R and row role one uses $\alpha_C\alpha_R\alpha_C$, with the cyclic analogues for columns and symbols. These statements follow immediately by writing the three cells of the adjacent flag rectangles.

Across α_V the potential changes by m_V ; across the displayed conjugate it changes by $m_W + m_V + m_W = m_V$. If $m_V = 1$, the two signs in each role pair are opposite while their selected cells lie on the same V -line. Each role sum, and hence the full line marginal, vanishes. $\square$

Unlike Theorem 16.12, this statement applies to every component and therefore reaches the multiplicity-one surface. It remains a componentwise linear consequence of exactness, not a joint X/Y compatibility theorem.

Theorem 16.14 (Cellwise holonomy Fourier transform). *Let the exact colour-mask space of a connected flag component have basis $m_1, \dots, m_d$, and choose compatible binary potentials p_i . Label a flag by $q(f) = (p_1(f), \dots, p_d(f)) \in \mathbb{F}_2^d$, and let $N_x(a)$ count the flags of label a incident with a Latin cell x . For $c \in \mathbb{F}_2^d$, let u_c be the cell pushforward of the potential $\sum_i c_i p_i$. Then*

$$u_c(x) = \sum_{a \in \mathbb{F}_2^d} (-1)^{c \cdot a} N_x(a).$$

Consequently,

$$N_x(a) = 2^{-d} \sum_c (-1)^{c \cdot a} u_c(x), \quad \sum_c u_c(x)^2 = 2^d \sum_a N_x(a)^2.$$

In particular, every inverse signed sum is a nonnegative multiple of 2^d .

Proof. Group the flags in the definition of $u_c(x)$ by their label a . This gives the first formula. The other two are Fourier inversion and Parseval on the elementary abelian group $\mathbb{F}_2^d$. A basis change relabels the group, while complementing a potential translates its labels and changes character signs, so the asserted constraints are intrinsic. $\square$

For an intercalate component, $d = 2$ and a supported cell sees three of the four flag labels. Parseval becomes $3^2 + 1^2 + 1^2 + 1^2 = 4 \cdot 3$, recovering the oriented intercalate vectors of Theorem 16.12. In general the theorem couples all exact masks at a cell, but it does not yet relate different connected components.

Theorem 16.15 (Affine transport of cell roles). *In the notation of Theorem 16.14, let $N_{x,j}(a)$ count label- a flags in which x has cell role j , and let $b_V = ((m_1)_V, \dots, (m_d)_V)$. Then*

$$N_{x,2}(a + b_R) = N_{x,0}(a), \quad N_{x,2}(a + b_C) = N_{x,1}(a), \quad N_{x,1}(a + b_S) = N_{x,0}(a).$$

Hence the three role totals are equal. With $\delta = b_R + b_C + b_S$, one has $N_{x,0}(a + \delta) = N_{x,0}(a)$. If $\delta \neq 0$, the total component incidence at x is divisible by six. This holds in particular for every orientable component.

Proof. The dual involution α_R exchanges shared roles $(0, 2)$ and adds b_R to the flag label. Likewise, α_C exchanges $(1, 2)$ and adds b_C , while α_S exchanges $(0, 1)$ and adds b_S . This gives the three displayed bijections. Their two routes from role zero to role two yield invariance under δ . If $\delta \neq 0$, its translation orbits have size two, so each role total is even. There are three equal role totals. Finally, orientability makes mask 111 exact by Theorem 16.9; therefore not every exact mask has even weight and $\delta \neq 0$. $\square$

Theorem 16.16 (Cross-component cell-parity cover). *For a flag-surface component F , let $t_F(x)$ be its common role mass from Theorem 16.15 at the Latin cell x , and put $p_F(x) = t_F(x) \bmod 2$. If $e_F^R(r), e_F^C(c), e_F^S(s)$ count the correspondingly labelled dual edges, then*

$$\sum_{x \in r} t_F(x) = 2e_F^R(r), \quad \sum_{x \in c} t_F(x) = 2e_F^C(c), \quad \sum_{x \in s} t_F(x) = 2e_F^S(s).$$

Thus every p_F has even sum on each row, column, and symbol class. Moreover,

$$\sum_F t_F(x) = n - 1, \quad \sum_{F: \delta_F=0} p_F = (n - 1)\mathbf{1} \quad \text{over } \mathbb{F}_2.$$

For every colour V and corresponding point label a , the component edge counts also satisfy $\sum_F e_F^V(a) = n(n - 1)/2$. If $\bar{t}$ is the cellwise intercalate-degree vector modulo two, then

$$\sum_{\substack{F \text{ nonintercalate} \\ \delta_F=0}} p_F = (n - 1)\mathbf{1} + \bar{t} = (E_R \otimes E_C \otimes E_S)X|_L \pmod{2}.$$

Consequently, at even order every cell lies in an odd number of $\delta_F = 0$ components with odd role mass. If its intercalate degree is even, an odd number of these components are nonintercalate.

Proof. For a fixed row, count t_F using role zero. The corresponding faces are exactly the faces incident with the α_R edges carrying that row label, and every such edge has two incident faces. This proves $\sum_{x \in r} t_F(x) = 2e_F^R(r)$. Using role two and α_C gives the column identity; using role zero and α_S gives the symbol identity. Their reductions modulo two give the asserted even-line conditions.

Every cell occurs exactly $n - 1$ times in each role globally, which gives the first displayed equality. Summing any one of the three line identities over F and then over the n cells of the fixed line gives $2 \sum_F e_F^V(a) = n(n - 1)$, which proves the global edge budget. Theorem 16.15 makes $t_F(x)$ even when $\delta_F \neq 0$, giving the second displayed equality. An intercalate component has $\delta_F = 0$ and contributes its four-cell incidence vector, so their sum is $\bar{t}$. Finally, the multiplicity-one formula in Theorem 16.8 is $((E_R \otimes E_C \otimes E_S)X)(x) = 3(n - 1 - \iota(x))$ on Latin cells. Reduction modulo two proves the last identity and its parity consequences. $\square$

Theorem 16.17 (Gauge selection and cubic cell lift). *Let $M_F \leq \mathbb{F}_2^3$ be the exact-mask space of a connected flag component. Choose mask potentials compatibly, and let $v_{m,j}(x)$ be the signed pushforward at the Latin cell x restricted to cell role j . Then*

$$v_{m,2} = (-1)^{m_R} v_{m,0}, \quad v_{m,2} = (-1)^{m_C} v_{m,1}, \quad v_{m,1} = (-1)^{m_S} v_{m,0}.$$

Consequently every odd-weight exact mask has identically zero full cell pushforward $u_m = \sum_j v_{m,j}$.

A compatible gauge change is a linear functional $\lambda : M_F \rightarrow \mathbb{F}_2$ and sends $u_m \mapsto (-1)^{\lambda(m)} u_m$. The formal monomial $\prod_m u_m^{e_m}$ has trivial transformation character exactly when

$$\sum_m (e_m \bmod 2) m = 0.$$

For an evaluated monomial that is not identically zero, this is also equivalent to gauge invariance. An identically zero evaluated monomial is invariant regardless of its formal charge. In particular, independent component gauges admit no nonzero linear signed cancellation across components without anchors.

If the full even-mask plane $\{0, 3, 5, 6\}$ is exact, then

$$K_F(x) = -u_3(x)u_5(x)u_6(x)$$

is gauge invariant. With $t_F(x)$ the common role mass,

$$|K_F(x)| \leq t_F(x)^3, \quad K_F(x) \equiv t_F(x)^3 \pmod{4},$$

and $8 \mid K_F(x)$ when $t_F(x)$ is even. On every intercalate component, $K_F = 1$ on its four cells.

Proof. The three displayed role identities are the character transforms of the three affine bijections in Theorem 16.15. Comparing the first with the other two reverses the sign of $v_{m,0}$ exactly when $|m|$ is odd, forcing that integer and hence all role sums to vanish.

Two compatible potential families differ by a componentwise constant linear functional λ . A monomial therefore acquires the sign obtained by pairing λ with its displayed mask charge. If that charge is nonzero, some linear functional takes value one on it; a nonzero evaluated monomial then changes sign. This proves the character criterion and its qualified value-level converse. Since the gauges of distinct components are independent, a charged linear contribution cannot be cancelled intrinsically by another component.

For the three nonzero even masks the role formula gives $u_3 = v_{3,0}$, $u_5 = -v_{5,0}$, and $u_6 = v_{6,0}$. Thus K_F is the product of the three nontrivial Walsh coefficients of the four-point role-zero label distribution. If its total is t , write these coefficients as $t - 2s_i$. Their absolute values are at most t . For even t all three are even. For odd t , every nonzero four-point label belongs to the negative half of exactly two nontrivial characters, so $\sum_i s_i$ is even; expanding $\prod_i (t - 2s_i)$ modulo four proves the congruence. The intercalate value is the compatible-sign specialization of Theorem 16.12. $\square$

The cubic invariant contains genuine nonintercalate information and does not sum merely to the intercalate degree. For example, in the reduced FFF square

$$\begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 0 & 3 & 1 \\ 3 & 2 & 1 & 0 \end{pmatrix}$$

the cell $(0, 0, 0)$ has one nonintercalate contribution 8 and one intercalate contribution 1, so $\sum_F K_F = 9 \neq 1 = \iota$. Thus the cross-component parity cover in Theorem 16.16 remains the canonical binary cover; unanchored linear signs and this naive cubic sum do not provide a stronger gluing law.

Definition 16.18 (Equivariant gauge-selection rule). A *gauge-selection rule* assigns, to every coordinate-labelled Latin square and every connected flag component, one compatible potential family from Definition 16.10. It is *isotopy-equivariant* if every colour-preserving isotopy transports each selected potential to the selected potential on the image component. A rule constructed only from the complete endpoint-lift data is understood to use no additional root flag, distinguished cell, or external ordering.

Theorem 16.19 (Quadratic Gram lift and equivariant gauge obstruction). *For an exact mask m on a flag component F , put*

$$Q_{F,m} = u_{F,m} u_{F,m}^T, \quad G_m = \sum_{F:m \in M_F} Q_{F,m}.$$

These integral cell kernels are independent of all compatible potential gauges, symmetric, and positive semidefinite. If the V bit of m is set, every V -line indicator lies in their kernel. In particular, for each nonzero even mask $m \in \{3, 5, 6\}$,

$$\text{rank } G_m \leq (n-1)^2, \quad G_m(x, y) \equiv \sum_{F: m \in M_F} p_F(x) p_F(y) \pmod{2}.$$

More generally, if $d = \dim M_F$ and $N_x(a)$ is the label multiplicity from Theorem 16.14, then for all cells x, y ,

$$\sum_{m \in M_F} u_{F,m}(x) u_{F,m}(y) = 2^d \sum_{a \in \mathbb{F}_2^d} N_x(a) N_y(a).$$

Consequently,

$$\sum_{m \in M_F \cap \{3, 5, 6\}} u_{F,m}(x) u_{F,m}(y) = 2^d \sum_a N_x(a) N_y(a) - 9t_F(x)t_F(y). \quad (*)$$

There is no gauge-selection rule of the preceding kind that is defined for all Latin squares and is isotopy-equivariant, even when the complete endpoint-lift data of Theorem 16.31 are supplied.

Proof. A gauge change multiplies $u_{F,m}$ by one sign, which occurs twice in $Q_{F,m}$. The kernel assertion follows from Theorem 16.13. The two relevant families of line indicators span a $(2n-1)$ -dimensional space: each family has dimension n , and their intersection consists only of the constant vector because the two Latin partitions meet once in every pair of blocks. This gives the rank bound. For even m , the role formula in Theorem 16.17 gives $u_{F,m}(x) \equiv t_F(x) \pmod{2}$, proving the congruence.

For the Fourier identity, insert the character expansions at x and y . The sum over all characters of $\mathbb{F}_2^d$ is 2^d when the two labels agree and zero otherwise. Mask zero contributes $u_{F,0}(x)u_{F,0}(y) = 9t_F(x)t_F(y)$, while every odd-weight exact mask vanishes by Theorem 16.17; subtracting the zero mask gives $(*)$.

For the final assertion, use the Klein-four table

$$\begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 1 \\ 3 & 2 & 1 & 0 \end{pmatrix}.$$

Fix the intercalate component on rows, columns, and symbols $\{0, 1\}$. The autotopism that fixes rows and applies $(0\ 1)(2\ 3)$ simultaneously to columns and symbols stabilizes this component. In a flag order for which it acts as $(f_0\ f_1)(f_2\ f_3)$, normalized potentials for masks 3, 5, 6 are

$$(0, 1, 1, 0), \quad (0, 1, 0, 1), \quad (0, 0, 1, 1).$$

The autotopism complements the first two potentials and fixes the third. Adding a gauge constant does not change these shifts. An equivariant selection for mask 3 or 5 would therefore have to be both fixed and complemented by the same autotopism, a contradiction. Theorem 16.31 reconstructs the endpoint lifts functorially from the table and conversely reconstructs the table from them, so supplying those lifts cannot remove this symmetry. $\square$

The theorem gives the canonical degree-two replacement for an impossible equivariant linear lift. It also exposes its exact boundary: components with zero-dimensional exact-mask space have no nontrivial $u_{F,m}$ and remain invisible to $(*)$ even when their cross-component parity support from Theorem 16.16 is nonzero. Thus the quadratic kernel is not an order-10 obstruction by itself.

Theorem 16.20 (Component cell-edge partition). *For every connected flag component F and every $V \in \{R, C, S\}$ there is a simple graph A_F^V on the Latin cells such that*

$$\bigsqcup_F E(A_F^V) = E(A^V),$$

where A^V is the disjoint union of the n copies of K_n on the V -lines. The three graphs belonging to one component have the same degree vector:

$$\deg_{A_F^R}(x) = \deg_{A_F^C}(x) = \deg_{A_F^S}(x) = t_F(x). \quad (1)$$

If B_F is the flag-by-cell incidence matrix restricted to F and $A_F = A_F^R + A_F^C + A_F^S$, then

$$B_F^T B_F = 3 \operatorname{diag}(t_F) + 2A_F. \quad (2)$$

Moreover, each $\Lambda_F^V = \operatorname{diag}(t_F) - A_F^V$ is positive semidefinite and

$$\sum_F \Lambda_F^V = (n-1)I - A^V. \quad (3)$$

On every V -line the entries of t_F form the degree sequence of a simple graph and hence satisfy all Erdos–Gallai inequalities. The union A_F is connected on its nonisolated cell support. For an intercalate component the three A_F^V are the three perfect matchings of K_4 ; consequently

$$\sum_{F \text{ nonintercalate}} \deg_{A_F^V}(x) = n-1 - \iota(x) \quad (4)$$

in every view. Summing (4) over the views recovers the multiplicity-one endpoint value $((E_R \otimes E_C \otimes E_S)X)(x) = 3(n-1 - \iota(x))$ from Theorem 16.8.

Proof. Two flags joined across a dual edge of colour V share exactly the two Latin cells forming the corresponding V -edge. Conversely, any two cells on one V -line lie together in exactly two flags, and those two flags are paired by that dual involution. Dual edges therefore map bijectively to Latin-cell edges. Restricting this bijection to connected components gives the stated edge partitions.

For a fixed cell x , the colour- R involution pairs its occurrences in roles 0 and 2, the colour- C involution pairs roles 1 and 2, and the colour- S involution pairs roles 0 and 1. Each role occurs $t_F(x)$ times by Theorem 16.15. Thus there are $2t_F(x)$ incident flag sides of any fixed colour and two sides above every cell edge, proving (1).

The diagonal entry of $B_F^T B_F$ at x counts the $3t_F(x)$ flags of F containing x . Two distinct cells occur together in two flags precisely when their shared cell edge belongs to A_F , and otherwise never. This proves (2). Each Λ_F^V is an ordinary graph Laplacian. Summing (1) over components gives $\sum_F \operatorname{diag}(t_F) = (n-1)I$, while the edge partition gives $\sum_F A_F^V = A^V$, proving (3).

All edges of A_F^V stay within V -lines, so (1) restricted to one line is an actual simple-graph degree sequence. Its Erdos–Gallai inequalities are therefore necessary; this is strictly stronger than even degree sum, since $(n-1, n-1, 0, \dots, 0)$ has even sum but violates the inequality at $k=2$ for $n \geq 4$. Every flag spans a triangle in A_F , and consecutive flags along a connected dual path share a full cell edge, proving connectedness on the support.

An intercalate component has four flags and four cells. In every view (1) gives degree one at each cell, hence a perfect matching; the three disjoint matchings exhaust K_4 . Exactly $\iota(x)$ intercalate components contain x . Removing their unit contribution from the global degree $n-1$ proves (4), and the final identity follows by summing the three views. $\square$

This integral lift also applies to components whose exact-mask space has dimension zero and therefore reaches the blind layer of Theorem 16.19. It remains a necessary projection: graphical line degrees alone do not ensure compatible rainbow flag triangles or a common Latin-square realization.

Theorem 16.21 (Orientable cell-companion surface). *Glue the rainbow Latin-cell triangle of every flag in a connected component F along the component cell edges of Theorem 16.20. At a Latin cell x , its link is a disjoint union of cycles. Every link cycle has length divisible by three, with edge-type vertices appearing periodically in the orders R, C, S or R, S, C .*

Let $k_F(x)$ be the number of these link cycles. Splitting x into one vertex for each link cycle gives a connected closed orientable surface Σ_F with

$$\chi(\Sigma_F) = \sum_x k_F(x) - \frac{|F|}{2} = 2 - 2g_F. \quad (5)$$

In particular,

$$\sum_x k_F(x) \equiv \frac{|F|}{2} \pmod{2}, \quad \sum_x k_F(x) \leq \frac{|F|}{2} + 2. \quad (6)$$

For an intercalate component, Σ_F is the sphere.

Proof. At x , take the incident component cell edges as link vertices. Every flag containing x joins its two incident cell edges and hence supplies one link edge. A component cell edge belongs to two flags, so each link vertex has degree two. More precisely, an R -edge sees x in roles 0 and 2 in its two flags; its two link neighbours are therefore of types S and C . The cyclic statements hold for C - and S -edges. Along any link cycle the three types must consequently repeat in one of the two cyclic orders, proving the divisibility assertion.

Orient each flag triangle by its cell-role order $x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow x_0$. Across α_S the common roles 0, 1 are exchanged, across α_C the roles 1, 2 are exchanged, and across α_R the roles 2, 0 are exchanged. Adjacent faces therefore induce opposite orientations on every common side. Splitting vertices by link components yields a closed orientable surface. It remains connected because the dual face-adjacency graph is the connected dual graph of F .

The normalized surface has $|F|$ faces, $3|F|/2$ edges, and $\sum_x k_F(x)$ vertices, giving (5). The parity and inequality in (6) follow from integral nonnegative genus. An intercalate has four faces and four length-three cell links, so (5) gives genus zero. $\square$

The cell-companion topology differs from the atom-vertex topology of Theorem 16.1: an intercalate is a sphere in the former and $\mathbb{RP}^2$ in the latter. The new genus is a joint cross-view invariant, but its necessary conditions still do not characterize Latin realizability.

Theorem 16.22 (Normalized cell-incidence rank). *Let V_F be the normalized vertices of the cell-companion surface Σ_F , and let C_F be its flag-by-vertex incidence matrix. Across each glued triangle edge, join the two vertices opposite that edge. If the resulting graph D_F on V_F has q_F connected components, then*

$$\ker_{\mathbb{Q}} C_F = \{z : z \text{ is constant on the components of } D_F \text{ and one face sum is zero}\}, \quad (7)$$

and hence

$$\text{null}_{\mathbb{Q}} C_F = q_F - 1, \quad \text{rank}_{\mathbb{Q}} C_F = |V_F| - q_F + 1. \quad (8)$$

For every nonempty connected flag component, $1 \leq q_F \leq 3$; no FFF assumption is needed for this bound. If $d_F(v)$ is the vertex-link length and M_F is the edge-multiplicity adjacency matrix of Σ_F , then

$$C_F^T C_F = \text{diag}(d_F) + 2M_F. \quad (9)$$

Finally, let R_F forget the normalized sheet over each Latin cell. For the component flag-by-cell incidence matrix B_F from Theorem 16.20,

$$B_F = C_F R_F, \quad B_F^\top B_F = R_F^\top (C_F^\top C_F) R_F. \quad (10)$$

Proof. The three corners of a flag lie over distinct Latin cells, and normalization preserves the underlying cell, so every row of C_F has three distinct ones. Subtract the incidence equations of two faces adjacent across one edge. Their two shared vertices cancel, leaving equality of the two vertices opposite the edge. A kernel vector is therefore constant on every component of D_F .

Conversely, crossing a face edge replaces only its opposite vertex by a vertex in the same D_F -component. Thus a vector constant on those components has the same three-vertex sum on every face. The dual face graph is connected, so all equations reduce to one face equation. The resulting linear functional on the q_F component constants is nonzero over $\mathbb{Q}$: the all-one vector has face sum 3. This proves (7) and (8).

More precisely, the multiset of D_F -component labels on the three corners of a face is unchanged across any dual edge: the shared corners remain, and the opposite corners are joined in D_F . Dual connectedness propagates this same multiset to every face. Every vertex occurs in a face, so at most three components can occur; nonemptiness gives $1 \leq q_F \leq 3$.

The diagonal Gram entry counts the faces at a normalized vertex and equals its link length. Two distinct vertices occur together in exactly the two faces incident with each surface edge between them, proving (9). Each normalized vertex lies over one Latin cell; forgetting that sheet in every face gives $B_F = C_F R_F$, and the Gram aggregation in (10) follows. $\square$

Thus Theorem 16.20 is the coarse cell aggregation of a canonical positive-semidefinite surface Gram. The exact rank need not be full: the frozen controls contain components with $q_F = 2$ and $q_F = 3$. Those finite distributions are not a general classification, and the theorem remains a necessary componentwise identity rather than an order-10 decision.

The atom and normalized-cell models also admit an exact mixed-rank decomposition.

Theorem 16.23 (Joint atom/cell incidence rank decomposition). *For a connected flag component F , let P_F be its flag-by-atom incidence, let C_F be its normalized flag-by-vertex incidence, and put*

$$a_F = \#\{\text{atoms of } F\}, \quad v_F = |V_F|, \quad s_F = \dim_{\mathbb{Q}}(\text{im } P_F \cap \text{im } C_F).$$

For $J_F = [P_F \ C_F]$ one has

$$\text{rank}_{\mathbb{Q}} J_F = a_F + v_F - q_F - 1 - s_F, \quad \text{null}_{\mathbb{Q}} J_F = q_F + 1 + s_F. \quad (11)$$

Moreover $s_F \geq 1$. Thus, with $\epsilon_F = s_F - 1$,

$$\text{rank}_{\mathbb{Q}} J_F = a_F + v_F - q_F - 2 - \epsilon_F, \quad \text{null}_{\mathbb{Q}} J_F = q_F + 2 + \epsilon_F. \quad (12)$$

The joint Gram $J_F^\top J_F$ is positive semidefinite and has the same rank and nullity. The integer ϵ_F is invariant under relabelling flags, atoms, and normalized vertices.

Proof. Write $U_F = \text{im } P_F$ and $W_F = \text{im } C_F$. The exact atom projection on one connected component gives $\dim U_F = a_F - 2$, while Theorem 16.22 gives $\dim W_F = v_F - q_F + 1$. Therefore

$$\text{rank } J_F = \dim(U_F + W_F) = \dim U_F + \dim W_F - \dim(U_F \cap W_F),$$

which is the rank formula in (11); subtracting it from $a_F + v_F$ gives the nullity formula.

Every flag contains exactly one row atom, so assigning potential one to all row atoms and zero to the other atoms gives the all-one face function in U_F . Every flag also has three normalized vertices, so assigning potential $1/3$ to every normalized vertex gives the same all-one face function in W_F . Hence $s_F \geq 1$, and (12) follows. Finally $\ker(J_F^\top J_F) = \ker J_F$ over $\mathbb{Q}$. Relabellings only permute rows or columns of the two incidence matrices, proving invariance. $\square$

The exhaustive structural audit certifies the rational intersection dimension for all 104645 components in the frozen 9777-table control corpus. In the 9768 order-8 FFF components, the values $s_F = 1, 2, 3, 4, 13$ occur with multiplicities 9755, 5, 5, 2, 1; the last case has $\epsilon_F = 12$. The sorted component profile $(|F|, a_F, v_F, q_F, s_F)$ takes 201 values on the 230 frozen order-8 FFF representatives, with 176 singleton values and maximum collision size three. Combining it with the canonical holonomy profile of Computational Theorem 16.11 gives 202 values and 178 singletons, still with maximum collision size three. These are exact finite-corpus invariants, not a main-class classification or a universal bound on s_F .

The intersection in Theorem 16.23 has a direct cycle-space description. For a flag f and a colour V , write $a_V(f)$ for its V -atom, let $\alpha_V(f)$ be the adjacent flag across the edge of colour V , and let $v_V(f)$ be the normalized vertex opposite that edge. Define the atom-transition multigraph Γ_F^V on the V -atoms by joining $a_V(f)$ to $a_V(\alpha_V(f))$ for every dual V -edge. Loops and parallel edges are retained.

Theorem 16.24 (Gradient characterization of common modes). *Each Γ_F^V is connected. For a normalized-vertex potential z , put on an oriented dual V -edge (f, g)*

$$\delta_z^V(f, g) = z(v_V(f)) - z(v_V(g)).$$

Then

$$C_F z \in \text{im } P_F \iff \delta_z^R, \delta_z^C, \delta_z^S \text{ are exact gradients on } \Gamma_F^R, \Gamma_F^C, \Gamma_F^S. \quad (13)$$

Equivalently, the signed sum of each cochain vanishes around every cycle of its transition multigraph.

Let X_F be the set of Latin cells used by F , let B_F^X be the flag-by-cell incidence restricted to these columns, and form the coarse opposite-cell graph E_F by joining the two underlying cells opposite each dual edge. If p_F is the number of components of E_F , then

$$\text{null}_{\mathbb{Q}} B_F^X = p_F - 1, \quad \text{rank}_{\mathbb{Q}} B_F^X = |X_F| - p_F + 1. \quad (14)$$

The sheet-forgetting projection gives $1 \leq p_F \leq q_F \leq 3$ and $\text{im } B_F^X \subseteq \text{im } C_F$. Hence, with

$$t_F = \dim_{\mathbb{Q}}(\text{im } P_F \cap \text{im } B_F^X), \quad \eta_F = s_F - t_F,$$

one has

$$1 \leq t_F \leq s_F, \quad 0 \leq \eta_F \leq |V_F| - |X_F| - q_F + p_F. \quad (15)$$

Thus $t_F - 1$ is the nonconstant common-mode dimension already visible on unsplit cells, while η_F is the quotient dimension contributed only by the normalization sheets.

Proof. Project a dual path in F to its V -atoms. Crossing a non- V edge leaves the V -atom fixed, while crossing a V -edge traverses an edge of Γ_F^V ; hence each transition graph is connected.

If $C_F z = P_F x$, subtract the equations of adjacent flags $g = \alpha_V(f)$. The two shared normalized vertices and the two shared atoms cancel, leaving

$$z(v_V(f)) - z(v_V(g)) = x_V(a_V(f)) - x_V(a_V(g)).$$

Thus all three cochains are gradients. Conversely choose atom potentials x_V realizing the three gradients. The residual

$$h(f) = (C_F z)(f) - \sum_V x_V(a_V(f))$$

is unchanged across every dual edge. It is therefore constant because the dual graph is connected; adding that constant to all atom potentials of one colour makes the residual zero. This proves (13), and the cycle criterion is the standard characterization of exact cochains on a connected graph.

The same subtraction argument for $B_F^X w = 0$ says that w is constant on every component of E_F . Conversely such a potential has the same three-cell sum on every flag, so only one face-sum equation remains. This functional is nonzero over $\mathbb{Q}$, because the all-one potential has sum three, proving (14). Forgetting the sheet maps every edge of the normalized opposite-vertex graph onto the corresponding edge of E_F , so $p_F \leq q_F$; it also gives $B_F^X = C_F R_F$ after restricting to X_F . The constant face mode gives $t_F \geq 1$. Finally, for $U = \text{im } P_F$, $T = \text{im } B_F^X$, and $W = \text{im } C_F$, the natural map $(U \cap W)/(U \cap T) \hookrightarrow W/T$ is injective. Taking dimensions and using (14) and Theorem 16.22 proves (15). $\square$

If Z_F^{norm} and Z_F^{cell} denote the normalized and coarse potential spaces satisfying the corresponding three gradient conditions, then the exact sequences induced by C_F and B_F^X give

$$s_F = \dim Z_F^{\text{norm}} - (q_F - 1), \quad t_F = \dim Z_F^{\text{cell}} - (p_F - 1). \quad (16)$$

There is also a rational Schur formulation. For the orthogonal projectors Π_C, Π_B onto $\text{im } C_F$ and $\text{im } B_F^X$, respectively,

$$\text{null}(P_F^\top(I - \Pi_C)P_F) = 2 + s_F, \quad \text{null}(P_F^\top(I - \Pi_B)P_F) = 2 + t_F.$$

This positive-semidefinite projector statement is asserted over $\mathbb{Q}$ (or $\mathbb{R}$), not over arbitrary finite fields.

The complete frozen audit checks all 104645 components of the 9777 control tables and certifies 1111 nonconstant integer coarse witnesses with no error. All 1080 exceptional order-6 components have $(s_F, t_F, \eta_F) = (2, 2, 0)$. Among the 9768 order-8 FFF components, $t_F = 1, 2, 3, 4, 13$ occurs with multiplicities 9756, 5, 4, 2, 1. Exactly one observed component has positive normalization-only quotient: source 1955, component 4, has $(s_F, t_F, \eta_F) = (3, 1, 2)$. In particular, the source 114 component with $s_F = 13$ is entirely coarse. Every observed component has $p_F = q_F$, but neither this equality nor uniqueness of the sheet-only case is asserted universally. The partial order-10 records remain diagnostic only.

The componentwise gradient conditions have a particularly rigid global intersection.

Theorem 16.25 (Global common modes and exact Schur nullity). *Let the flag surface have c connected components, let a be its total number of atoms, and let P, B, Q be the global flag-by-atom, flag-by-cell, and cell-by-atom support matrices. Put*

$$Z_L = \{w \in \mathbb{Q}^{n^2} : Bw \in \text{im } P\}, \quad r_L = \dim Z_L, \quad \kappa_L = r_L - 1.$$

For $n > 3$, the map $w \mapsto Bw$ identifies Z_L with $\text{im } P \cap \text{im } B$, and

$$\text{rank}_{\mathbb{Q}}[P \ B] = a - 2c + n^2 - r_L, \quad \text{null}_{\mathbb{Q}}[P \ B] = 2c + r_L. \quad (17)$$

If Π_B is the rational orthogonal projector onto $\text{im } B$, then

$$\text{null}_{\mathbb{Q}}(P^{\top}(I - \Pi_B)P) = 2c + r_L = 2c + 1 + \kappa_L. \quad (18)$$

Every common pair $P\xi = Bw$ has one common line sum λ on every row, column, and symbol class, and satisfies

$$Q\xi = (n - 3)w + 2\lambda\mathbf{1}. \quad (19)$$

After setting

$$w_0 = nw - \lambda\mathbf{1}, \quad \xi_0 = n\xi - 3\lambda e_R,$$

where e_R is one on all row atoms and zero elsewhere, one has

$$P\xi_0 = Bw_0, \quad N^{\top}w_0 = 0, \quad Q\xi_0 = (n - 3)w_0. \quad (20)$$

Consequently $\kappa_L \leq (n - 1)(n - 2)$. Finally, for the atom-only operator

$$\mathcal{D}_L = (n - 3)P - BQ$$

one has the exact formula

$$\text{null}_{\mathbb{Q}} \mathcal{D}_L = 2c + \kappa_L. \quad (21)$$

Thus the forced Schur lower bound $2c + 1$ is exact precisely when the stacked global gradient system has only the constant cell mode.

Proof. The Gram spectrum in Theorem 16.1 makes B injective for $n > 3$. The image-sum formula, together with $\text{rank } P = a - 2c$, then gives (17). The map $\ker(P^{\top}(I - \Pi_B)P) \rightarrow \text{im } P \cap \text{im } B$, $x \mapsto Px$, has kernel $\ker P$ and is onto, proving (18).

Write a cell as $x = (r, c, s)$. Sum $P\xi = Bw$ over each of the three families of $n - 1$ flags in which x occupies one fixed cell-corner role. On the atom side, every family enumerates once all atoms incident with x , and so has sum $(Q\xi)_x$. On the cell side the three sums are

$$(n - 3)w_x + R_r + C_c, \quad (n - 3)w_x + R_r + T_s, \quad (n - 3)w_x + C_c + T_s,$$

where R, C, T are the row, column, and symbol line sums of w . Equality forces $R_r = C_c = T_s$ at every cell. Since every row-column pair is a cell, all $3n$ sums equal one value λ , and (19) follows. The identities $Pe_R = \mathbf{1}$ and $Qe_R = (n - 1)\mathbf{1}$ now give (20).

The line-incidence Gram $N^{\top}N$ has diagonal blocks nI and off-diagonal blocks J , hence $\text{rank } 3n - 2$. Therefore $\dim \ker N^{\top} = n^2 - (3n - 2) = (n - 1)(n - 2)$, proving the bound.

Equation (20) maps centered common modes into $\ker \mathcal{D}_L$. Conversely, if $\mathcal{D}_L\xi = 0$, put $w = Q\xi/(n - 3)$. Then $P\xi = Bw$; comparison with (19) forces $\lambda = 0$. The resulting map from $\ker \mathcal{D}_L$ onto the centered common-mode space has kernel $\ker P$, because $Q = \frac{1}{3}B^{\top}P$. Its kernel has dimension $2c$ and its image dimension κ_L , proving (21). $\square$

The frozen global audit evaluates both descriptions of r_L over two large prime fields and supplies independent integer lower-bound witnesses. All 230 order-8 FFF representatives have $r_L = 1$, so their global Schur nullity is exactly $2c + 1$. In the complete order-6 corpus, $r_L = 1, 2, 3$ occurs with multiplicities 8088, 1080, 240. All 1560 nonconstant integer basis witnesses satisfy (19)–(20), and the atom-only nullity (21) agrees over both prime fields on all 9777 audited tables. These distributions are finite-corpus statements; the theorem is general. The 135 tracked order-10 tables all have $r_L = 1$, but that corpus is partial and gives no order-10 decision.

The phrase “trade space” in Theorem 16.25 refers to the integer dependencies among the occupied Latin triples. It is useful to separate this object from a classical Latin bitrade:

a nonempty bitrade mate changes symbols at fixed row-column positions and therefore uses ambient triples outside the occupied set of the original square.

For a Latin square L , write

$$\mathcal{T}_L = \ker_{\mathbb{Z}} N^{\top}.$$

If disjoint sets A, B of four occupied Latin cells have the same incidence with every row, column, and symbol line, call $\mathbf{1}_A - \mathbf{1}_B$ a balanced $4 + 4$ cell exchange and let $\mathcal{E}_4(L)$ be the lattice generated by all such exchanges.

Computational Theorem 16.26 (Order-6 common-mode exchange lattice). *Across all 9408 reduced Latin squares of order 6, the lattice $\mathcal{E}_4(L)$ is saturated in its rational span. It has rank 20 for 5448 squares and rank 19 for 3960 squares. Every balanced $3 + 3$ cell exchange belongs to $\mathcal{E}_4(L)$.*

For all 1080 global-common-mode exceptions with $r_L = 2$, and for 60 of the 240 exceptions with $r_L = 3$, one has $\mathcal{E}_4(L) = \mathcal{T}_L$. In the remaining 180 cases,

$$\mathcal{T}_L / \mathcal{E}_4(L) \cong \mathbb{Z},$$

and the saturation in $\mathcal{T}_L$ of the full rational centred common-mode space maps onto the index-two subgroup. In each of these 180 cases, a balanced $6 + 6$ cell exchange completes $\mathcal{E}_4(L)$ to $\mathcal{T}_L$.

Proof. For each table, the program groups all three- and four-cell subsets by their complete row/column/symbol incidence profile and pairs disjoint subsets in the same group. The line-incidence matrix has rank 16, so $\mathcal{T}_L$ has rank 20. Smith and Hermite normal forms in saturated kernel coordinates determine the rank, saturation, containment, and quotient indices. The rational centred common-mode basis is saturated inside $\mathcal{T}_L$ before the index-two comparison, so the conclusion is independent of the particular integer certificates produced by the global common-mode audit.

An independent implementation recomputes ambient Smith and Hermite forms with Python-FLINT and checks every rank over $\mathbb{F}_2$, $\mathbb{F}_3$, and $\mathbb{F}_{1000003}$. It also verifies all 1560 centred modes and all 180 stored $6 + 6$ completions. Both implementations report zero errors; the frozen commands, outputs, and hashes are listed in Section 21. $\square$

This theorem exhibits an integral layer that rational Schur rank does not see. It is a complete order-6 census result, not a universal quotient theorem and not an order-10 obstruction.

Reduce $\mathcal{E}_4(L)$ modulo two and let Λ_L be the binary span of the row-, column-, and symbol-line indicators. Since every balanced exchange has even intersection with every line, $\Lambda_L \subseteq \text{Ann}(\mathcal{E}_4)$. If the nonzero Smith factors of $N^{\top}$ are odd and $\mathcal{E}_4(L)$ is a saturated codimension-one sublattice of $\mathcal{T}_L$, with rank preserved modulo two, then

$$\text{Ann}(\overline{\mathcal{E}_4}) / \Lambda_L \cong \mathbb{F}_2. \tag{22}$$

The unique nonzero class evaluates on $\mathcal{T}_L$ independently of its representative, because line indicators pair trivially with line-balanced vectors. It is the unique binary character of $\mathcal{T}_L / \mathcal{E}_4(L)$. Isotopies and parastrophes transport cells, lines, and balanced exchanges, so this class is relabelling covariant. In particular, the weight enumerator of its covector coset and the line-degree profiles of minimum representatives are main-class invariants.

Computational Theorem 16.27 (Order-6 intrinsic exchange parity). *The complete order-6 census has 12 main classes. Exactly three, containing 3960 reduced squares, have $\text{rank } \mathcal{E}_4(L) = 19$. Their triples*

(reduced-class size, minimum covector weight, number of minima)

are respectively

$$(3240, 7, 12), \quad (540, 9, 80), \quad (180, 9, 160).$$

The last class is exactly the rank-19, $r_L = 3$ family of Computational Theorem 16.26; all its representatives are TTT and non-group-isotopic. Every saturated centred common mode has parity zero, while every stored 6 + 6 completion has parity one.

Proof. Canonical reduction under all isotopies and six parastrophes partitions the 9408 reduced squares into class sizes

$$20, 40, 60, 108, 180, 360, 540, 1080, 1080, 1080, 1620, 3240.$$

For every rank-19 table, exact binary elimination computes (22), enumerates its 2^{16} covector coset, and records minimum weights and line profiles. The three resulting signatures are constant inside, and distinct between, the three classes. An independent implementation rebuilds the balanced 4 + 4 supports directly and recomputes all three representative signatures; it also checks the full source partition and reports zero errors. The parity statements for common modes and completions follow from the index-two and primitive-completion conclusions of Computational Theorem 16.26 and are checked on all 180 tables. $\square$

This identifies the index-two phenomenon as one precise order-6 main-class structure. It does not force such a quotient for an FFF square of order 10.

For an arbitrary integer matrix N with m columns and an integral sublattice $E \subseteq \ker_{\mathbb{Z}} N$, put

$$\rho_{\mathbb{Q}} = \text{rank}_{\mathbb{Q}} N, \quad \rho_2 = \text{rank}_{\mathbb{F}_2} \overline{N}, \quad e_{\mathbb{Q}} = \text{rank}_{\mathbb{Q}} E, \quad e_2 = \dim_{\mathbb{F}_2} \overline{E}.$$

Proposition 16.28 (Binary exchange-defect decomposition). *The quotient*

$$Q_2(N, E) = \ker_{\mathbb{F}_2} \overline{N} / \overline{E}$$

has dimension

$$\dim Q_2(N, E) = (m - \rho_{\mathbb{Q}} - e_{\mathbb{Q}}) + (\rho_{\mathbb{Q}} - \rho_2) + (e_{\mathbb{Q}} - e_2). \quad (23)$$

All three summands are nonnegative. Moreover

$$\text{Ann}(\overline{E}) / \text{row}(\overline{N}) \cong Q_2(N, E)^*.$$

Proof. Rank-nullity gives $\dim Q_2 = m - \rho_2 - e_2$; adding and subtracting $\rho_{\mathbb{Q}} + e_{\mathbb{Q}}$ gives (23). Reduction modulo a prime cannot increase the rank of an integer matrix, and $E \subseteq \ker_{\mathbb{Q}} N$, proving nonnegativity. The annihilator statement is the standard duality between a subspace of a nondegenerate binary pairing and the dual of the corresponding quotient. $\square$

For a Latin square, take $N = N_L$ and $E = \mathcal{E}_4(L)$, and denote the dimension in (23) by $q_4(L)$. Proposition 16.28 separates three genuinely different sources of a binary character: rational exchange codimension, line-incidence rank loss modulo two, and exchange-rank loss modulo two. These quantities are main-class invariants, because isotopies and parastrophes merely permute occupied cells and line rows.

Computational Theorem 16.29 (Order-8 FFF exchange-parity controls). *Across the complete frozen set of 230 order-8 FFF representatives,*

$$\#\{L : q_4(L) = 0, 1, 2, 3\} = (39, 147, 35, 9).$$

The binary line ranks are distributed as

$$19^1 20^{10} 21^{156} 22^{63},$$

and the binary $\mathcal{E}_4$ ranks as

$$39^1 40^8 41^{46} 42^{175}.$$

In particular, FFF does not force a nonzero or a codimension-one binary exchange quotient. One explicit non-group-isotopic FFF witness is frozen local source line 170 (ANU zero-based record 169), for which

$$\text{rank}_{\mathbb{F}_2} \bar{N}_L = 22, \quad \dim_{\mathbb{F}_2} \bar{\mathcal{E}}_4(L) = 42, \quad q_4(L) = 0.$$

Proof. For each table, the audit enumerates all four-cell subsets, groups them by their complete row/column/symbol incidence profile, and pairs exactly the disjoint subsets with equal profile. Direct binary elimination computes the two ranks and verifies both inclusions defining the quotient. A separate packed-profile implementation rebuilds all exchanges and ranks for all 230 tables and reports zero discrepancies. It also stores the full local-source-line-170 table and independently confirms its FFF census identity. $\square$

On the separate, nonexhaustive corpus of 135 tracked order-10 diagnostic tables, every record has $\rho_2 = 28$, $e_2 = 72$, and hence $q_4 = 0$; four pattern representatives were independently recomputed. Those records are not FFF examples and do not contribute to the order-10 nonexistence proof. Their role is diagnostic: the exchange-parity layer is already saturated on the best available near-models, so it supplies no additional obstruction there.

Definition 16.30 (Fibres, contractions, and coherent edge lifts). For a finite set A , write $E(A)$ for its unordered two-subsets. Given $T : E(A) \times E(B) \times E(C) \rightarrow \{0, 1, 4\}$, a two-label fibre is obtained by fixing two edge labels and varying the third; its type is the multiset of its positive values. For $a \in A$ and $f \in E(B)$, the A -endpoint contraction is

$$C^a(f, g) = \sum_{\substack{e \in E(A) \\ a \in e}} T(e, f, g) \quad (g \in E(C)),$$

with cyclic analogues in the other two colours. Its dominant output is the unique g at which the contraction has value at least two. The dominant outputs for a fixed endpoint a are a coherent two-subset lift if there is a point permutation $\lambda_a : B \rightarrow C$ such that the dominant output for every $f \in E(B)$ is $\lambda_a(f) = \{\lambda_a(b) : b \in f\}$. The corresponding families in the other colours are denoted $\mu_b : A \rightarrow C$ and $\nu_c : A \rightarrow B$.

Theorem 16.31 (Endpoint characterization). *Let A, B, C be sets of cardinality $n \geq 3$. A function*

$$T : E(A) \times E(B) \times E(C) \longrightarrow \{0, 1, 4\}$$

is the pair-label tensor of a Latin square if and only if the following hold: every two-label fibre is empty, $[1, 1]$, or $[4]$; every endpoint contraction in all three colours is $[2, 1, 1]$ or $[4]$; and, for each fixed endpoint, its dominant-output map on all input edges is coherently the two-subset lift of one point permutation. The three recovered permutation families are automatically sharply transitive and define the same Latin square. Adding the link-divisibility condition of Proposition 16.6 characterizes the tensors of FFF Latin squares.

Proof. Necessity is Proposition 16.6. For sufficiency, let X and Y denote the tensor entries of multiplicity one and four. Fix a B -edge f . If p_f fibres $T(*, f, g)$ have type $[1, 1]$ and q_f have type $[4]$, double counting their incidences with endpoints of A gives $p_f + 2q_f = n$. There are also exactly n dominant endpoint contractions. A $[4]$ fibre contributes two dominant endpoints, whereas the two distinct A -edges in a $[1, 1]$ fibre contribute at most their one common endpoint. Equality forces those two edges to meet. Cyclically, every $x \in X$ therefore has unique dominant endpoints $a(x), b(x), c(x)$ in its three edge labels.

Let $\lambda_a : B \rightarrow C$ and $\mu_b : A \rightarrow C$ be the point permutations supplied by the first two endpoint families. For fixed (a, b) count

$$D_{ab} = \#\{x \in X : a(x) = a, b(x) = b\} + \#\{(e, f, g) \in Y : a \in e, b \in f\}.$$

Every X -entry is counted once and every Y -entry four times, so $\sum_{a,b} D_{ab} = n^2(n-1)$. An object counted by D_{ab} has a C -edge g containing both $\lambda_a(b)$ and $\mu_b(a)$. The map to g is injective, since (a, b, g) recovers f and e through the two bijective edge lifts. Two vertex stars of K_n intersect in $n-1$ edges when their centres agree and in one edge otherwise. Hence $D_{ab} \leq n-1$ for all a, b . The total attains the sum of these n^2 upper bounds, and $n-1 > 1$; therefore

$$\lambda_a(b) = \mu_b(a) \quad (a \in A, b \in B).$$

The cyclic count for (a, c) similarly gives $\nu_c(a) = \lambda_a^{-1}(c)$. Thus $L(a, b) = \lambda_a(b) = \mu_b(a)$ has permutational rows and columns and is Latin; all three families are compatible and sharply transitive.

Finally, a multiplicity-one entry with dominant corner (a, b, c) has $c \neq L(a, b)$: equality would make the corresponding rectangle an intercalate and force the second edge in a $[1, 1]$ fibre to equal the first. Each multiplicity-four entry supplies its four nonincidence corners. Indeed, both endpoint rows map its column edge bijectively onto its symbol edge, and column uniqueness makes the resulting rectangle an intercalate. Conversely, a multiplicity-one entry cannot reconstruct an intercalate: its second row endpoint would then also dominate the same symbol edge, so the two distinct support edges in the relevant $[1, 1]$ fibre would contain both row endpoints and hence coincide. The preceding star injection is a bijection, so the resulting corners exhaust exactly the $n^2(n-1)$ triples (a, b, c) with $c \neq L(a, b)$. The recovery map

$$(a, b, c) \mapsto (\{a, \mu_b^{-1}(c)\}, \{b, \lambda_a^{-1}(c)\}, \{L(a, b), c\})$$

returns each selected corner to its original three pair labels by the dominant edge-lift equations. The flag bijection of Theorem 16.1 and the preceding dichotomy therefore give one flag over each entry of X , four over each entry of Y , and none elsewhere. Hence T is exactly the flag tensor of L . The FFF statement now follows from the link criterion in Proposition 16.6. $\square$

The restriction $n \geq 3$ is sharp for faithful recovery of the witnessing point permutations and their automatic compatibility. At $n = 2$ every edge set is a singleton, so the two-subset action cannot distinguish the two point permutations. The sole tensor value four is nevertheless the flag tensor of either labelled order-2 Latin square; these form one main class.

Proposition 16.32 (Intrinsic recognition of endpoint edge lifts). *Let U, W be n -element sets and let $M : E(U) \rightarrow E(W)$ be a bijection. If $n = 3$ or $n \geq 5$, then M is the two-subset action of a unique point bijection $U \rightarrow W$ if and only if it preserves edge intersection. If $n = 4$, the intersection-preserving maps are the point-induced maps and their compositions*

with target-edge complementation $e \mapsto W \setminus e$; requiring vertex stars to map to vertex stars selects exactly the point-induced maps.

Consequently the point-lift axiom in Theorem 16.31 may be replaced, for $n = 3$ or $n \geq 5$, by bijectivity and preservation of intersection on pair labels. At $n = 4$ one adds preservation of the family of vertex stars.

Proof. In the line graph of a complete graph, cliques are pairwise-intersecting edge families. Such a family either has a common endpoint or has at most three members: after choosing $\{a, b\}$ and $\{a, c\}$, any member avoiding a must be $\{b, c\}$. Hence, for $n \geq 5$, the cliques of size $n - 1$ are precisely the vertex stars. An intersection-preserving edge bijection maps the source stars onto the target stars, defining a point bijection $p : U \rightarrow W$. Since $\{\{u, v\}\} = \text{Star}(u) \cap \text{Star}(v)$, it follows that $M(\{u, v\}) = \{p(u), p(v)\}$. Uniqueness is immediate. At $n = 3$, there are six point bijections and six edge bijections, and the two-subset action is faithful.

For $n = 4$, edge complementation preserves intersection but sends each vertex star to the complementary triangle, so it is not point-induced. The three complementary edge pairs are exactly the nonedges of $L(K_4)$. Every graph automorphism permutes these three pairs and may swap within them, yielding at most $3!2^3 = 48$ maps. The 24 point lifts and their 24 distinct complement compositions attain this bound. Exactly the first coset preserves the vertex stars. $\square$

The order-4 exception is also a sharp warning against replacing endpoint recognition by local completion rules alone. Complementing one pair-label coordinate of a genuine order-4 tensor preserves its fibre trichotomy and all coordinatewise edge-intersection relations, but exhaustive comparison with all 576 labelled order-4 Latin squares computes that the transformed tensor is not Latin-realizable.

Definition 16.33 (Cell-local cycle-incidence signature). For a Latin cell $x = (r, c, L(r, c))$, a view $V \in \{R, C, S\}$, and $2 \leq \ell \leq n$, let $a_\ell^V(x)$ count the other lines whose induced permutation with the distinguished V -line through x has an ℓ -cycle containing x . Put

$$A^V(x) = (a_2^V(x), \dots, a_n^V(x)).$$

The *cell-local cycle-incidence signature* is the multiset

$$\{(A^R(x), A^C(x), A^S(x)) : x \text{ a Latin cell}\},$$

canonically minimized over the six simultaneous permutations of the three view coordinates. This convention makes the signature invariant under isotopy and parastrophy. Pointwise $\sum_\ell a_\ell^V(x) = n - 1$, and the three values $a_2^R(x), a_2^C(x), a_2^S(x)$ all equal the number of intercalates through x .

Computational Theorem 16.34 (Order-8 rainbow-square classifier). *On the 230 order-8 FFF main-class representatives, the pair consisting of the signature in Definition 16.33 and*

$$w_6(L) = \text{Fix}((\alpha_R \alpha_C \alpha_S)^2)$$

takes 230 distinct values.

The cell-local signature alone has 226 values and four collision pairs. Their respective w_6 values are (352, 288), (448, 192), (448, 192), and (288, 352). The complete flag identities were audited with no error on all 9408 reduced order-6 squares and all 283657 order-8 representatives. A separate implementation recomputed w_6 on all 230 FFF representatives with zero mismatches. This is a finite census classifier, not an injectivity theorem for higher orders.

Computational Theorem 16.35 (Pair-label triangle classifier). *On the same 230 representatives, the pair consisting of the cell-local cycle-incidence signature from Definition 16.33 and $t(L)$ takes 230 distinct values.*

The scalar $t(L)$ alone takes 100 values and $(I(L), t(L))$ takes 188. The four cell-local-signature collision pairs [4, 22], [11, 114], [13, 19], and [117, 118] have respective t -values (2112, 1856), (2560, 1536), (2048, 1536), and (1792, 1920). The pair matrices were reconstructed both from flag codegrees and from the induced edge lifts, with no mismatch. As above, this is an exact classifier only on the frozen order-8 FFF census.

17 The order-10 computational exclusion

The strict local condition studied by Kobayashi and Nakamura requires every pair of factors in a 1-factorization of $K_{n,n}$ to form only 4-cycles; such a 4-semiregular factorization exists precisely in power-of-two orders [KN94]. In one Latin-square view, that condition forces every induced two-line cycle to have length 2. Our F condition is weaker: induced cycles may have any even length, and FFF requires the weaker condition simultaneously in all three coordinate views. The nearby strict classification historically suggested a stronger existence conjecture, now known to be false.

Equivalently, view a Latin square as a triangle decomposition of $K_{n,n,n}$. The links of the vertices in each part form a 1-factorization between the other two parts. A two-line permutation cycle of length ℓ corresponds to an alternating link-union cycle of length 2ℓ . Thus FFF asks that all pair-union components in all three coupled link factorizations have length divisible by four. Moreover, the total number of 4-cycles in each of the three link families is the same: in every view it is exactly the number of intercalates of the Latin square. This identity is a baseline three-view coupling, although it is not by itself an obstruction.

Remark 17.1 (Former power-of-two existence conjecture). The assertion that FFF Latin squares of order $n > 1$ exist exactly at powers of two is false. Computational Theorem 8.1 supplies an explicit FFF square of order 12.

The power-of-two positive direction follows from Corollary 3.3. Nonexistence at odd n follows from Proposition 3.1. The specific even non-power-of-two orders 6 and 10 are excluded by the computational theorems below. Orders 12, 14, and 36 have positive witnesses; the first undecided small even non-power-of-two order is 18.

Corollary 4.6 was the first structural reduction for order 10: every order-10 FFF candidate must be congruence-simple in every isotope. Thus central, binary-extension, and other congruence-imprimitive constructions cannot produce a counterexample. This reduction alone is structural rather than an existence decision; the complete palette computation below closes the case.

Computational Theorem 15.1 sharpens this again at the level of each line action: every normalized view must generate A_{10} or S_{10} , and Corollary 15.2 forces at least one S_{10} view. This is a separate necessary-condition reduction, not an assumption used to discard candidates in the two master computations below. Those computations cover their full normalized permutation families without a primitive-group filter.

There is also an exact cycle-type reduction. The seven possible cycle types of a fixed-point-free permutation of degree 10 with no odd cycle are

$$\mathcal{T} = \{10, 8 + 2, 6 + 4, 6 + 2 + 2, 4 + 4 + 2, 4 + 2 + 2 + 2, 2 + 2 + 2 + 2 + 2\}.$$

For one view, call the set of types occurring among its 45 relative two-line permutations its *exact palette*.

Computational Theorem 17.2 (Order-10 cycle-type palette exclusions). *Every view of a hypothetical order-10 FFF Latin square has an exact palette containing at least six members of $\mathcal{T}$. None of the 21 five-member subsets of $\mathcal{T}$ can occur as the exact palette of any view:*

$$\begin{aligned}
&\{10, 2^5, 4 + 2 + 2 + 2, 4 + 4 + 2, 6 + 2 + 2\}, \\
&\{10, 2^5, 4 + 2 + 2 + 2, 4 + 4 + 2, 6 + 4\}, \\
&\{10, 2^5, 4 + 2 + 2 + 2, 4 + 4 + 2, 8 + 2\}, \\
&\{10, 2^5, 4 + 2 + 2 + 2, 6 + 2 + 2, 6 + 4\}, \\
&\{10, 2^5, 4 + 2 + 2 + 2, 6 + 2 + 2, 8 + 2\}, \\
&\{10, 2^5, 4 + 2 + 2 + 2, 6 + 4, 8 + 2\}, \\
&\{10, 2^5, 4 + 4 + 2, 6 + 2 + 2, 8 + 2\}, \\
&\{10, 2^5, 4 + 4 + 2, 6 + 4, 8 + 2\}, \\
&\{10, 2^5, 6 + 2 + 2, 6 + 4, 8 + 2\}, \\
&\{10, 4 + 2 + 2 + 2, 4 + 4 + 2, 6 + 2 + 2, 6 + 4\}, \\
&\{10, 4 + 2 + 2 + 2, 4 + 4 + 2, 6 + 2 + 2, 8 + 2\}, \\
&\{10, 4 + 2 + 2 + 2, 4 + 4 + 2, 6 + 4, 8 + 2\}, \\
&\{10, 4 + 2 + 2 + 2, 6 + 2 + 2, 6 + 4, 8 + 2\}, \\
&\{10, 4 + 4 + 2, 6 + 2 + 2, 6 + 4, 8 + 2\}, \\
&\{10, 2^5, 4 + 4 + 2, 6 + 2 + 2, 6 + 4\}, \\
&\{2^5, 4 + 2 + 2 + 2, 4 + 4 + 2, 6 + 2 + 2, 6 + 4\}, \\
&\{2^5, 4 + 2 + 2 + 2, 4 + 4 + 2, 6 + 2 + 2, 8 + 2\}, \\
&\{2^5, 4 + 2 + 2 + 2, 4 + 4 + 2, 6 + 4, 8 + 2\}, \\
&\{2^5, 4 + 2 + 2 + 2, 6 + 2 + 2, 6 + 4, 8 + 2\}, \\
&\{2^5, 4 + 4 + 2, 6 + 2 + 2, 6 + 4, 8 + 2\}, \\
&\{4 + 2 + 2 + 2, 4 + 4 + 2, 6 + 2 + 2, 6 + 4, 8 + 2\}.
\end{aligned}$$

Here 2^5 abbreviates $2 + 2 + 2 + 2$.

The lower bound combines complete exclusions of every exact palette of size at most four. Each of the twenty-one additional exclusions uses a complete root-orbit cover, a fully audited compatibility graph, reduced and unreduced searches, and an independent clique check. Several compatibility graphs contain Latin completions: f18 and f19 each have 1920, f20 has 23040, including 19200 with the exact f20 row palette, f21 has 1920 with no exact f21 row palette, and f02 has 101760, including 99840 with the exact f02 row palette, while f07 has 1920 with no exact f07 row palette. The f03 graph has 768 cliques, split as 384 FFT and 384 FTF, with no exact f03 row palette; the f04 graph has no bucket-transversal 8-clique. The f05 graph has 8832 exact-f05 tables, split as 3840 FTT, 2496 FTF, and 2496 FFT. The f11 graph has 384 exact-f11 tables, all with pattern FTT. The f06 graph has 2880 candidate tables, none with the exact f06 row palette, split as 1152 FTT, 864 FTF, and 864 FFT. The f12 graph has 1920 candidate tables, of which 1152 have the exact f12 row palette; in aggregate all 1920 candidates split as 1152 FTT, 384 FTF, and 384 FFT. The f08 graph has 21120 candidate tables, none with the exact f08 row palette; all 21120 candidates split as 7680 FTT, 6720 FTF, and 6720 FFT. The f13 graph has 17280 candidate tables, of which 13632 have the exact f13 row palette; all 17280 candidates split as 9600 FTT, 3840 FTF, and 3840 FFT. Direct three-view validation, not pairwise compatibility alone, excludes FFF. The f14 graph has 21312 candidate tables, of which 18432 have the exact f14 row palette; all 21312 candidates split as 13056 FTT, 4128 FTF, and 4128 FFT, with zero FFF. The f09 graph has 23040 candidate tables, of which 21120 have the exact f09 row palette; all 23040 candidates split equally as 7680 FTT, 7680 FTF, and 7680 FFT, with zero FFF. The f10 graph has 55680 candidate tables, of which

34560 have the exact f10 row palette; all 55680 candidates split as 26880 FTT, 14400 FTF, and 14400 FFT, with zero FFF. Only f15 among the exact five-type palettes remained at that stage. The f15 graph has 32896 candidate tables, including 13504 with the exact f15 row palette; all 32896 candidates split as 6208 FTT, 13344 FTF, and 13344 FFT, with zero FFF. Thus every exact five-type palette is excluded.

The remaining palettes reduce to two disjoint normalized master families. If the involution type 2^5 occurs, one such line pair can be normalized and the master permits all seven types. This covers the six exact six-type palettes containing 2^5 and the exact seven-type palette. If 2^5 does not occur, the lower bound in Computational Theorem 17.2 forces the unique six-type palette

$$\{10, 4 + 2 + 2 + 2, 4 + 4 + 2, 6 + 2 + 2, 6 + 4, 8 + 2\},$$

and an occurring $4 + 2 + 2 + 2$ pair is normalized instead. Simultaneous conjugation preserves all three F conditions, and exact residual-group orbit covers reduce the two searches from 18300 and 23760 root tasks to 386 and 506 representatives.

Computational Theorem 17.3 (Nonexistence at order 10). *There is no FFF Latin square of order 10.*

For the noninvolution master, a complete semantic audit checks all 14855277528 unordered pairs among 172368 candidates and reconstructs 1650487056 graph edges with zero errors. Two complete decompositions of the 386 orbit representatives agree exactly at 2514104197 search nodes and 1769153203 cross-view prunes, with no FFF completion. For the involution/all-seven master, the corresponding figures are 190080 candidates, 18065108160 audited pairs, 2050329600 edges, 506 orbit representatives, 5001728282 nodes, and 3586770865 prunes, again with no FFF completion. The orbit identities cover every root task, and both package validators and fail-closed mutation audits pass. The two decompositions in each package use the same frozen search binary and control slicing and merge determinism; independent model checks are supplied by separate inventory implementations, complete graph auditors, orbit reconstruction, and positive order-8 controls.

By the lower bound, every hypothetical FFF view lies in exactly one of these two masters. Their complete exclusion proves the theorem. This establishes nonexistence at order 10 independently of the false global assertion in Remark 17.1.

Completeness bridge. The two-master reduction fixes the identity row and a canonical second row ρ : an involution of type 2^5 , or a permutation of type $4 + 2 + 2 + 2$. Every further row is a permutation of ten symbols. The candidate list enumerates all such permutations, retaining exactly those whose relative row types lie in the master palette and whose three-row partial table has no already closed odd column or symbol cycle. Candidate vertices are adjacent exactly when their relative row type is allowed and their joint four-row partial table passes the same cross-view check. The eight parts are the possible first-column images $2, \dots, 9$. A completion therefore selects one candidate from each part, with pairwise graph compatibility.

Write $q_r(c) = L(r, c)$ for the permutation of row r . For columns $c \neq d$, each chosen row contributes the partial map on symbols $L(r, c) \mapsto L(r, d)$. For symbols $u \neq v$, it contributes the partial map on columns $q_r^{-1}(u) \mapsto q_r^{-1}(v)$. Latin compatibility makes every such partial map injective. A closed odd cycle already present in one map cannot be broken by adding rows: all its incoming and outgoing edges are fixed. Rejecting it is consequently sound. At

completion, the column maps are conjugate to the usual two-column permutations and the symbol maps are the usual two-symbol permutations, so these tests are exact.

The search branches over every remaining candidate in a selected part and intersects the remaining parts with its graph neighbours. An empty part or a closed odd cross-view cycle cannot contain a completion. Induction on the number of unfilled parts shows that all other transversal choices are visited. At depth eight, the two fixed rows and eight selected rows form a Latin square, with every row, column and symbol test enforced. Finally, the recorded residual conjugation actions preserve these tests and the chosen root part; their complete orbit covers account for every initial task. This establishes the mathematical reduction from full FFF to the recorded exhaustive searches. Executable correctness and completed coverage remain computational evidence obligations, not consequences of the candidate counts alone.

18 An exact position–value coloring formulation

This section gives an equivalent finite constraint model, not a decision of any new order. In one view let each line a be a bijection $f_a : P \rightarrow Q$. For distinct a, b the permutation $\pi = f_b^{-1}f_a$ is fixed-point-free. A permutation has only even cycles if and only if its points admit a binary coloring that toggles along every edge: alternating a coloring around a cycle is consistent exactly when its length is even.

Theorem 18.1 (Position–value channel). *Introduce colors $e_i, h_v \in \{0, 1\}$ for positions $i \in P$ and values $v \in Q$. The equations*

$$e_i = h_{f_a(i)}, \quad e_i = 1 - h_{f_b(i)} \quad (i \in P) \quad (14)$$

are satisfiable if and only if π has only even cycles. Every solution has exactly $n/2$ ones in each color family. One may add $e_{i_0} = 0$ at any chosen position without changing satisfiability.

Proof. For $j = \pi(i)$, equality $f_a(i) = f_b(j)$ makes (14) imply $e_j = 1 - e_i$. Conversely, from a toggling coloring set $h_v = e_{f_a^{-1}(v)}$. The first equation is immediate. The second follows because $f_a^{-1}f_b = \pi^{-1}$ and toggling also holds on inverse edges. Since the lines are bijections, summing the two equations gives $\sum_i e_i = \sum_v h_v = n - \sum_v h_v = n/2$. Complementing all e and h simultaneously preserves the equations and permits the single gauge condition. $\square$

For a Latin cell variable X_{rcs} asserting $L(r, c) = s$, the antecedents A, B for $f_a(i) = v$ and $f_b(i) = v$ are:

View	Position	Value	A	B
row	column i	symbol v	X_{aiv}	X_{biv}
column	row i	symbol v	X_{iav}	X_{ibv}
symbol	column i	row v	X_{via}	X_{vib}

Writing $e = e_i$ and $h = h_v$, four clauses implement the equations:

$$(\neg A \vee \neg e \vee h), \quad (\neg A \vee e \vee \neg h), \quad (\neg B \vee e \vee h), \quad (\neg B \vee \neg e \vee \neg h).$$

With Latin exactly-one constraints, imposing these clauses independently for every line pair in every view is equivalent to FFF. The reverse implication extends each FFF table

by its pairwise alternating colorings; no coupling of arbitrary gauges is required. Isotopy invariance permits reduced tables in an existence search.

Each line pair uses $2n$ colors, $4n^2$ conditional clauses, and one optional gauge unit. Balance follows from the equations and requires no cardinality auxiliaries. Frozen positive controls at orders 8, 12, and 14 and checked negative order-6 controls validate the implementation. In one fixed order-18 row context, the recorded model has 22356 variables and 745065 clauses, compared with 270216 and 3190617 in the prior physical-arc representation. Both bounded 60-second solver probes timed out. This is a size reduction, not evidence of faster solution or of nonexistence at order 18.

19 The precisely bounded order-18 frontier

None of the results in this section decides unrestricted order-18 FFF existence. They distinguish a lossless normalization from exclusions of particular construction families and local contexts.

Proposition 19.1 (A free-column same-sign normalization). *Every row-F Latin square of even order $n \geq 4$ is isotopic to a reduced square whose second row is a canonical even-cycle permutation of positive sign, with zero in a longest cycle. At order 18 this gives a complete family of 14 row-1 cases.*

Proof. Among the row bijections $\rho_r : C \rightarrow S$, two, say ρ_a, ρ_b , have the same sign. Their relative permutation $\phi = \rho_a^{-1}\rho_b$ has positive sign, no fixed point, and only even cycles. Choose a column c_0 on a longest cycle, and a column labelling κ taking its successive points to $0, 1, \dots$. Label the other cycles on consecutive intervals in decreasing order of length. Put

$$\sigma = \kappa\rho_a^{-1}, \quad \alpha(r) = \sigma(L(r, c_0)), \quad L'(\alpha(r), \kappa(c)) = \sigma(L(r, c)).$$

All maps are bijections. The first row and column of L' are identity lines, $\alpha(a) = 0$, $\alpha(b) = 1$, and row 1 is $\kappa\phi\kappa^{-1}$. This isotopy preserves all three pattern bits. Conversely, a square in any retained case is an unrestricted square. The even partitions of 18 are twice the partitions of 9; exactly 14 of the 30 have an even number of parts and therefore positive permutation sign. $\square$

The first column is chosen freely before reduction. A model with previously pinned cells or a distinguished column cannot use this normalization without transporting those anchors. None of the 14 complete cases is excluded by the bounded model-size probes.

Completed construction exclusions. The following are exact scoped results in the frozen research archive; they are not substitutions for the 14 unrestricted cases. One dependency needed for arbitrary transversals deserves an explicit statement. It is not the standard-prolongation criterion of Section 10.

Proposition 19.2 (The cyclic-base dependency, C225). *No standard one-point prolongation of the addition table of $\mathbb{F}_{17}$ along any transversal is FFF.*

Proof. Write a transversal as $(r, f(r), r + f(r))$, so both f and $r + f(r)$ are permutations. For two old rows r, s , put $t = (f(s) - f(r))/(r - s)$ in $\mathbb{F}_{17}$, using its representative in $\{2, \dots, 16\}$. Cutting the two selected arrows in the original 17-cycle and inserting the new point gives cycles of lengths t and $18 - t$. Thus row-F forces t even for every pair.

All secant directions of the graph of f lie among the eight values $-\{2, 4, \dots, 16\}$. The Rédei–Megyesi direction theorem requires at least ten directions for a noncollinear set of 17 points [Som22, Theorem 1.1]. Consequently $f(r) = ar + b$ with $a \notin \{0, -1\}$.

Translations remove b . The old-row and old-column even-cycle tests require the least positive residues of $-a$ and $-a^{-1}$ to be even; they leave only $a = 1, 5, 7$. In the symbol view the base operation is $r = s - c$. Replacing c by $-c$ turns it into addition and gives transversal coefficient $-a/(a + 1)$. The same old-line cycle test therefore requires the residue of $a/(a + 1)$ to be even. For $a = 1, 5, 7$ those residues are respectively 9, 15, 3, all odd. Already these old-line pairs exclude FFF. No claim about an arbitrary order-18 square is used. $\square$

Translation-equivariant one-point family (C221). For complete maps $f : \mathbb{F}_{17} \rightarrow \mathbb{F}_{17}$ with $f(0) = 0$, the standard prolongation of $x + f(y - x)$ contains no FFF square. A complete parastrophe/scaling reduction has three nonaffine cases. The final case-2 conditional-lex CNF is UNSAT with an independently checked DRAT trace, joining the previously checked cases 3 and 4 and the finite affine exclusion. Translation equivariance is an additional hypothesis.

Arbitrary transversals of odd-map bases (C235). Among the 12513 odd complete mappings, 12450 bases fail a necessary old-line-pair cycle screen, 15 affine survivors reduce to Proposition 19.2, and all 48 nonaffine survivors have individually checked UNSAT certificates for the necessary transversal model. This covers every transversal for these odd-map bases, not nonodd maps or arbitrary order-17 bases. Indeed, for an affine complete map $f(t) = at$, relabelling rows by $(1 - a)x$ and columns by ay turns $x + f(y - x)$ into addition. These relabellings extend over the new point and transport every transversal prolongation. The C225 proof and its finite controls are retained explicitly in the companion snapshot.

Fixed-remainder contraction of represented FFF20 classes (C251). For the 514 represented classes of Computational Theorem 11.3, delete an order-two Latin sub-square and refill only holes left by its deleted symbols, keeping all other surviving triples fixed. This produces no FFF18 square. The finite screen covers all 1024 trade masks and the two quadratic controls. All 9728 base completions are TTT; proper-mask completion spaces coincide with a base case, and the quadratic controls have no Latin completion. The rule is invariant under isotopy and parastrophy because it is stated on the three sorts and their retained triples. It excludes neither other FFF20 classes nor repairs that change retained entries.

A genuinely three-view local conflict. One frozen reduced ten-row prefix in case 14 has the five further conditions

$$L(11, 6) = 2, \quad L(11, 9) = 7, \quad L(11, 10) = 6, \quad L(11, 11) = 13, \quad L(11, 16) = 3,$$

with zero-based indexing. At the thirteen-row stage, each of the seven proper subsets of the three view constraints has a separately validated partial Latin solution with no closed odd cycle in its active views. All three views together are UNSAT with a verified DRAT trace. The seven solutions need not be the same table. This is an exact joint obstruction for that prefix and those five cells, not a whole-case exclusion or a universal claim that two views never suffice.

The full row-fibred and affine-orbit constructions do not resolve these open searches. In particular, the explicit FFF98 control already rules out any universal obstruction based

solely on $n \equiv 2 \pmod{8}$. More computation on a selected source is not coverage of all Latin squares; an eventual negative decision must close an exhaustive, justified search space.

20 Current search boundaries and research priorities

Order 18 remains undecided. A fixed input, permitted transformation, and complete finite search space must accompany every local negative result. The following developments are not a cumulative percentage of an unrestricted search.

Proposition 20.1 (Intercalates forced by paired blocks). *If a normalized line family in any view of an order- $2m$ Latin square preserves a partition into m pairs, then its intercalate count is $I = m^2 + 2K$ for some nonnegative integer K .*

Proof. Normalize a row view by its base row, identifying columns and symbols. Every row has the form $L(g, (x, t)) = (\phi_g(x), t + f_g(x))$ on $[m] \times \mathbb{F}_2$. Let J flip every symbol pair. The two column maps of block x satisfy $C_{x,1} = JC_{x,0}$. Thus their relative permutation is conjugate to J and has m transpositions. The m within-block column pairs contribute exactly m^2 intercalates. Between two distinct blocks, the relative maps for column bits $(0,0), (1,1)$ are identical, as are those for $(0,1), (1,0)$. Their two-cycle contributions are therefore even. Parastrophy gives the other views without changing the intercalate count. $\square$

At order 18 a paired-block view therefore requires at least 81 intercalates and an odd total. This is a necessary condition, not a proof that a hypothetical FFF18 square cannot have such a view.

Local distance certificates. One frozen FFT18 source has a certified same-label lower bound of 35 changed cells for even a symbol-F Latin target. A different frozen FTF18 source has a bound of 37 cells for even a column-F target. The supporting reductions use physical odd-cycle supports and Latin replacement constraints; their retained UNSAT records have independently checked DRAT traces. This paper update does not rerun those historical proof checkers. Neither bound constructs a target at the stated distance or excludes a more distant target. A small number of failing pairs is not evidence of a small cell-edit distance to FFF.

Distinct normalizations with different coverage proofs. The 14-case longest-zero-cycle normalization in the full report is one complete unrestricted split. A later dual-sign argument gives an alternative 33-case split with positive row-1 and column-1 signs. These are alternatives: the extra column-sign constraint must not be added to the 14 cases without another coverage proof. A further three-positive-sign shortcut is false even on a frozen FFF8 control. The paired baseline/overlay order-18 tests time out; they establish neither a case exclusion nor a runtime improvement.

Other finite-family exclusions. The fixed-remainder contraction audit now covers the entire frozen 22-row-pair trade-mask family and the two quadratic controls, with no FFF18 output. A separate theorem excludes FFF completions of principal fixed-remainder two-point contractions of Steiner loops of order $n \geq 8$ divisible by four: their Latin completions are symmetric of order $n - 2 \equiv 2 \pmod{4}$ and greater than two. Neither statement covers arbitrary parents, nonprincipal Steiner contractions, or repairs changing retained cells.

The new unique-intercalate family has its different, direct noncontraction obstruction in Corollary 13.2.

Free-order-three incidence conditions and affine quotient no-goods constrain specified symmetry subclasses only. The recent Mendelsohn, four-transversal and repair pilots do not decide them. These model derivations can be useful without being a breakthrough on unrestricted existence. A shorter equivalent CNF is not, by itself, evidence of a faster solver.

How further searches will be evaluated. Before extending a computation, specify its mathematical hypothesis, covered space, baseline, resource limit and decisive outcome. Separate general theorems, complete finite exclusions and heuristic observations. Use positive and negative physical controls, compare genuinely different methods, and require evidence of benefit before expanding a sequence of similar timeout pilots. Literature comparison and external expert review remain independent tasks. CPU utilization, file counts and claim identifiers are not measures of distance to an order-18 solution.

21 Computational reproducibility

The repository accompanying this manuscript separates proofs, frozen data, programs, commands, logs, and output manifests. The principal packages are:

For a compact command-to-output map, the source archive includes `manuscript/REPRODUCIBILITY_QUICKSTART.md`. It gives one exact rerun command and one authoritative output hash for each computational theorem. The command

```
python3 repro_runs/2026-08-10_paper_symbol_sign_audit/scripts/audit_manuscript_computational_claims.py
```

cross-checks the thirteen computational theorem environments in the historical September 16 manuscript against its frozen outputs and must report `overall_pass=true`. It predates the explicit positive non-power-of-two witnesses below and does not audit their new theorem environment.

This edition's additional evidence

The dated source package `manuscript/candidates/2026-09-28_research_review/` contains build commands, bounded auditors, selected frozen data and proof notes, and a theorem-to-evidence map. Its intake stops at C271. The public copy is a curated selection; its own manifests identify the bytes actually shipped. Earlier editions and their evidence are retained as history, not silently resealed.

The preceding review's 1026 direct binary-rank certificates and larger-field scope control remain included. Its auditor reconstructs those fixed tables, checks all eight order-8 patterns and the stated order-12, 14, 36, 48, 98, and 160 controls. This update separately reconstructs 1703 order-20 representatives, checks each table by independent permutation and matching-component methods, compares exact spectra, and constructs and independently verifies a 58-square binary minor for every table. The new class-growth bound therefore does not inherit C157. Table hashes and minor cell indices are retained, and malformed certificates are rejected by regression tests.

The update also checks every physical line pair of the two order-12 block-substitution examples and all 253 affine corner-trade parameters over primes through 43. Its 216 generic cases test both the intercalate formula and the empty contraction-hole domains. Compact

CRT-return checks use separate physical probes for the two line-label coefficients; they do not materialize the enormous example table.

The source-specific order-18 distance certificates, larger trade-mask enumerations and earlier solver results are retained with their original scope. A fresh small representative audit is not a rerun of every historical enumeration. Neither a solver timeout nor a hash agreement is a proof of a larger nonexistence statement.

`ORDER10_REPRODUCIBILITY_GATE.md` records the unresolved durable external route for the full C157 computation. The paper update does not rerun its master searches or establish public reconstruction of their large graph payloads. This remains a separate reproducibility gate.

The two PDF variants have the same evidence boundary and overlapping content; they are not independent studies. Internal AI proof review, independent implementations and machine proof checking are recorded separately from external human refereeing. SHA-256 certifies bytes, not correctness, completeness, novelty, author responsibility or a rights grant. Literature priority and expert review remain open.

Small orders `repro_runs/2026-04-26_small_order_witness_reruns/`. The exact-cover generator freshly enumerates the 1, 4, and 9408 reduced squares of orders 2, 4, and 6. The generation validator confirms Latin, reduced, and uniqueness properties and full scan coverage.

Order-8 census `repro_runs/2026-04-26_order8_core/`. The full scan completed in 9.67 wall-clock seconds in the recorded environment. The frozen 5,140,393-byte input has SHA-256 `e5980ff5bc4ac46cbbefd1f1333258d0905724c42a976c4afe61ea501a4ccf87`. The theorem-relevant output is `results/order8_recheck_fullscan_results.json`, with SHA-256 `10f2f43c8dcaa8ec93f83abf6b768efa47f909a9c8623d4bc648e6da097ccd39`.

Group-isotopy closure `repro_runs/2026-04-26_order8_closure_reconstruction/`. It reconstructs the basis-family closure scan and matches the previously delivered result objects, including the 5/225 split.

Direct products `repro_runs/2026-04-26_direct_product_fff/`. In addition to the proof note, the validation checks product permutations and least-common-multiple cycle lengths in all views, verifies the OR formula on selected products, and checks the group-isotopy equivalence on group/group, group/non-group, and non-group/non-group examples.

Explicit non-power-of-two witnesses `repro_runs/2026-09-23_fff_nonpower_witness_intake/`. A standard-library validator reads five frozen tables from the supplied source archive, checks Latin and reduced status, and traverses all 198, 273, or 1890 two-line permutations in the three views at orders 12, 14, or 36, respectively. Every input and output is SHA-256-bound. The reduced table in Table 1 is exactly `data/order12_first.json` (SHA-256 `8b71d6ecaa6ab86a806a0660529b404d8f3667db2b58b82bf5c72fed010fa416`). The source ZIP has SHA-256 `103956a071d72701e4641334bb57e204debc0d20d0efff7c847b4b0fd4f5d892`; its internal byte manifest is verified, but this does not establish external mathematical review.

Three-point finite-field lift controls `repro_runs/2026-09-23_fff_three_point_lift_audit/`. The companion checks all relevant coset moments for $m = 5, 7, 9, 15, 21$

and independently scans all three views at orders 56, 80, and 240. Its standard-library script, commands, and SHA-256 manifests bind the controls. Theorem 8.3 rests on its written proof, not on extrapolation from these finite checks.

Subsquare and congruence obstructions `repro_runs/2026-08-09_fff_subsquare_congruence_obstructions/`. The validator scans subsquares of all 230 frozen FFF representatives, exhausts all 512 binary twists over the cyclic quotient of order 3, and checks all 9408 reduced order-6 squares for congruences. Exactly 660 order-6 representatives are congruence-imprimitive, and all 660 have pattern TTT, in agreement with Theorem 4.5. The same package validates the explicit twisted order-8 loop as Latin, FFF, nonassociative, and not closed at the identity basis.

Cycle-parity obstruction `repro_runs/2026-08-09_fff_cycle_parity_obstruction/`. It scans all 9408 reduced order-6 squares and all 283657 frozen order-8 census representatives, checks the degree-10 cycle-partition classification, and independently validates a frozen low-Q-defect order-10 table. The heuristic origin of that table is not treated as exhaustive evidence.

Primitive degree-10 reduction `repro_runs/2026-08-09_fff_primitive_group_reduction/`. The package constructs the seven small primitive degree-10 actions and checks their exact orders, closure, transitivity, and primitivity. It computes their maximum F-cliques including the identity as 1, 2, 1, 2, 2, 2, 4. The five nontrivial upper bounds are CaDiCaL-UNSAT and all five DRAT traces are independently verified. The two remaining entries have no eligible nonidentity element. Completeness of the nine-group primitive classification remains a cited external mathematical dependency.

Primitive degree-10 catalogue freeze `repro_runs/2026-08-27_fff_primitive_group_classification_freeze/`. A GAP 4.16.0/PrimGrp 4.0.3 query records all nine catalogue entries, their orders and structure descriptions, together with source-archive hashes and dependency commits. This freezes the invoked catalogue version; correctness and completeness of PrimGrp remain the explicit external classification assumption.

Order-10 cycle-type palettes The lower bound of five types uses the complete palette-exclusion chain through `repro_runs/2026-08-14_fff_four_type_p20_search/`. The 21 exact five-type palettes and all 105 root choices are independently audited in `repro_runs/2026-08-14_fff_five_type_palette_inventory/`. The twenty-one excluded palettes have separate packages `repro_runs/2026-08-14_fff_five_type_f01_search/`, `repro_runs/2026-08-14_fff_five_type_f16_search/`, `repro_runs/2026-08-14_fff_five_type_f17_search/`, `repro_runs/2026-08-14_fff_five_type_f18_search/`, `repro_runs/2026-08-14_fff_five_type_f19_search/`, `repro_runs/2026-08-20_fff_five_type_f20_search/`, `repro_runs/2026-08-20_fff_five_type_f21_search/`, `repro_runs/2026-08-21_fff_five_type_f02_search/`, `repro_runs/2026-08-21_fff_five_type_f07_search/`, `repro_runs/2026-08-21_fff_five_type_f04_search/`, `repro_runs/2026-08-21_fff_five_type_f03_search/`, `repro_runs/2026-08-21_fff_five_type_f05_search/`, `repro_runs/2026-08-21_fff_five_type_f11_search/`, `repro_runs/2026-08-21_fff_five_type_f06_search/`, `repro_runs/2026-08-21_fff_five_type_f12_search/`, `repro_runs/2026-08-22_fff_five_type_f08_search/`, `repro_runs/2026-08-22_fff_five_type_f13_search/`, `repro_runs/2026-08-23_fff_five_type_f14_search/`, `repro_runs/2026-08-24_fff_five_type_f15_search/`.

e_f09_search/, repro_runs/2026-08-24_fff_five_type_f10_search/, and repro_runs/2026-08-25_fff_five_type_f15_search/. Every package binds its root-orbit cover, graph audit, reduced and unreduced searches, and independent clique census to SHA-256 manifests. Large graph and checkpoint files remain outside Git and are hash-bound in the tracked reports. These computations exclude all exact five-type palettes; all exact six- and seven-type palettes are handled by the following master packages.

Order-10 master exclusions repro_runs/2026-08-26_fff_six_type_master_inventory/ independently inventories the two disjoint normalized families in C++ and Python. The dedicated packages repro_runs/2026-08-26_fff_six_type_noninvolution_master_search/ and repro_runs/2026-08-26_fff_six_type_involution_master_search/ bind the dense graph bytes, complete semantic graph audits, residual-group orbit covers, two differently sliced full searches of each representative file, exact merge comparisons, positive controls, and fail-closed validators. The noninvolution package checks 14855277528 candidate pairs and searches 386 representatives; the involution package checks 18065108160 pairs and searches 506 representatives. Both are complete negative. Their central validators pass 15/15 checks and their mutation audits 13/13 checks. The external graphs and checkpoints are retained outside Git and bound by SHA-256 manifests.

Local Hamming-neighborhood exclusion repro_runs/2026-08-09_fff_hamming_neighborhood_obstruction/. The exact V9 FFF encoding is augmented by a sequential counter for cell Hamming distance from the best tracked 131/135 near-model. The radius-20 formula is CaDiCaL-UNSAT and its DRAT trace is independently verified. This excludes only one fixed-coordinate neighborhood; it is not an order-10 nonexistence result.

Residual column-1 symmetry repro_runs/2026-08-09_fff_col1_lexleader_symmetry/. After a canonical second row is fixed, the remaining simultaneous conjugations act on column 1. Lex-leader constraints reduce the previously enumerated 27634 row/column orbit formulas to twelve equisatisfiable fixed-row formulas. An exhaustive comparison checks all 132300 compatible columns and recovers exactly the archived orbit representatives. All four order-6 parent formulas are UNSAT with independently verified DRAT traces, and a fixed order-8 FFF table remains SAT and validates independently. Two bounded order-10 pilots time out, so no performance or existence claim is drawn from this reduction.

Conditional stabilizer refinement The exhaustive S_4 truth audit, four checked order-6 DRAT gates, and bounded order-10 comparison are in repro_runs/2026-08-09_fff_conditional_row2_lexleader/.

Cell-local incidence and component overlaps repro_runs/2026-08-09_fff_cell_local_incidence_census/. The package checks the cell-local cycle-incidence identities and the line-pair sign cut on all 9408 reduced order-6 squares, all 283657 frozen order-8 representatives, and the tracked 135-table order-10 corpus. It then evaluates cycle-component support-overlap invariants on all 230 order-8 FFF representatives. The weighted structural 1-WL fingerprint has 229 values; this is an exact finite invariant, not a complete graph-isomorphism or order-10 result.

Three-coloured flag surface repro_runs/2026-08-09_fff_flag_surface_audit/. The package checks all flag-edge, link, Euler, Gram, and dual-word identities on

all 9408 reduced order-6 squares, all 283657 frozen order-8 representatives, and the tracked 135-table order-10 partial corpus. The order-8 census uses eight independent workers and reports zero identity errors. A second implementation independently recomputes the rainbow-square trace on all 230 FFF representatives with zero mismatches. Together with the cell-local cycle-incidence signature, this trace gives 230 distinct values on the frozen FFF census; no higher-order completeness is inferred.

Exact flag-realizability projection `repro_runs/2026-08-09_fff_flag_realizability_projection/`. The package checks the exact atom Gram rank controls, integral Schur formula, forced kernels, one- and two-atom PSD consequences, mixed-word topology, and FFF Euler/sign-cut consequences. It covers all four reduced order-4 squares, all 9408 reduced order-6 squares, all 230 frozen order-8 FFF representatives, and 135 tracked order-10 partial tables: 37871017 atom pairs give zero errors. The associated size audit stops the naive atom-support SAT architecture. A separate exact constructor gives the decoupled 900-flag aggregate consistency witness following Corollary 16.4. It lacks compatible line-pair/flag-incidence labels as well as common cell-coordinate support. The witness is not a Latin square and the package is not an order-10 decision.

Global common modes and exact Schur nullity `repro_runs/2026-08-11_fff_global_common_mode/`. The package computes the global atom/cell intersection independently from the joint incidence matrix and the stacked component-gradient system over two large prime fields. Integer witnesses certify every nonconstant rational direction. It also checks the endpoint-contraction, line-balance, trade-space centering, and atom-only kernel identities on all stored basis witnesses and verifies the predicted atom-only nullity on every audited table. The 135 order-10 inputs are a tracked diagnostic corpus, not an exhaustive census.

Labelled line-pair incidence `repro_runs/2026-08-09_fff_labelled_flag_incidence/`. The package reconstructs 1918172 flag-label triples on all four reduced order-4 squares, all 9408 reduced order-6 squares, all 230 frozen order-8 FFF representatives, and 135 tracked order-10 partial tables. It checks 7711917 pair-matrix entries against an independent edge-lift construction, 921954 two-factor profiles, and selected direct Gram ranks and traces, with zero errors. It also computes the exact 230-class pair-triangle classifier in Computational Theorem 16.35. Neither the tracked order-10 sample nor the pair-label conditions constitute an order-10 decision.

Common-mode occupied-cell exchange lattice `repro_runs/2026-08-11_fff_common_mode_trade_lattice/`. The package enumerates all balanced $3 + 3$ and $4 + 4$ exchanges on all 9408 reduced order-6 Latin squares. SymPy computes exact Smith/Hermite forms in saturated kernel coordinates; an independent Python-FLINT implementation recomputes ambient normal forms and three modular ranks. A separate saturation audit removes dependence on the stored global-common-mode certificate basis and verifies the 180 index-two common-mode quotients. All 180 stored $6 + 6$ completions are checked independently. This is an order-6 census, not an order-10 decision.

Intrinsic occupied-cell exchange parity `repro_runs/2026-08-11_fff_intrinsic_exchange_parity/`. The package canonically partitions all 9408 reduced order-6 squares into 12 main classes and audits the intrinsic binary covector on all 3960 rank-19 tables using eight workers. The three rank-19 main classes have distinct minimum-weight/count signatures $7/12$, $9/80$, and $9/160$. A separate implementation rebuilds

the balanced $4 + 4$ supports and binary elimination on each class representative, checks the complete source partition, and reports zero errors. The 180 common-mode exceptions form exactly the final class. This is an order-6 classification, not an order-10 decision.

Exchange-parity controls `repro_runs/2026-08-11_fff_exchange_parity_controls/`. The primary implementation enumerates every balanced $4 + 4$ occupied-cell exchange on all 230 frozen order-8 FFF representatives and all 135 tracked order-10 diagnostic tables. It uses eight workers for the order-8 jobs and two workers for the measured memory-heavy order-10 jobs. A separate packed-profile implementation recomputes all 230 order-8 records and four order-10 pattern representatives with independent binary elimination and zero discrepancies. The order-8 quotient distribution is $q_4 = 0 : 39, 1 : 147, 2 : 35, 3 : 9$; source line 170 is stored as an explicit FFF $q_4 = 0$ witness. The order-10 records are diagnostic only.

Binary line kernel and half-order subsquares `repro_runs/2026-08-11_fff_binary_homotopy_subsquares/`. With eight independent table workers, the audit constructs every nonzero binary line-kernel quotient class and its four predicted half-order subsquares, then independently enumerates all half-order subsquares by row/column subsets. The two sets agree exactly on all 9778 tested tables: the complete reduced orders 2, 4, and 6, all 230 frozen order-8 FFF representatives, and the 135 tracked order-10 diagnostics. A separate semantic verifier reproduces the earlier order-8 subsquare distribution and reports zero errors. The theorem is proved independently; the order-10 corpus remains partial and supplies no existence decision.

Prime-field line quotients `repro_runs/2026-08-11_fff_prime_field_line_quotients/`. The eight-worker audit computes ranks over five prime fields on 9778 tables and checks all 1160 nonuniversal quotient classes for balanced fibres, occupied-cell equations, and the predicted normalized-view block transports. A separately implemented sparse elimination rebuilds all 48890 modular ranks. Both implementations report zero errors. All 135 tracked order-10 diagnostics are already saturated; this is a control, not an existence or nonexistence result.

Full pair-label tensor `repro_runs/2026-08-09_fff_pair_label_tensor/`. The audit checks 7711917 complete two-label fibers against the exact $0/[1, 1]/[4]$ completion law and compares 460977 pair-label vertex links with independently reconstructed two-line cycle profiles. A further 2877258 endpoint-contraction fibers reconstruct 1100916 named row, column, and symbol line entries exactly. It covers all four reduced order-4 squares, all 9408 reduced order-6 squares, all 230 frozen order-8 FFF representatives, and the tracked 135-table order-10 partial corpus, with zero errors. The final corpus is diagnostic, not exhaustive.

Endpoint-contraction characterization `repro_runs/2026-08-09_fff_endpoint_contraction_gates/`. The package encodes the abstract fibre and endpoint axioms without Latin-cell variables. Free orders 3, 4, and 5 decode to a common Latin square and reproduce their tensor exactly. Canonical negations of sharp transitivity, row/column compatibility, and row/symbol compatibility are UNSAT through order 6; their DRAT traces are independently checked. These finite checks audit Theorem 16.31; the theorem itself is proved combinatorially, and the dense small-order formulas are not an order-10 search architecture.

Pair-tensor Smith obstruction `repro_runs/2026-08-10_fff_pair_tensor_smith_obstruction/`. The package independently computes every active tensor-slice rank over $\mathbb{F}_2$ and $\mathbb{F}_{1000003}$ on 9777 frozen tables. All 460977 slices satisfy the Smith rank-gap formula and reproduce the direct three-view failure pattern, with zero errors. The global binary flattening ranks are diagnostic only, and the tracked order-10 sample is not exhaustive.

Triple endpoint contraction `repro_runs/2026-08-11_fff_triple_endpoint_parity/`. The package constructs D independently from the aggregated pair-label tensor of Proposition 16.6 and from a direct ordered-flag enumeration. Across 9777 frozen tables it checks 1100916 endpoint lines, 366972 Latin-cell values, and 366972 complementary masses, with zero errors. The binary support and mode-rank distributions of the tracked order-10 partial corpus are diagnostic only.

Sparse endpoint-record overlay `repro_runs/2026-08-09_fff_sparse_endpoint_record_overlay/`. The package extracts the complete local fibre records from all three conjugate Latin tables and verifies their common pair-label tensor. The order-6 negative and fixed order-8 negative gates have independently checked DRAT traces; two fixed order-8 positive gates are decoded and validated in all three orientations, including a cyclic example that exercises the multiplicity-one completion branch. Its exact size audit shows that this definitional layer is larger than the direct line-pair colouring encoding, so the package contains no order-10 solver run.

Intrinsic endpoint edge-lift recognition `repro_runs/2026-08-10_fff_intrinsic_edge_lift_recognition/`. The package enumerates every automorphism of $L(K_n)$ through order 8, checks 178914 endpoint maps reconstructed directly from pair-label-tensor contractions, and verifies the sharp order-4 complement exception against all 576 labelled order-4 Latin squares. The proof of Proposition 16.32 is combinatorial. A transparent size comparison shows that the naive intrinsic order-10 one-hot translation has many more clauses than the explicit point-lift layer, so no solver run is started from it.

Odd-subset collision characterization `repro_runs/2026-08-10_fff_odd_subset_collision_encoding/`. The corpus audit checks Theorem 14.2 on 9773 Latin squares, 29319 views, and 2998860 induced subset images, with no equivalence mismatch. The plain and redundant-colour order-6 formulas are CaDiCaL-UNSAT and both DRAT traces are independently verified. A fixed order-8 FFF table is SAT in both formulations and independently validates as Latin, reduced, and FFF; a fixed non-FFF table is UNSAT with an independently verified DRAT trace. In one fixed order-10 row/column split, the plain and redundant-colour formulas both time out under CaDiCaL and Kissat at 300 seconds. These four timeouts are engineering measurements, not mathematical evidence for unsatisfiability.

The same package computes the full scalar irreducible character table of S_{10} and audits the associated normal-Cayley relaxation. The standard character gives only the sharp-transitivity bound $|S| \leq 10$, attained by every Latin line family. An exact one-view profile from the frozen order-10 diagnostic corpus remains feasible when copied into all three views and combined with the known sign-parity and intercalate-equality aggregates. Thus those scalar character inequalities and aggregates do not decide order 10; the witness is not an FFF Latin square.

Colour-parity holonomy `repro_runs/2026-08-11_fff_color_parity_holonomy/`.

The package constructs the dual flag cellulation on all four reduced order-4 squares, all 9408 reduced order-6 squares, all 230 frozen order-8 FFF representatives, and all 135 tracked order-10 partial tables. It checks 784315 atoms, 1918172 flags, 1568630 boundary colour counts, and 93662 FFF-component holonomy quotients with no error. The same run computes the canonical main-class holonomy profile: it has 152 values on the 230 order-8 FFF representatives, of which 109 are singletons. The order-10 records are diagnostic only.

Holonomy-to-cell intercalate coupling `repro_runs/2026-08-11_fff_holonomy_cell_intercalate/`. The package independently reconstructs all 92472 intercalate components in the same 9777 frozen tables. It checks 277416 oriented cell vectors, 369888 pointwise product values, and 366972 cell-degree identities with no error. This audits the complete multiplicity-four layer of Theorem 16.12; the order-10 records remain diagnostic only.

Exact-mask cell pushforwards `repro_runs/2026-08-11_fff_exact_mask_cell_pushforward/`. Across all 104645 flag components in the frozen 9777 tables, the package checks 8631774 cell-role involutions, 285399 exact-mask pushforwards, and 571632 forced zero marginals with no error. The order-10 records remain diagnostic only.

Cellwise holonomy Fourier transform `repro_runs/2026-08-11_fff_cellwise_holonomy_fourier/`. The package checks 390044 compatible potentials, 756704 component-cell incidences, 2039916 transforms, the same number of inversions, and 756704 Parseval identities with no error. The order-10 records remain diagnostic.

Affine cell-role transport `repro_runs/2026-08-11_fff_cell_role_affine_transport/`. The package checks 6119748 affine role transports, 2039916 orientation-displacement invariances, and 756704 local divisibility conditions with no error. The order-10 records remain diagnostic.

Cross-component cell parity `repro_runs/2026-08-11_fff_cross_component_cell_parity/`. The package checks 2011440 exact component-line/dual-edge identities and their parity consequences, plus 366972 each of the global cell cover, intercalate refinement, and multiplicity-one endpoint parity decomposition, with no error. The order-10 records remain diagnostic.

Gauge-invariant cell lift `repro_runs/2026-08-11_fff_gauge_invariant_cell_lift/`. The package checks 6119748 role-Fourier identities, 71672 pointwise odd-mask zero values, 403122 cubic congruences, and 1612488 gauge transformations with no theorem error. It also records 25828 exact counterexamples to the naive global cubic sum; the order-10 records remain diagnostic.

Quadratic Gram lift and gauge obstruction `repro_runs/2026-08-11_fff_component_gram_lift/`. The package checks 15876164 off-diagonal Fourier identities, 43457616 Gram-entry parity identities, 13710864 line-kernel identities, and the explicit Klein-four equivariant-anchor obstruction with no error. It records 179048 odd dimension-zero component-cell incidences as the exact blind layer of nontrivial mask lifts; the order-10 records remain diagnostic.

Component cell-edge partition `repro_runs/2026-08-11_fff_component_edge_partition/`. The package checks 15876164 component cell-Gram entries, 13230696 Erdős–Gallai inequalities, 2877258 dual-edge/global edge-partition incidences, and

1100916 triple-endpoint residual degrees on all 9777 frozen control tables with no error. It includes all 8788 exact-mask dimension-zero components; the order-10 records remain diagnostic.

Orientable cell-companion surfaces `repro_runs/2026-08-11_fff_cell_companion_surface/`. The package checks 2877258 opposite face orientations, 5754516 local link degrees and colour-neighbour sets, 858970 three-periodic link cycles, 104645 normalized genus formulas, and 92472 intercalate spheres on the 9777 frozen control tables with no error. The order-10 records remain diagnostic.

Normalized cell-incidence rank `repro_runs/2026-08-11_fff_normalized_cell_rank/`. The package independently compares normalized corner orbits with the cell-link components of Theorem 16.21 and checks 104645 exact binary ranks, 14142187 normalized Gram entries, 1918172 row factorizations, and 8316434 Gram aggregations on all 9777 frozen control tables with no error. It records 2160 order-6 components with $q_F = 3$ and one order-8 FFF component with $q_F = 2$; these are corpus observations, while the rank theorem is proved independently.

Joint atom/cell incidence rank `repro_runs/2026-08-11_fff_joint_atom_cell_rank/`. The package certifies the rational intersection dimensions of the exact atom and normalized-cell face-signal spaces from Theorems 16.3 and 16.22 on all 104645 components of the 9777 frozen tables. It stores 1113 explicit integer nonconstant common face-mode witnesses and reports no certification error. On the 230 order-8 FFF representatives its component profile has 201 values, 176 singletons, and maximum collision size three; combining it with the canonical holonomy profile of Computational Theorem 16.11 gives 202 values and 178 singletons. The 135 tracked order-10 tables are diagnostic only.

Common-mode gradient characterization `repro_runs/2026-08-11_fff_common_mode_gradient/`. The package constructs the three atom-transition multigraphs and the normalized and coarse opposite-potential edge systems on all 104645 components of the 9777 frozen tables. It checks 2877258 dual edges, certifies 1111 nonconstant integer coarse common-mode witnesses, and reports no error. All 1080 exceptional order-6 components are coarse; exactly one observed order-8 component has positive sheet-only quotient dimension, namely source 1955, component 4, with $(s_F, t_F, \eta_F) = (3, 1, 2)$. These finite uniqueness and $p_F = q_F$ observations are not universal claims, and the 135 order-10 tables remain diagnostic only.

External-input release audit `repro_runs/2026-08-09_paper_release_audit/`. It downloads the three cited ANU/Wanless inputs afresh, compares them with the frozen files, and compares the independently generated order-6 set with the official 9408-square file.

Paper sign and theorem audit `repro_runs/2026-08-10_paper_symbol_sign_audit/`. It exhaustively checks 1028307 two-line trades on all reduced Latin squares of orders 2, 4, and 6, with zero sign-table failures. It also supplies an explicit odd symbol-view trade that preserves the conventional row-column sign, and independently checks the computational theorem statements in its historical September 16 scope against frozen result objects. The additional statements in this edition use the separate snapshot audit, not a retroactive extension of that historical gate.

Every package contains a README, exact commands, an environment record, input and output manifests, and a package-level SHA-256 manifest. Paths and elapsed times are

treated as volatile comparison fields; pattern counts, witness sets, group-isotopy flags, and input hashes are not normalized away.

The computations use ordinary Python integer and permutation operations; no random sampling enters Computational Theorems 7.1 or 7.2. The repository also contains direct SAT/CP existence investigations of order 10, but those particular runs are deliberately excluded from the theorem set because they end in timeouts rather than a validated FFF model or a complete unsatisfiability certificate. Order-10 nonexistence is established instead by the two complete cycle-palette master searches in Computational Theorem 17.3. The bounded primitive-group clique computations are different again: their small finite formulas are decided and independently proof-checked.

Data and code availability

The local research repository contains the custom source programs, small inputs, commands, validation outputs, proof-check logs, and SHA-256 records for the stated results. This snapshot does not assert that every local artifact is already public or tracked. The large C221 traces and parts of the C235 proof collection remain outside Git; their hash-bound custody and regeneration instructions are recorded in their dated runs. Durable external access remains a publication gate. The two large generated order-10 graph files and their checkpoint trees are retained outside Git; their hashes, sizes, builders, complete semantic graph audits, and path-configurable rebuild commands are recorded in the corresponding repro packages. The external ANU and Wanless census files remain available from the cited data pages; the exact frozen copies used here are identified by cryptographic hashes and by the release audit above. Generated PDFs, solver binaries, and third-party software environments are not part of the source archive. A persistent DOI archive containing or durably linking the large external graph artifacts has not yet been deposited and remains a pre-submission task.

22 Discussion and open problems

Order 10 is excluded by the complete master searches in Computational Theorem 17.3. The explicit order-12 witness refutes the former power-of-two existence conjecture. Neither fact decides order 18. The earlier subset-action, sign, incidence, and flag-surface investigations remain structural results and diagnostic boundaries; their unresolved solver runs are not additional proofs of nonexistence.

The most concrete recent advance is constructive rather than an order-18 obstruction. Theorems 11.1 and 11.5 give an exact pattern formula and a within-family equivalence criterion. Their numerical consequences distinguish many main classes, but count only a specified family. They do not imply that every FFF square has a recoverable binary quotient or that arbitrary direct products cancel.

The position-value channel of Theorem 18.1 reduces the formula size without removing any view. The finite order-18 exclusions in Section 19 instead close specified construction routes. These are different forms of progress: an equivalent model is not a solved search, and exclusion of a route is not a complete order classification.

1. Decide existence at order 18, without replacing unrestricted Latin squares by an unjustified affine, quotient, symmetry, or trade ansatz.
2. Determine minimal dyadic exponents $h(m)$ beyond the cases already settled, and identify which upper bounds can be sharpened.

3. Find a structural explanation of the order-10 exclusion that does not require its exhaustive master graphs.
4. Determine which constructions produce classes outside the fixed affine and row-fibred families, and when weaker hypotheses still recover their quotient layers.
5. Complete a systematic priority comparison and independent expert review, including the computer-assisted dependencies.

These questions concern the even-cycle condition in three line views. They are not a resolution of the global Alon–Tarsi conjecture [AT92]. The research archive retains the historical investigation sequence; this edition separates that history from the currently supported theorem statements.

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